

Monetary Policy Surprises: A Cautionary Tale¹

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Abstract

Impulse responses to monetary policy shocks identified from high-frequency interest-rate surprises are large: larger than any conventional identification scheme delivers on the same reduced form, and larger still when the surprise is first orthogonalised on pre-announcement information. [Bauer and Swanson \(2023b\)](#) read that enlargement as the removal of an attenuation bias. I argue that it is residual contamination magnified by a weaker first stage. The correction is necessary, since the surprise is predictable from pre-announcement data, but at meeting frequency it removes proportionally more of the first stage than of the contamination. Because a proxy-SVAR divides whatever contamination survives by the first stage, the bias can be larger after the correction than before it, on impact and at every horizon. The constructive response is to widen the instrument window. The shock is the part of the policy decision not owed to economic conditions, and the announcement window reveals only the slice the market had not already priced. Instruments built from the decision itself, an extended [Romer and Romer \(2004\)](#) shock and a Cholesky-based instrument, carry contamination of their own but keep their first stage under the same purge, and their responses are not enlarged. When no single proxy is both strong and clean, a pool-within, annihilate-across estimator replaces a clean instrument with a restriction on the *structure* of the contamination: pooling *within* families of proxies that share a contamination direction buys strength, annihilating *across* families buys validity, and the direction is identified by the dispersion of relevance within families. On a panel of sixteen U.S. monetary proxies in five families, the estimator delivers a first stage of 24 and a standard monetary contraction.

Keywords: monetary policy shocks; high-frequency identification; predictability of monetary surprises; weak instruments; instrument pooling; proxy-SVAR.

JEL classification codes: E43, E52, E58, C32, C36

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1 Introduction

The workhorse for measuring the effects of monetary policy is now high-frequency identification: the change in interest-rate futures in a tight window around an FOMC announcement is treated as the surprise component of policy and used to instrument a VAR or a local projection (Gertler and Karadi, 2015; Gürkaynak et al., 2005; Kuttner, 2001; Nakamura and Steinsson, 2018). The responses it delivers are enlarged twice over. They are larger than what the same reduced form delivers under any conventional identification scheme. And they are larger still once the surprise is corrected for its known defect, although the correction leaves a weaker instrument: the estimated magnitude is largest exactly where the instrument is weakest.

The correction is necessary. The Fed responds to news the market has already seen (Bauer and Swanson, 2023a), so the announcement bundles the systematic response to the state of the economy with the policy innovation, and the surprise inherits a component that is predictable ex post from pre-announcement data (Bauer and Swanson, 2023b; Miranda-Agrippino and Ricco, 2021). The received remedy is to project the surprise on pre-announcement controls and keep the residual. However, that remedy has a non-negligible cost. The forecastable component contains a sizable portion of the first-stage strength of the instrument, so orthogonalising at meeting frequency leaves a weak instrument. The weaker, orthogonalised instrument delivers the larger responses. Bauer and Swanson (2023b) read this enlargement as the removal of an attenuation bias.

This paper argues that any residual contamination is magnified by a weaker first stage. A proxy-SVAR identifies the response to the mixture of shocks the proxy tracks, normalised so that the policy rate moves by one. The purge removes part of the contamination but generically not all of it, and since it also lowers the first stage, whatever contamination survives is magnified and displaced into every other variable, at every horizon. Given that the correction leaves the instrument weak, how can the identification of monetary policy shocks be pushed further?

This paper first makes the contamination–strength trade-off explicit. The response a proxy-SVAR identifies is the true response plus a fixed contamination direction divided by the first stage, and the first stage of any proxy is its relevance times the policy rate’s response to its own shock, plus the contamination the rate reacts to within the month (Section 3.1). The purge can therefore lower the first stage only by removing contamination, and the share of the first stage it removes is observable. It multiplies whatever bias survives by the ratio of the two first stages, and the bias is magnified when orthogonalising the surprise removes a smaller share of the contamination than of the first stage (Section 3.3). Both readings of the enlargement predict a larger purged response; what separates them is what the enlargement depends on, and the evidence points to the first stage. Across identification schemes run on the same reduced form, the high-frequency surprise delivers the largest responses. Across the two ways of orthogonalising it, the enlargement is ordered by first-stage loss: purging after monthly aggregation lowers the first stage from $F = 14.5$ to 7.1 and deepens the output trough from -1.9% to about -5% , while purging meeting by meeting lowers it to 3.6 and deepens the trough to -6.6% (Section 2.1). Weak-instrument-robust inference restores coverage but not the magnitude: it covers the displaced estimand rather than the truth (Section 3.4).

The paper then turns to instruments that remain strong once orthogonalised. Since the bias scales with the inverse of the first stage, the productive move is an instrument whose cleaning costs less first stage. A policy shock is a deviation from a rule given an information set, and the announcement window sees only the slice of it the market had not already priced: a decision signalled in advance reads zero on announcement day. Widening the window from the announcement to the decision lifts

that floor. An extended [Romer and Romer \(2004\)](#) shock removes the Fed’s own Greenbook forecasts from the intended target change at the meeting, and a Cholesky-based instrument removes the month’s output and price innovations from the policy-rate innovation; each is then orthogonalised on the same pre-announcement high-frequency controls as the surprise. Both keep their first stage under the purge that collapses the futures surprise, and their responses are not enlarged (Section 4). Neither is assumed to be clean and free from any residual contamination, but they are contaminated in different directions, and that difference is what the rest of the paper exploits.

Because most proxies are individually weak, the paper’s main step is to pool proxies that identify the same shock, so that their strength adds up. Simple pooling is not enough: averaging removes the contamination specific to each proxy, but not the contamination that a group of proxies has in common, which the pooled instrument inherits whatever its size. The paper therefore replaces the assumption of a perfectly exogenous instrument with a restriction on the *structure* of the contamination. Proxies that share a construction, a window and a purge form a family whose members share the mean of their contamination, a common component that the estimator must remove. Demeaning the identifying moments within families removes it exactly, without any clean instrument, and the target impact vector is the leading eigenvector of the within-demeaned moments in the metric of the reduced-form covariance (Section 5). What identifies the direction is the dispersion of relevance within families; what adds strength is a single pooled instrument whose weights are free within families and whose family totals annihilate the common contamination across them (Section 5.6). Hence the name of the estimator: pool within, annihilate across. Two order conditions say when it works: with m proxies in C families and n variables, at least one within-family contrast must survive, $m - C \geq 1$, and more families than non-policy shocks, $C > n - 1$, guarantees that a combination that is both strong and clean exists.

Finally, the paper applies the estimator to a panel of sixteen purged U.S. monetary proxies grouped on economic grounds into five families. The pooled instrument has a first stage of $F = 24$, against 3.6 for the orthogonalised surprise, and identifies a standard monetary contraction: after a +1 percentage-point funds-rate shock, output falls by 5.3% at a fourteen-month trough, the headline CPI falls at every horizon while core prices respond later, and the credit spread widens by 0.9 points on impact. The immediate fall in prices disappears: the price level falls by 0.1% in the month of the shock, against more than half a percent under the orthogonalised surprise, in line with a gradual transmission of monetary policy to prices. Relative to the orthogonalised surprise alone, pooling also cuts the price response at two years by two thirds and the credit response by half, and moderates the output response.

1.1 Related literature

The paper relates to three strands of the literature. The first is *high-frequency identification* of monetary policy: surprises measured in a tight window around FOMC announcements ([Acosta et al., 2025](#); [Gürkaynak et al., 2005](#); [Kuttner, 2001](#)), used as external instruments in proxy SVARs ([Gertler and Karadi, 2015](#); [Mertens and Ravn, 2013](#); [Stock and Watson, 2018](#)) or local projections ([Nakamura and Steinsson, 2018](#); [Plagborg-Møller and Wolf, 2021](#)). This paper adopts that machinery and asks what happens to the identified responses once the weakness of the instrument, and the contamination that produced it, are confronted directly.

The second is the *predictability of the surprises*. [Bauer and Swanson \(2023a\)](#) show that the evidence for a Fed information effect is equally consistent with the Fed and forecasters both responding

to economic news; [Bauer and Swanson \(2023b\)](#) document that surprises are predictable from pre-announcement data, propose the orthogonalisation studied here, and read the larger responses it produces as removing an attenuation bias. [Miranda-Agrippino and Ricco \(2021\)](#) condition on central-bank forecasts, [Jarociński and Karadi \(2020\)](#) separate policy from information shocks by the sign of the stock-price response, and [Bolhuis et al. \(2024\)](#) purge by elastic net. This paper states the bias-removal reading precisely enough to be tested against magnification, and shows that the credibility of a purged instrument must also be judged on its surviving first stage, not on the purge regression alone.

The third is *identification with weak, many or imperfect proxies*. Reaction-function residuals ([Coibion, 2012](#); [Romer and Romer, 2004](#)), recursive identification ([Bernanke and Blinder, 1992](#); [Christiano et al., 1999](#); [Sims, 1980](#)) and a growing menu of alternatives ([Bu et al., 2021](#); [Lewis, 2021](#); [Swanson, 2021](#)) are usually compared in a horse race; this paper reads them as the same construction at different windows, and treats the menu as a panel whose joint structure is itself identifying information. Weak-instrument diagnostics and inference ([Andrews et al., 2019](#); [Montiel Olea and Pflueger, 2013](#); [Staiger and Stock, 1997](#); [Stock and Yogo, 2005](#)) have been adapted to proxy-SVARs ([Jentsch and Lunsford, 2022](#); [Montiel Olea et al., 2021](#)).

Three recent papers relax the assumption that a single strong and valid proxy is available. [Angelini et al. \(2024\)](#) keep the proxies valid but, when those for the target shock are weak, identify it indirectly through proxies for the non-target shocks. [Huang and Neri \(2026\)](#) drop exact exogeneity in a zoo of proxies and bound each proxy’s correlation with the non-target shocks, which set-identifies the responses. Closest to the method of this paper, [Lin et al. \(2024\)](#) study a linear IV model with many individually weak instruments of unknown validity: identification rests on the rule that the valid instruments form the largest group of instruments that agree on the effect, and a penalised estimator selects them. This paper departs from all three in the same way: no proxy is assumed valid, for the target shock or for any other. Its contribution is to identify the target from the *structure* of the contamination rather than from its absence. The proxies of a family share the mean of their contamination, so differencing within families removes that common component exactly; every proxy is kept rather than selected out, and the dispersion of relevance within families point-identifies the target. The restriction is closest in spirit to [Kolesár et al. \(2015\)](#), who also allow every instrument to be invalid and identify the effect, as the number of instruments grows, when the instruments’ direct effects are uncorrelated with their first-stage effects. Assumption 1 is a within-family version of that condition: each family may carry a common contamination, related or not to its relevance, which differencing removes, and only the deviations from the family mean must be unrelated to relevance. Pooling within families then turns mostly weak proxies into a strong instrument, and annihilating across families keeps it clean. Because the proxies enter a single instrument and the direction is an eigenvector rather than an overidentified IV estimate, the many-instrument bias of overidentified estimators ([Bekker, 1994](#); [Mikusheva and Sun, 2022](#); [Newey and Windmeijer, 2009](#)) does not arise. For the heterogeneity across proxies, the estimator adapts the heterogeneity test of [Cochran \(1954\)](#) and the random-effects estimator of [DerSimonian and Laird \(1986\)](#).

The remainder is organised as follows. Section 2 presents the motivating fact and the received remedy. Section 3 develops the contamination–strength trade-off and the magnification it induces. Section 4 constructs the two instruments that remain strong under the purge, and Section 5 develops the pool-within, annihilate-across estimator. Section 6 reports the empirical application, Section 7 the robustness analysis, and Section 8 concludes.

2 The motivating fact and the received remedy

This section documents the two enlargements described in the introduction on the reduced form used throughout the paper, and defines the purge that the rest of the paper studies.

2.1 The motivating fact: the responses are large, and larger from the weaker instrument

The motivating evidence is assembled in Figure 1, which holds one reduced form and one normalisation fixed and varies the identification. The system is the monthly VAR of the empirical application (Section 6.1: industrial production, the consumer price index, the excess bond premium and the funds rate, six lags, estimated on 1973–2019), and the figure reports the responses of the four variables to a +1 percentage-point funds-rate shock.² Three of the identifications use the Bauer and Swanson (2023b) high-frequency surprise as an external instrument: once raw, and twice orthogonalised on pre-announcement information, by the two orderings of the purge and the monthly aggregation defined in Section 4.3. Purge-then-aggregate (PTA), the construction of Amodeo (2026) and the one used throughout the paper, purges each surprise at the meeting frequency and then aggregates to the month; aggregate-then-purge (ATP), the construction of Bauer and Swanson (2023b), aggregates the surprises to the month and then purges the monthly series.³ The shaded region is the range spanned by five conventional schemes: a recursive ordering, sign restrictions, sign restrictions disciplined by the narrative record, restrictions on the systematic component of policy, and a heteroskedasticity-based scheme (Appendix C.1 describes each). That region is a range across identifications rather than a confidence band, and it measures disagreement between schemes.

The instrument-identified responses are larger than any conventional scheme delivers. Under the raw surprise, a one-point move in the funds rate lowers output by about two percent at a twenty-month trough, already beyond every conventional scheme, although the price level barely moves. Orthogonalising the surprise enlarges every response, and the more first stage the purge removes, the larger the enlargement. The PTA surprise (orange dashed), which loses the most ($F = 14.5 \rightarrow 3.6$), takes output to -6.6% , prices to -3.0% after two years and the credit spread to two percentage points on impact, against -1.9% , -0.1% and a quarter of a point under the raw surprise; the ATP surprise (green dash-dotted), which loses less ($F = 7.1$), lies between the two in every panel, with an output trough close to -5% . Both orthogonalised surprises also reverse the funds rate, from about the sixth month under PTA and the seventh under ATP, which no conventional scheme does. Across the three lines strength and magnitude are therefore ordered in opposite directions: the correction that was meant to make the instrument credible made it weaker, and the responses it identifies grew. Nor is the enlargement a property of high-frequency measurement: three further high-frequency instruments are all more contractionary than the raw surprise, yet the orthogonalised one lies outside their range as well (Appendix Figure 9).

²One reduced form is used throughout the paper: for the single-instrument diagnostics of Sections 2.2 and 4 (Table 1 and Appendix Table 2), for this figure, and for the panel of Section 6. The VAR is estimated on 1973m1–2019m12 and each instrument’s identifying moment on the months it is observed within 1988m1–2019m12, the window over which the pre-announcement controls exist. On this system the surprise has $F = 14.5$ raw, $F = 3.6$ orthogonalised by purge-then-aggregate and $F = 7.1$ by aggregate-then-purge.

³Amodeo (2026) shows that purging at the monthly frequency can bias the purged surprise in months with more than one announcement, since aggregate-then-purge is then equivalent to orthogonalising with respect to the information available before the first announcement of the month.

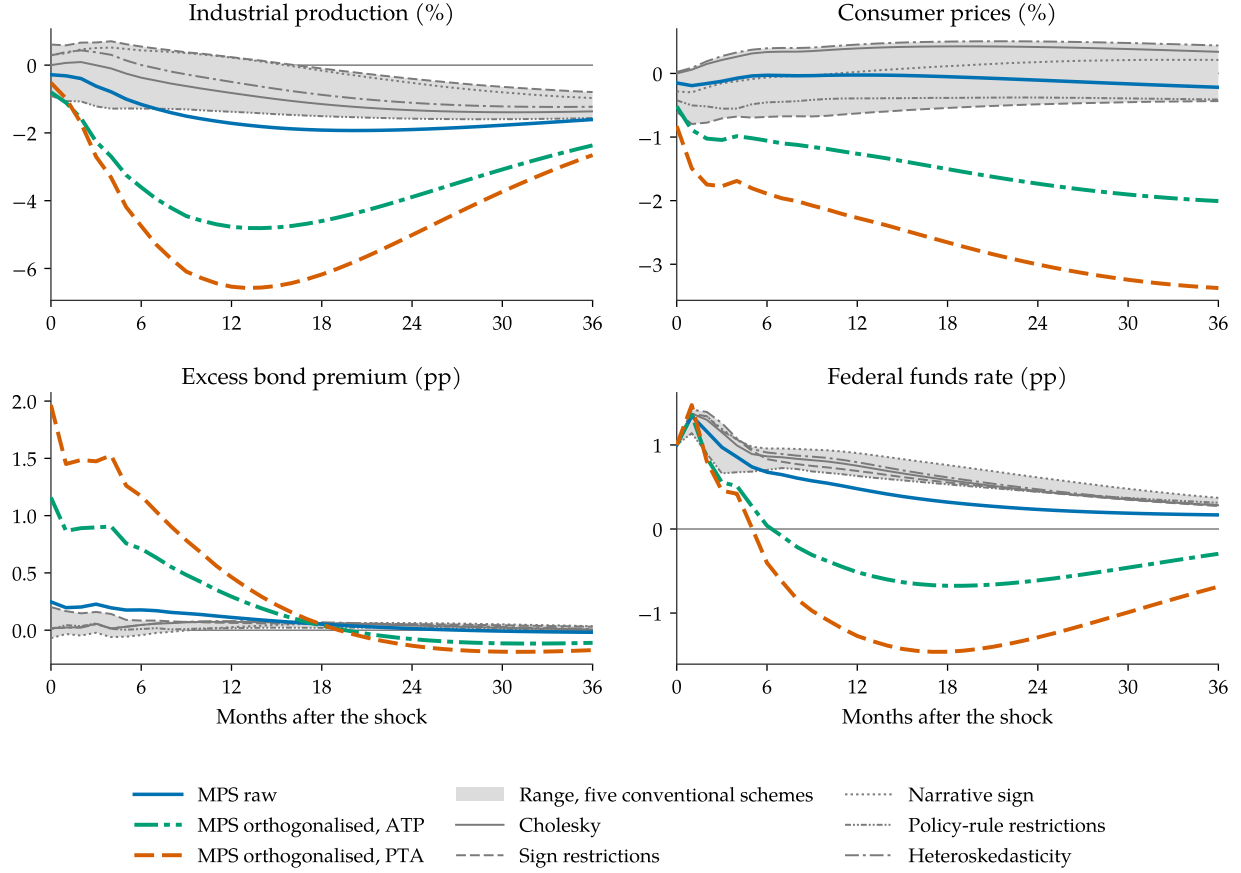


Figure 1: The largest responses are obtained with the high-frequency instruments. Responses to a +1 percentage-point funds-rate shock, on the same reduced form and the same normalisation throughout. Shaded: the range spanned by five conventional schemes, drawn individually as grey lines, namely a recursive (Cholesky) ordering, sign restrictions, sign restrictions with a narrative filter, restrictions on the systematic component of policy, and a heteroskedasticity-based scheme (Appendix C.1). Blue: the [Bauer and Swanson \(2023b\)](#) surprise, raw ($F = 14.5$). Green dash-dotted: the same surprise orthogonalised on pre-announcement information after monthly aggregation (ATP, $F = 7.1$). Orange dashed: orthogonalised announcement by announcement and then aggregated (PTA, $F = 3.6$). Both orthogonalised instruments lie outside the conventional range in every panel, the weaker one furthest from it; the raw surprise lies outside it for output, the credit spread and the funds rate, and inside it for prices. The shaded region is a min-max range across identification schemes, not a confidence band; set-identified schemes are shown at the pointwise median of the identified set.

The enlargement is visible already on impact: under PTA the price level falls by more than half a percent, and output by about half a percent, in the month of the announcement. So immediate a fall in prices sits uneasily with recent evidence on the timing of transmission, in which the aggregate price level takes years to respond (Aruoba and Drechsel, 2024a) and consumer prices decline only gradually even as spending reacts within weeks (Buda et al., 2025).

2.2 The remedy, and the bind it creates

High-frequency surprises are predictable from the Fed’s own Greenbook forecasts (Miranda-Agrippino and Ricco, 2021) and from public data dated before the announcement (Bauer and Swanson, 2023b). Appendix Table 2 reproduces the second finding on the sample of this paper: the six pre-FOMC controls jointly predict the surprise with $R^2 = 0.17$ over 1988–2019 ($F = 5.2$). The reading adopted throughout is that the Fed responds to news (Bauer and Swanson, 2023a): the forecastable component is the systematic response of policy to economic conditions that the market had not fully priced before the announcement.

The remedy of Bauer and Swanson (2023b) projects the surprise on a set $\mathbf{W}$ of pre-announcement controls (the nonfarm-payrolls surprise relative to the survey median, employment growth over the year to the latest release, the S&P 500 over the sixty-five trading days to the day before, the second principal component of the one- to ten-year yield curve over three months, commodity prices over three months, and the skewness implied by ten-year Treasury options over the prior month), and keeps the residual. Bauer and Swanson (2023b) apply the projection after aggregating the surprises to the month; this paper applies it meeting by meeting, before aggregation, following Amodeo (2026) (Section 2.1),

$$mps_m^\perp = mps_m - \text{Proj}(mps_m | \mathbf{W}), \quad (1)$$

where m indexes FOMC meetings and $\text{Proj}(y | \mathbf{X})$ denotes the linear projection of y on $\mathbf{X}$ and a constant. Throughout the paper $\mathbf{W}$ denotes these six controls, each measured before the announcement of the meeting it is paired with, so that the residual is orthogonal to pre-announcement information by construction. The purged surprise, summed within the month, is the object Section 3 calls $z_t^\perp$, and every instrument in the paper is the residual of a projection of this kind, differing only in its left-hand side and its conditioning set.

The controls are dated before the announcement but within the month, and in the framework of Section 3 they are combinations of the month’s non-policy shocks that the market had not yet priced and that the surprise therefore still carries. Purging removes the part of these shocks that the controls span. The result is an instrument orthogonal to the controls, not an exogenous one: it is clean only if the contamination lies entirely in the span of the controls, the condition stated in Corollary 2(i).

The bind is that richer controls reduce residual contamination but also strip first-stage variation: a proxy keeps its first stage under the purge only to the extent that its relevance did not rest on the contamination the controls see (Section 3.2). For the high-frequency surprise it did. Against the funds rate its first stage falls from $F = 14.5$ to $F = 3.6$, and it stays below 4 against every candidate policy rate (Table 1, Section 4.3). Whatever contamination survives the purge is then divided by a much smaller first stage; Section 3 makes this precise.

3 The contamination–strength trade-off and the magnification of impulse responses

Any proxy that loads on non-policy shocks identifies a contaminated impulse response. The proxy-SVAR recovers the response to the mixture of shocks the proxy tracks, which is the true response plus a bias equal to the contamination divided by the proxy’s first stage. The purge of Section 2.2 targets the numerator: orthogonalising the surprise on pre-announcement controls removes the contamination those controls span, but a finite set of controls need not span all of it, and whatever lies outside their span survives. The purge also reduces the denominator, because part of the first stage was carried by the contamination it removes. What survives is then divided by a smaller first stage, and when the purge removes proportionally more of the first stage than of the contamination, the bias after the purge exceeds the bias before it: the correction magnifies the endogeneity it was meant to remove. The fall in the first stage that accompanies the gain in the plausibility of exogeneity is the contamination–strength trade-off. Section 3.1 derives the bias of a contaminated proxy, Section 3.2 shows that the first stage the purge removes is the part carried by contamination, and Section 3.3 shows that the purge multiplies whatever bias survives by the ratio of the first stages, at every horizon, and states when it magnifies. Section 3.4 explains why weak-instrument-robust inference does not resolve the problem: it restores coverage for the estimand the purged surprise identifies, but that estimand is displaced by the surviving contamination, and a wider confidence set corrects neither the displacement nor the point estimate. The derivations are in Appendix A.

3.1 Identification with a contaminated proxy

Because the surprise carries the Fed’s response to news, it loads on the non-policy structural shocks. This subsection sets out the proxy-SVAR notation used throughout the paper and derives what that loading does to the identified impulse responses. Let $y_t \in \mathbb{R}^n$ follow a stationary monthly VAR with s lags, $y_t = c + \sum_{\ell=1}^s \Pi_\ell y_{t-\ell} + u_t$, with Wold representation $y_t = \mu + \sum_{k \geq 0} \Phi_k u_{t-k}$, where $\Phi_0 = I_n$ and $\Phi_k = \sum_{\ell=1}^{\min(k,s)} \Pi_\ell \Phi_{k-\ell}$, so that the response at horizon h to an impact vector b is $\Phi_h b$. The innovations are linked to the structural shocks by

$$u_t = B \varepsilon_t, \quad \varepsilon_t = \begin{pmatrix} \varepsilon_t^p \\ \varepsilon_t^{-p} \end{pmatrix}, \quad \mathbb{E} [\varepsilon_t \varepsilon_t'] = I_n, \quad B = [b_p \ b_{-p}], \quad (2)$$

where ε_t^p is the policy shock, with impact vector b_p , and ε_t^{-p} collects the non-policy shocks, with impact matrix b_{-p} ; the shocks are orthogonal with unit variance, so that $\Sigma_u := \mathbb{E}[u_t u_t'] = BB'$. With s_p the vector that selects the policy rate in y_t ,

$$u_t^p := s_p' u_t = b_{pp} \varepsilon_t^p + \lambda' \varepsilon_t^{-p}, \quad b_{pp} := s_p' b_p > 0, \quad \lambda' := s_p' b_{-p}, \quad \beta := b_p / b_{pp}, \quad (3)$$

where b_{pp} is the response of the policy rate to its own shock, λ its within-month response to the non-policy shocks, and β , with $s_p' \beta = 1$, the impact of a policy shock that raises the policy rate by one unit; the responses reported in the paper are $\Phi_h \beta$. The normalisation is a choice of units: a proxy identifies the impact vector only up to scale, and any variable with a non-zero impact response could serve, but normalising on the policy rate makes the normalising coordinate of the identifying moment the proxy’s first stage. The variance of the policy innovation is $\text{Var}(u_t^p) = b_{pp}^2 + \|\lambda\|^2$.

A potentially contaminated proxy is

$$z_t = \phi \varepsilon_t^p + \psi' \varepsilon_t^{-p}, \quad (4)$$

with relevance $\phi > 0$ and *contamination loading* $\psi \in \mathbb{R}^{n-1}$: ψ_j is how much of a one-standard-deviation non-policy shock j the surprise carries.⁴ This is the proxy-SVAR of [Mertens and Ravn \(2013\)](#) and [Stock and Watson \(2018\)](#) with the exogeneity condition $\psi = \mathbf{0}$ possibly failing. Because u_t depends on the month's shocks alone, only this contemporaneous condition matters: a correlation of the proxy with past or future shocks does not enter the identifying moment, and any part of the proxy predictable from lagged information is absorbed by the VAR lags. The identifying moment and the first stage are

$$\mathbb{E}[u_t z_t] = \phi b_p + b_{-p} \psi, \quad \pi := s_p' \mathbb{E}[u_t z_t] = \text{Cov}(z_t, u_t^p) = \phi b_{pp} + \lambda' \psi. \quad (5)$$

The first stage has two sources: ϕb_{pp} because the surprise contains the policy shock, and $\lambda' \psi$ because the non-policy news it carries also moves the policy rate within the month. The proxy-SVAR estimand is the moment normalised to a unit policy coordinate,

$$\tilde{\beta} := \frac{\mathbb{E}[u_t z_t]}{s_p' \mathbb{E}[u_t z_t]} = \frac{\phi b_p + b_{-p} \psi}{\pi}, \quad s_p' \tilde{\beta} = 1, \quad (6)$$

the probability limit of the proxy-SVAR estimator and the impact vector of the composite shock $\phi \varepsilon_t^p + \psi' \varepsilon_t^{-p}$ the surprise tracks. A valid proxy ($\psi = \mathbf{0}$) delivers β ; otherwise $\tilde{\beta} - \beta$ is an asymptotic bias. Sampling variability is taken up in Section 3.4. The bias follows from (6) in one step. The policy-shock part of the moment is $\phi b_p = \phi b_{pp} \beta$, and by (5) $\phi b_{pp} = \pi - \lambda' \psi$, so that $\phi b_p = \pi \beta - \beta \lambda' \psi$. Substituting,

$$\tilde{\beta} = \frac{\pi \beta + b_{-p} \psi - \beta \lambda' \psi}{\pi} = \beta + \frac{d}{\pi}, \quad d := D \psi, \quad D := b_{-p} - \beta \lambda', \quad s_p' d = 0, \quad (7)$$

and hence $\Phi_h \tilde{\beta} = \Phi_h \beta + \Phi_h d / \pi$ at every horizon.⁵ Because B is nonsingular, $D \psi = \mathbf{0}$ would require $B(-\lambda' \psi / b_{pp}, \psi')' = \mathbf{0}$, so $d = \mathbf{0}$ iff $\psi = \mathbf{0}$: any contamination biases the impact vector. The direction d has a direct reading. $b_{-p} \psi$ is the impact of the non-policy news the surprise carries; that news moves the policy rate within the month by $\lambda' \psi$, and $\beta \lambda' \psi$ is what a policy shock of that size would do. So d is the impact of the contamination *net of the policy move it induces*, and $s_p' d = 0$: the normalisation fixes the policy rate's impact response, so the contamination is displaced entirely into the other variables.

Measuring the bias. Comparing biases across proxies requires a size for the vector $\tilde{\beta} - \beta = d / \pi$. Let $\|\cdot\|$ be any norm on $\mathbb{R}^n$. Because a norm is homogeneous, $\|\tilde{\beta} - \beta\| = \|d\| / \pi$ whichever one is

⁴Classical measurement noise $\omega_t \perp \varepsilon_{t+j}$ for all j can be added to (4); it changes no covariance with a structural object and lowers the first-stage F and nothing else. The scale of z_t is free, since the IV estimand is invariant to rescaling the instrument. Measured in the units of the policy rate, a proxy that registers a fraction γ of the policy rate's response to its own shock has $\phi = \gamma b_{pp}$.

⁵The same estimand comes out of LP-IV with the VAR lags as controls ([Plagborg-Møller and Wolf, 2021](#); [Stock and Watson, 2018](#)).

used, and the same holds at horizon h with $\Phi_h d$ in place of d , so every comparison below can be read on impact or at any horizon.⁶

Corollary 1 (Ranking proxies). *Let $z_t^{(1)}$ and $z_t^{(2)}$ be two proxies of the form (4) for the same policy shock in the same VAR, with contamination directions $d^{(1)}, d^{(2)}$ and first stages $\pi^{(1)}, \pi^{(2)} > 0$. Then*

$$\|\tilde{\beta}^{(1)} - \beta\| \leq \|\tilde{\beta}^{(2)} - \beta\| \iff \frac{\|d^{(1)}\|}{\pi^{(1)}} \leq \frac{\|d^{(2)}\|}{\pi^{(2)}}. \quad (8)$$

The less contaminated proxy, $\|d^{(1)}\| < \|d^{(2)}\|$, is the more biased one whenever $\pi^{(1)} / \pi^{(2)} < \|d^{(1)}\| / \|d^{(2)}\|$.

The corollary shows that neither contamination nor strength alone ranks proxies: the ratio does. Exogeneity is usually maximised at any cost in precision, on the reasoning that a valid instrument is at worst noisy. That is correct only when the instrument is exactly valid: with any residual contamination the same defect is divided by π , so buying a small reduction in ψ at the price of a large reduction in π leaves the estimate further from the truth.

3.2 What orthogonalisation removes

Orthogonalisation is the operation (1) of Section 2.2: it regresses the surprise on pre-announcement controls $\mathbf{W}_{t-\Delta}$, dated before the announcement, with $\mathcal{W}$ the span of the controls and a constant and $P_{\mathcal{W}}$ the projection on it, so that $\text{Proj}(z_t \mid \mathbf{W}) = P_{\mathcal{W}} z_t$, and keeps the residual $z_t^\perp = (I - P_{\mathcal{W}})z_t = z_t - \text{Proj}(z_t \mid \mathbf{W})$, discarding the fitted value $\hat{z}_t = P_{\mathcal{W}} z_t$. The controls are taken to be orthogonal to the policy shock, $P_{\mathcal{W}} \varepsilon_t^p = 0$, so that the purge acts only on the contamination, which becomes $\psi^\perp := \mathbb{E}[\varepsilon_t^{-p} z_t^\perp]$; the purged first stage is $\pi^\perp := \text{Cov}(z_t^\perp, u_t^p)$, and $d^\perp := b_{-p} \psi^\perp - \beta \lambda' \psi^\perp$, $\tilde{\beta}^\perp = \beta + d^\perp / \pi^\perp$ are (7) evaluated at them. Throughout, $\pi > 0$ and $\pi^\perp > 0$: the raw surprise has a positive first stage by construction, and a purged one does whenever $\lambda' \psi^\perp > -\phi b_{pp}$, which holds automatically in the case of Corollary 2(ii), where $\pi^\perp = \phi b_{pp}$.

Proposition 1 (What the purge does to the pair). *If $P_{\mathcal{W}} \varepsilon_t^p = 0$:*

- (i) $\|\psi^\perp\|^2 \leq \|\psi\|^2 - \text{Var}(P_{\mathcal{W}} \psi' \varepsilon_t^{-p})$: *validity improves relative to the raw proxy, strictly if the menu predicts any contamination.*
- (ii) $\pi^\perp = \phi b_{pp} + \lambda' \psi^\perp$: *the purged first stage has the same two-term form, so every unit of first stage above ϕb_{pp} is bought with contamination, and a fully valid instrument has $\pi^\perp = \phi b_{pp}$ exactly.*
- (iii) *The share of the raw first stage lost to the purge is observable:*

$$\frac{\pi^\perp}{\pi} = 1 - \delta_\pi, \quad \delta_\pi := \frac{\text{Cov}(\hat{z}_t, u_t^p)}{\text{Cov}(z_t, u_t^p)}. \quad (9)$$

Part (i) says the purge cannot make the surprise less valid than it was, but only in the comparison with no purge at all: it orders a menu against no menu, not menus against one another. The proposition needs nothing about the controls beyond their orthogonality to the policy shock. Part

⁶Only homogeneity is used, so the statements also hold for a single response, measured by $|e_i' \Phi_h d|$ with e_i the i -th unit vector, wherever it is non-zero.

(ii) is the contamination–strength trade-off: the purge can lower the first stage *only* by removing contamination the policy rate reacts to, and it reaches the floor ϕb_{pp} exactly when it has removed all of it. The floor has a direct reading: a valid first stage cannot fall below ϕb_{pp} , the proxy’s relevance times the policy rate’s response to its own shock. A purge that leaves the first stage above ϕb_{pp} has left contamination the rate responds to, and one that drives it to ϕb_{pp} has paid for validity in the rate’s direction with the whole of the first stage above the floor: what the controls strip from the first stage is the part carried by contamination. Part (iii) measures that cost: δ_π is the share of the raw covariance with the policy innovation that lies in the fitted value of the purge regression.

The predictability–strength correspondence. Raw monetary policy surprises are predictable from information available before the announcement (Section 2.2), so the raw surprise decomposes into its predictable part and the purged residual, $z_t = \hat{z}_t + z_t^\perp$. Its predictability is measured by the R^2 of the regression of z_t on the controls, $R^2 := \text{Var}(\hat{z}_t) / \text{Var}(z_t)$, and its strength by the correlation with the policy innovation, $\rho := \text{Corr}(z_t, u_t^p)$ for the raw surprise, $\rho^\perp := \text{Corr}(z_t^\perp, u_t^p)$ for the purged one and $\hat{\rho} := \text{Corr}(\hat{z}_t, u_t^p)$ for the predictable part. The more predictable the surprise, the more first-stage strength the purged instrument loses. Since $\hat{z}_t$ and $z_t^\perp$ are orthogonal, the purge removes a share R^2 of the variance of the instrument and, by (9), a share δ_π of its covariance with u_t^p , so that

$$\rho^\perp = \frac{1 - \delta_\pi}{\sqrt{1 - R^2}} \rho, \quad \delta_\pi = \sqrt{R^2} \frac{\hat{\rho}}{\rho}, \quad \frac{F^\perp}{F} \approx \frac{(1 - \delta_\pi)^2}{1 - R^2}, \quad (10)$$

where the last expression uses $F = T\rho^2 / (1 - \rho^2)$ for a single instrument (Stock and Yogo, 2005) and holds when both correlations are small. The first expression in (10) records the two effects: the numerator falls by the share δ_π of the covariance that the predictable part carried, the denominator by the factor $\sqrt{1 - R^2}$ because the instrument has less variance left. The second expression says where δ_π comes from. The covariance the purge removes is the size of the predictable part, $\sqrt{R^2}$ in standard-deviation units, times how closely that part moves with the policy rate relative to the surprise as a whole, $\hat{\rho} / \rho$. For a given $\hat{\rho}$, a more predictable surprise therefore loses a larger share of its first stage. The third expression restates the first in the units of the first-stage F , which is approximately proportional to the squared correlation.

Whether strength rises or falls depends on which of the two effects dominates:

$$\rho^\perp < \rho \iff \delta_\pi > 1 - \sqrt{1 - R^2} \approx \frac{1}{2} R^2. \quad (11)$$

The covariance falls one for one with δ_π , whereas the standard deviation falls only with the square root of $1 - R^2$, so the variance effect is small: removing a fifth of the variance lowers the standard deviation by about ten percent. A surprise with $R^2 = 0.2$ therefore loses strength as soon as its predictable part accounts for more than about ten percent of its covariance with the policy rate. If the predictable part were instead noise, $\hat{\rho} = 0$, then $\delta_\pi = 0$ and the purge would raise strength by $1 / \sqrt{1 - R^2}$. Given the raw and purged F and the R^2 of the purge regression, (10) returns δ_π ; Appendix B applies it to the high-frequency surprise.

3.3 The magnification of the surviving bias

By (7) the bias is a numerator, the contamination direction d , divided by the first stage, and the purge acts on both: it removes part of the contamination, and by Proposition 1(ii) it lowers the first stage by exactly the covariance that the removed contamination carries. Whatever survives is then divided by a smaller number, which $\Phi_h \tilde{\beta}^\perp$ carries to every horizon. The ratio of the bias after the purge to the bias before it puts the two effects together.

Proposition 2 (Magnification). *Let $d^\perp = D\psi^\perp$, let $\|\cdot\|$ be any norm on $\mathbb{R}^n$ and $\psi \neq \mathbf{0}$, and write $\delta_d := 1 - \|d^\perp\|/\|d\| \leq 1$ for the share of the contamination the purge removes, the counterpart of $\delta_\pi = 1 - \pi^\perp/\pi$ for the numerator. The factor by which the purge scales the bias is*

$$\varrho := \frac{\|\tilde{\beta}^\perp - \beta\|}{\|\tilde{\beta} - \beta\|} = \frac{\|d^\perp\|/\|d\|}{\pi^\perp/\pi} = \frac{1 - \delta_d}{1 - \delta_\pi}, \quad (12)$$

and the denominator is observable by (9). Since $\pi, \pi^\perp > 0$, $\delta_\pi < 1$, of either sign, and

$$\varrho > 1 \iff \delta_d < \delta_\pi, \quad \varrho \leq \frac{1}{1 - \delta_\pi} \text{ whenever } \delta_d \geq 0, \quad (13)$$

with equality in the bound iff $\delta_d = 0$. The purge magnifies the bias iff it removes a smaller share of the contamination than of the first stage.

Equation (12) has two factors. The first, $1/(1 - \delta_\pi)$, is observable and is one number: it multiplies whatever bias survives the purge, and it is the same for every norm, every outcome and every horizon. The second, $1 - \delta_d$, is the share of the contamination that survives. It depends on *which* non-policy news the purge removes and, unless $d^\perp$ is collinear with d , on the outcome and the horizon, so the purge can reduce the bias in one response and magnify it in another while multiplying what survives in both by the same observable number. Equation (13) compares the two: removing contamination is not enough; the purge has to remove a larger share of the contamination than of the first stage, and since by Proposition 1(ii) the first stage above ϕb_{pp} is itself contamination, a purge that removes exactly what the policy rate reacts to can take out most of π while taking out only the part of d that happens to lie in that direction. The mechanism requires $\delta_\pi > 0$. The sign of δ_π is that of $\lambda'(\psi - \psi^\perp)$, the policy-rate response to the contamination removed, which is positive when the surprise and the rate load on the removed news with the same sign, as under the “Fed responds to news” reading (Bauer and Swanson, 2023a). If the removed component instead moved the rate against the surprise, the purge would raise the first stage and attenuate whatever survives; the estimate of δ_π says which case the data are in.

A purged response larger than the raw one admits two readings. In the reading of Bauer and Swanson (2023b), the purge removes contamination that attenuated the raw estimate, $\delta_d > \delta_\pi$, and the purged estimate is closer to β . In the reading developed here, the purge potentially removes part of the contamination but not its entirety. It lowers the first stage in the process, so that whatever contamination survives is magnified by up to $1/(1 - \delta_\pi)$. Under this reading, the enlargement is ordered by δ_π across instruments purged with the same controls (Section 4.4), and a treatment that addresses the same contamination without lowering the first stage does not enlarge the responses. A larger purged response is therefore not by itself a better estimate of β : it may be residual contamination magnified by a weaker first stage.

The corollary below makes the two shares explicit for the purge actually performed. It adds one assumption: the controls are exact functions of the month's pre-announcement structural shocks, $\mathbf{W} = A' \varepsilon_t^{-p}$ for a loading matrix A . The purge is then a projection within shock space: it removes the part of ψ in the span of the controls and leaves the rest as it was, so nested menus shrink $\|\psi^\perp\|$ monotonically.

Corollary 2 (Projection on a finite menu). *Let $\mathbf{W} = A' \varepsilon_t^{-p}$, $\mathcal{A} := \text{col}(A)$ and $P_A := A(A'A)^{-1}A'$, so that $P_W(\psi' \varepsilon_t^{-p}) = (P_A \psi)' \varepsilon_t^{-p}$ and $\psi^\perp = (I - P_A)\psi$.*

- (i) *The purge removes from the first stage and from the contamination direction exactly what the projected contamination $P_A \psi$ contributed to each,*

$$\pi - \pi^\perp = \lambda' P_A \psi, \quad d - d^\perp = D P_A \psi, \quad \text{so} \quad \delta_\pi = \frac{\lambda' P_A \psi}{\pi}, \quad \delta_d = 1 - \frac{\|D(I - P_A)\psi\|}{\|D\psi\|}, \quad (14)$$

and residual contamination is generic: $\psi^\perp = \mathbf{0}$ iff $\psi \in \mathcal{A}$.

- (ii) *If $\lambda \in \mathcal{A}$, so that the menu spans the direction the policy rate reacts to, then $\pi^\perp = \phi b_{pp}$ and*

$$\tilde{\beta}^\perp = \beta + \frac{b_{-p} \psi^\perp}{\phi b_{pp}}, \quad 1 - \delta_\pi = \frac{\phi b_{pp}}{\phi b_{pp} + \lambda' P_A \psi}, \quad (15)$$

so whatever survives is divided by the policy rate's response to its own shock, and $\varrho > 1$ iff $\delta_d < \delta_\pi = \lambda' P_A \psi / (\phi b_{pp} + \lambda' P_A \psi)$: the purge magnifies unless the share of the contamination it removes exceeds the share of the raw first stage that the contamination carried.

- (iii) *Adding one control direction $w \perp \mathcal{A}$, $\|w\| = 1$, is itself a purge, applied to the already purged proxy, and Proposition 2 applies to it with $(\psi^\perp, \pi^\perp, d^\perp)$ in the role of the raw triple. With $k_w := w' \psi^\perp$ the contamination along w and $\ell_w := \lambda' w$ the policy rate's response to it, the addition removes $\ell_w k_w$ from $\pi^\perp$ and the vector $k_w D w$ from $d^\perp$, so the shares it removes are*

$$\delta_\pi^{(w)} := \frac{\ell_w k_w}{\pi^\perp}, \quad \delta_d^{(w)} := 1 - \frac{\|d^\perp - k_w D w\|}{\|d^\perp\|}, \quad \varrho^{(w)} = \frac{1 - \delta_d^{(w)}}{1 - \delta_\pi^{(w)}}, \quad (16)$$

and by (13) the addition lowers the surviving bias iff $\delta_d^{(w)} > \delta_\pi^{(w)}$: it removes a larger share of the surviving contamination than of the surviving first stage.

Part (ii) is the empirically relevant case. A menu of macroeconomic news is built to span what the central bank reacts to, and once it does the purged first stage is at its floor ϕb_{pp} : the orthogonalised surprise is valid in the direction of the policy rate, and the identified response is the true response plus the outcome's response to the surviving non-policy news, divided by the policy rate's response to its own shock. The surviving news is the part of the month's non-policy shocks that lies outside the span of the controls. This is also the case in which the calibration of Appendix B is exact.

Part (iii) gives the effect of adding one control. A direction the policy rate does not react to, $\ell_w = 0$, costs no first stage and lowers the bias whenever it removes contamination, $\delta_d^{(w)} > 0$. A direction the policy rate reacts to strongly but the outcome does not can remove a larger share of the first stage than of the contamination, and then magnifies the bias. For a menu of macroeconomic

news the two kinds of direction coincide: what the central bank responds to within the month is also what output and prices respond to, so the controls that remove the most contamination also remove the most first stage. Once the menu spans λ the first stage is at its floor and further controls are free, but whatever contamination remains is already divided by ϕb_{pp} . Because $\delta_{\pi}^{(w)}$ has the sign of $\ell_w k_w$, the first stage need not fall monotonically as controls are added; only the endpoint ϕb_{pp} is ordered. The remedy the corollary points to is therefore a better instrument rather than a better menu of controls.

3.4 Why weak-instrument-robust inference does not rescue the magnitude

The purge leaves the surprise weak as well as contaminated, and the standard response to a weak first stage is weak-instrument-robust inference.⁷ An Anderson–Rubin-type confidence set has correct coverage uniformly in instrument strength (Montiel Olea et al., 2021), and moving-block-bootstrap critical values extend it to the proxy-SVAR with serially dependent data (Jentsch and Lunsford, 2022). When the first stage is weak the robust set can be unbounded, which reflects how little the instrument identifies. Robust inference does not, however, address contamination. Coverage is coverage of the estimand, and by (7) the estimand of a contaminated proxy is $\tilde{\beta}$, not β : a robust set built from $z_t^{\perp}$ covers $\beta + d^{\perp} / \pi^{\perp}$ with the stated probability whatever the strength of the first stage, and so inherits the magnification of Proposition 2 in full. Robust inference is robust to weakness, not to invalidity (Andrews et al., 2019). Contamination and weakness are distinct failures, and a wider confidence set repairs neither the bias nor its magnification.

4 Reviving the instrument: Romer–Romer and recursive proxies under the purge

Section 3 leaves the high-frequency surprise in a bind. Raw, it is contaminated; orthogonalised, it is weak, and by (7) whatever contamination the purge leaves is magnified by the smaller first stage. An ideal instrument would be unpredictable from pre-announcement information and strong at the same time: validity from the window, relevance from the size of the policy shock it captures. The surprise buys the first and, once orthogonalised, loses the second.

The reason it loses the second is what the window measures. A policy shock is a deviation from a rule given an information set. The policy shock of the VAR, ε_t^p , is the part of u_t^p not explained by the month’s non-policy shocks: it contains everything the Fed did in month t that was not a reaction to the state, whenever the market learned of it, whether in the announcement window, in a speech by the Fed Chair two weeks earlier, or from the minutes released afterwards. The futures surprise measures only the slice of ε_t^p revealed inside the window, and any deviation from the rule that was signalled in advance is priced by announcement day. This suggests one reading of the weak first stage after orthogonalisation. If part of the policy shock is conveyed to market participants before the announcement, it is already in the market’s expectation in the minutes before the window, and the surprise carries little of the shock.

One remedy for the weakness of the purged surprise proposed in the literature is to enlarge the *event set* rather than the window: adding Fed Chair speeches and other communication to the FOMC announcements, so that more monetary news survives the purge (Bauer and Swanson, 2023b). The trade-off of Section 3 suggests why this need not restore the first stage. A speech

⁷Weakness is a problem in its own right: under weak identification the IV estimate is biased toward its least-squares counterpart, which in a proxy-SVAR is the recursive response obtained by ordering the policy indicator first in a Cholesky decomposition of the residual covariance matrix (Stock and Watson, 2018).

rarely deviates from the rule. It conveys news about the rule itself, for instance that the Fed will respond more aggressively to a given level of inflation, or about the Fed’s view of the economic outlook: a shift in the reaction function or information about the state, not a policy shock. A policy shock conveyed in a speech would amount to announcing a change in the rate at the next meeting unrelated to the state of the economy.

This section widens the instrument window instead. Each instrument measures the deviation from the rule relative to a different information set: the futures surprise, the change in futures prices around the announcement, measures it relative to what the market expected at $t - \Delta$; the Romer–Romer residual relative to the Fed’s own forecasts; and the recursive innovation relative to what the VAR implies given the month’s output and price innovations. The two instruments below target the policy *decision* rather than the announcement, and both are built the same way: as the left-hand side of a reaction function, the rate change that a regression on the policymaker’s information set does not explain. The extended [Romer and Romer \(2004\)](#) shock takes the intended target change at the meeting and removes the Fed’s own Greenbook forecasts; the recursive instrument takes the monthly innovation in the funds rate and removes the contemporaneous output and price innovations. Each is then orthogonalised on the same pre-announcement controls as the surprise. A wider window admits other shocks, so both carry contamination of their own; but they better preserve their first stage under the purge.

4.1 The extended Romer–Romer instrument

The Romer–Romer instrument is a reaction-function residual ([Romer and Romer, 2004](#)). Its left-hand side is the FOMC’s *intended* change in the funds-rate target at meeting m , Δi_m , recovered from the FOMC record: the decision itself, rather than the market’s revision of its forecast of it. The systematic response is removed with the Fed’s own information set: the pre-meeting target level i_{m-} and the real-time Greenbook forecasts $\mathbf{G}_m$: output growth and inflation from the previous quarter to two quarters ahead, levels *and* revisions relative to the previous meeting’s Greenbook, and the current-quarter unemployment forecast. The Romer–Romer shock is what the projection on that information set does not explain,

$$\varepsilon_m^{\text{RR}} = \Delta i_m - \text{Proj}(\Delta i_m \mid i_{m-}, \mathbf{G}_m), \quad (17)$$

the part of the intended target change orthogonal to the Fed’s own information set (the regression is written out in Appendix [D](#), equation (57)). We then subject it to the *same* pre-FOMC purge applied to the high-frequency surprise: with $\mathbf{W}_{m-\Delta}$ the Bauer–Swanson controls dated before meeting m , the instrument is the residual of the projection on the Greenbook set *and* the controls jointly,

$$Z_m^{\text{RR},\perp} = \varepsilon_m^{\text{RR}} - \text{Proj}(\varepsilon_m^{\text{RR}} \mid \mathbf{W}_{m-\Delta}), \quad (18)$$

which by Frisch–Waugh–Lovell is a two-step residual: purge the Romer–Romer shock of the part of the controls that the forecasts do not already explain,

$$Z_m^{\text{RR},\perp} = \varepsilon_m^{\text{RR}} - \text{Proj}(\varepsilon_m^{\text{RR}} \mid \tilde{\mathbf{W}}_m), \quad \tilde{\mathbf{W}}_m = \mathbf{W}_{m-\Delta} - \text{Proj}(\mathbf{W}_{m-\Delta} \mid i_{m-}, \mathbf{G}_m). \quad (19)$$

The residual predictability that this purge removes has *two* distinct sources, and separating them is the point of the FWL construction. The first is *timing*. The Greenbook closes about six days

before the meeting, so information arriving in that window, an employment surprise, a market move, enters the public record but *not* the forecast set $\mathbf{G}$; if the Committee reacts to it at the meeting, the Romer–Romer residual $\varepsilon_m^{\text{RR}}$ of (17) still carries a systematic-response component, visible as predictability from the pre-FOMC controls $\mathbf{W}$. The second is that the two orthogonalisation are simply *not the same projection*: purging on the Greenbook $\mathbf{G}$ (the Fed’s internal forecast) and purging on the Bauer–Swanson controls $\mathbf{W}$ (market-based pre-announcement data) remove different contamination, because neither control set fully spans the other. The joint purge (18), in its two-step form (19), strips both the Greenbook-spanned systematic response and the systematic response to news that arrives after the Greenbook closes. The latter is contamination rather than shock, so removing it leaves the policy-shock part ϕb_{pp} of the first stage intact; empirically, the first stage does not fall.

The purge leaves the first stage essentially unchanged ($F = 29.1$ to 29.6), although the residual is predictable from $\mathbf{W}$ ($R^2 = 0.11$, against 0.17 for the raw surprise, mostly through the slope of the yield curve and Treasury skewness, Appendix Table 2). The identifying variation is the decision. It contains the part of the shock that reached the market before the announcement, which the futures surprise nets out, and the stable first stage indicates that the pre-announcement controls see little of that part.

4.2 The Cholesky-based instrument

The recursive instrument widens the window further, to the whole month. Its left-hand side is the policy-rate innovation of the VAR, u_t^p , the part of the month’s policy rate not predicted by the lags of the system: it contains the full announcement, and everything else the rate does within the month, rather than the slice of the decision revealed inside the futures window. Label the variables of the VAR output (o), prices (c , for the CPI), the policy rate (p) and the excess bond premium as the financial variable (f), with reduced-form innovations $(u_t^o, u_t^c, u_t^p, u_t^f)$, structural shocks $(\varepsilon_t^o, \varepsilon_t^c, \varepsilon_t^p, \varepsilon_t^f)$, and b_{ij} the impact response of variable i to shock j , so that b_{pp} is the policy rate’s response to its own shock of (3) and ε_t^f is the financial shock, the structural innovation to the excess bond premium. The instrument is the policy innovation residualised on the contemporaneous output and price innovations, as in the recursive identification of monetary policy shocks (Bernanke and Blinder, 1992; Christiano et al., 1999; Sims, 1980),

$$Z_t^{\text{CHOL}} = u_t^p - \text{Proj}(u_t^p \mid u_t^o, u_t^c) = b_{pp}\varepsilon_t^p + b_{pf}\varepsilon_t^f, \quad (20)$$

with the policy variable the effective federal funds rate. It is a reaction-function residual whose information set is what the VAR sees within the month: the policy rate is allowed to react to output and prices contemporaneously, and the instrument is what the rate does beyond that reaction. In the application the innovation is taken from the instrument database’s own recursive VAR, not from the headline VAR, and enters every specification as an external proxy.

The second equality in (20) uses the first two of the three sets of restrictions imposed by the recursive ordering (o, c, p, f) . On the policy shock, it does not move output and prices within the month of the decision, $b_{op} = b_{cp} = 0$. On the financial shock, it does not move output and prices within the month, $b_{of} = b_{cf} = 0$, which makes the projection remove exactly the output and price shocks, and the rate does not respond to it within the month, $b_{pf} = 0$. The instrument needs only the first. If $b_{op} = b_{cp} = 0$, the innovations u_t^o and u_t^c are orthogonal to ε_t^p .

The restriction on the policy shock is where the timing of the two conditioning sets matters. The pre-announcement controls are dated before the decision, whereas u_t^o and u_t^c are monthly innovations and include what happens in the rest of the month, after the decision. Conditioning on them removes none of the policy shock only if output and prices do not respond to it before the end of the month: the policy shock moves them from the start of the next month on. This is the monthly counterpart of the transmission lag. It is weaker than the recursive identification of the shock, since it restricts the response to the policy shock alone and leaves the responses to the non-policy shocks free.

Two things distinguish the object from the surprise. First, it is defined relative to the VAR's own information set: whatever the rate does in the month beyond its reaction to output and prices, whether the market learned of it before the announcement, in the window, or after. It registers the whole of the policy rate's response to its own shock by construction. Second, the pre-announcement controls stop the day before the meeting, so variation in u_t^p dated after the announcement is not in their span: the purge below removes what the controls see of the month, and what happens after the announcement survives it by construction.

The purge is the same projection with the controls added to the conditioning set, which by Frisch–Waugh–Lovell is the recursive instrument purged of the macro-residualised controls,

$$\begin{aligned} Z_t^{\text{CHOL},\perp} &= u_t^p - \text{Proj}(u_t^p \mid u_t^o, u_t^c, \mathbf{W}_{t-\Delta}) = Z_t^{\text{CHOL}} - \text{Proj}(Z_t^{\text{CHOL}} \mid \tilde{\mathbf{W}}_t), \\ \tilde{\mathbf{W}}_t &= \mathbf{W}_{t-\Delta} - \text{Proj}(\mathbf{W}_{t-\Delta} \mid u_t^o, u_t^c), \end{aligned} \quad (21)$$

with $\mathbf{W}_{t-\Delta}$ the monthly controls, dated before the month's announcement. The raw innovation is itself predictable from the controls ($R^2 = 0.06$, $p = 0.017$), through the payroll surprise and the slope of the yield curve (Appendix Table 2). Because the controls include financial variables (the S&P 500, the yield-curve slope, and Treasury-yield skewness), the purge removes the part of $b_{pf}\varepsilon_t^f$ they span, the financial exposure the ordering was meant to exclude; what they do not span survives, and so does the fact that $Z^{\text{CHOL},\perp}$ is defined only on 1988–2019, where the controls exist, a subsample on which the recursive zero restrictions need not hold. Since the monthly controls exist only in months with an FOMC announcement in the Bauer–Swanson sample, $Z^{\text{CHOL},\perp}$ is also zero in the other months, and those months carry the largest raw innovations of the window (Figure 10).

The three instruments of the paper are now on one footing. Each is $y - \text{Proj}(y \mid \mathcal{I})$ for a left-hand side y and an information set $\mathcal{I}$: the surprise has $y = mps_m$ and $\mathcal{I} = \mathbf{W}$, the Romer–Romer shock $y = \Delta i_m$ and $\mathcal{I} = (i_{m-}, \mathbf{G}_m, \mathbf{W}_{m-\Delta})$, the recursive instrument $y = u_t^p$ and $\mathcal{I} = (u_t^o, u_t^c, \mathbf{W}_{t-\Delta})$; the market's expectation, which the futures price nets out before the window opens, plays for the surprise the role the Greenbook and the macro innovations play for the other two.

4.3 First-stage strength after orthogonalisation

Table 1 reports heteroskedasticity-robust first-stage F for each instrument, raw and orthogonalised, across four candidate policy rates. Two orderings of the purge and the monthly aggregation are possible for an instrument observed at announcement dates: purge-then-aggregate (PTA) orthogonalises each announcement-level observation on the controls dated before that announcement and then sums within the month; aggregate-then-purge (ATP) sums first and orthogonalises the monthly series on the monthly controls. ATP is the construction of [Bauer and Swanson \(2023b\)](#). PTA, which [Amodeo \(2026\)](#) shows avoids the bias ATP introduces in months with more than one

announcement, is the construction used for the surprise throughout the paper. The Cholesky-based instrument is monthly by construction, so only ATP applies to it, and for the Romer–Romer shock the two are reported side by side. The pattern is the one Section 3 anticipates: the high-frequency surprise, reasonably strong when raw ($F = 14.5$ on the funds rate), falls to $F = 3.6$ under PTA; the Romer–Romer instrument stays strong on the funds and shadow rates ($F = 29.6$ on the funds rate, 21.8 on the shadow rate) *after* orthogonalisation; and the Cholesky-based instrument is strong throughout. Both revived instruments survive the very purge that collapses the high-frequency surprise.

*Table 1: Heteroskedasticity-robust first-stage F (HC0) by instrument, purge, and candidate policy rate: each instrument alone, for the reduced-form innovation of the policy rate. VAR in (IP, CPI, EBP, Policy Rate), the policy rate being the one of each column, with six lags and a constant, estimated on 1973m1–2019m12; each instrument on the months it is observed within 1988m1–2019m12 ($N = 383$ –384). **Bold:** $F > 10$ (strong single instrument, [Stock and Yogo, 2005](#)). ATP = aggregate-then-purge; PTA = purge-then-aggregate; FWL = Frisch–Waugh–Lovell purge on the Bauer–Swanson controls. The Cholesky rows use the recursive funds-rate innovation in every column (Appendix D).*

Instrument (statement window)	2-year	1-year	Fed funds	Shadow
<i>Monetary policy surprises (MPS)</i>				
Raw	9.4	18.6	14.5	8.0
Orthogonalised (PTA)	0.6	2.7	3.6	2.4
Orthogonalised (ATP)	2.8	6.4	7.1	5.7
<i>Romer–Romer</i>				
Raw	1.0	4.7	29.1	17.5
Orthogonalised (ATP)	0.9	1.8	17.5	10.4
Orthogonalised (PTA, FWL)	2.9	5.8	29.6	21.8
<i>Cholesky-based</i>				
Raw	24.3	55.5	427.4	390.7
Orthogonalised (ATP, FWL)	22.7	41.6	533.5	311.1

4.4 Strength divides the surviving contamination

Neither revived instrument is free of contamination. The Romer–Romer shock carries whatever neither the Greenbook nor the controls span, and the recursive instrument carries the financial exposure the controls do not span and what happens in the month after the announcement, so both have $\psi \neq 0$, each in a direction of its own. That is the price of the wider window. What makes them preferable is the identity (7): the bias is the surviving contamination divided by the first stage, and their first-stage F is roughly an order of magnitude larger than that of the purged surprise, so the contamination that survives weighs far less in the identified response. Proxies rank by that ratio rather than by contamination alone or by strength alone (Corollary 1).

Figures 11 and 12 of Appendix D report the responses to the two instruments, raw and purged, with 68% moving-block-bootstrap bands, on the reduced form of Section 6; the corresponding figure for the high-frequency surprise is Appendix Figure 13. Identified with either wide-window instrument, the output response builds gradually to a hump-shaped trough, and its magnitude is a fraction of the one reported by the purged surprise. Because the same control set is applied to all three instruments, the comparison also shows how the purge moves each of them, and the

order is that of the first-stage loss, as magnification predicts. The futures surprise loses most of its first stage ($F = 14.5 \rightarrow 3.6$; $\delta_\pi \approx 0.54$, Appendix B) and is enlarged most, its output trough more than threefold, from -1.9% to -6.6% . The Romer–Romer shock loses none of its first stage ($F = 29.1 \rightarrow 29.6$), and its output and price responses are essentially unchanged. The recursive instrument *gains* first stage ($F = 427 \rightarrow 534$) and its responses shrink, the output trough from -2.1% at nineteen months to -1.4% at thirty, which is consistent with (12) for a purge that raises the first stage ($\delta_\pi < 0$), since the surviving contamination is then divided by a larger first stage. Neither instrument delivers a price decline on its own: the Romer–Romer shock’s price response is close to zero, and under the purged recursive instrument prices rise, by 0.3% at two years.

The residual contamination is, moreover, not uniform across variables. When two proxies overidentify the same shock, an overidentification test asks equation by equation whether they agree; anticipating Section 5, the Romer–Romer and high-frequency instruments agree on the slow macro variables (industrial production and prices) but disagree on the fast financial variable (the excess bond premium), which is where the within-window financial contamination concentrates. Contamination, in other words, is *instrument-specific* in direction, and that structure is what the identification of Section 5 exploits: proxies contaminated in different directions are placed in different families, and what the members of a family share is annihilated.

5 Identification from many proxies: pool within, annihilate across

Section 4 ranked proxies by their contamination relative to their first stage and found instruments that are strong but not clean, and that disagree variable by variable. This section takes that logic to its conclusion. When every candidate proxy for the policy shock is contaminated and none can be assumed clean, the impact vector is still identified, not from the level of the identifying moments, where pooling looks, but from their cross-proxy dispersion, where the contamination structure lives. Proxies are grouped into families that share a contamination channel; differencing within a family removes what the family shares, and pooling within a family averages what it does not. Strength is then bought along the cleaned direction, without reintroducing bias.

Section 5.1 sets out the panel of contaminated proxies, whose moments alone do not separate the target from the contamination. Section 5.2 groups the proxies into families that share the mean of their contamination, so that demeaning within a family removes what the family shares; the identifying assumption is that the idiosyncratic part of the contamination is mean-independent of relevance. Section 5.3 establishes that the target is identified, as the leading eigenvector of the within-family scatter in the metric of the reduced-form covariance, when relevance varies within families, and Section 5.4 derives the estimator as its sample counterpart. Section 5.5 discusses the partition into families, the one input the researcher supplies. Section 5.6 contrasts standard pooling, which removes only the contamination specific to each proxy, with the strongest clean combination of the proxies, and shows that the latter delivers the same impulse responses as the eigenvector. Section 5.7 states the algorithm, and the proofs are in Appendix E.

5.1 The proxy panel

Keep the reduced form of (2), with unit-variance shocks and $\Sigma_u = BB'$. The estimator recovers the direction of b_p , and the unit-effect impact vector $\beta = b_p/b_{pp}$ of (3) is obtained at the end by

normalising the policy coordinate to one. We observe m proxies,

$$z_{jt} = \phi_j \varepsilon_t^p + \boldsymbol{\psi}_j' \varepsilon_t^{-p} + \omega_{jt}, \quad g_j := \text{Cov}(u_t, z_{jt}) = \phi_j b_p + b_{-p} \boldsymbol{\psi}_j \in \mathbb{R}^n, \quad j = 1, \dots, m, \quad (22)$$

with relevance ϕ_j (the ϕ of (4), one per proxy, and the relevances differ across the panel because the proxies read the policy shock over different windows and against different information sets, the futures surprise over thirty minutes, the Romer–Romer shock at the decision, the recursive innovation over the month, a dispersion the estimator below uses rather than assumes away), contamination loading $\boldsymbol{\psi}_j \in \mathbb{R}^{n-1}$ (the $\boldsymbol{\psi}$ of (4), one per proxy), and measurement noise ω_{jt} orthogonal to all shocks. Proxy j is observed on its own support $S_j \subseteq \{1, \dots, T\}$ with $T_j := |S_j|$ observations, so proxies of different vintages enter without alignment. The proxy-SVAR estimand of proxy j alone is $g_j / (s_p' g_j)$, the contaminated impact vector $\tilde{\beta}$ of (6): the true direction plus a tilt $b_{-p} \boldsymbol{\psi}_j / \phi_j$ in the $(n-1)$ -dimensional non-policy impact space $\text{range}(b_{-p})$. The maintained difficulty is that every $\boldsymbol{\psi}_j \neq \mathbf{0}$ and no proxy can be assumed clean, so the standard devices, imposing exogeneity on one instrument or testing one instrument against another, are unavailable; any such test compares two contaminated objects (Mertens and Ravn, 2013; Stock and Watson, 2018). Stacking the moments, $G := [g_1, \dots, g_m] \in \mathbb{R}^{n \times m}$, $\phi := (\phi_1, \dots, \phi_m)'$, $\Psi := [\boldsymbol{\psi}_1, \dots, \boldsymbol{\psi}_m]' \in \mathbb{R}^{m \times (n-1)}$:

$$G = b_p \phi' + b_{-p} \Psi'. \quad (23)$$

The moments alone do not say which direction in the column space of G is b_p ; the restriction on the contamination that does is the subject of Section 5.2.

5.2 Families and the within transform

Proxies fall into C families; write $j \in c$ when proxy j belongs to family c , and $|c|$ for the family's size. Within a family, write each loading as the family's mean loading plus a deviation,

$$\boldsymbol{\psi}_j = \bar{\boldsymbol{\psi}}_c + \boldsymbol{v}_j, \quad d_c := b_{-p} \bar{\boldsymbol{\psi}}_c, \quad j \in c, \quad (24)$$

so d_c is the family-common contamination direction (not the net direction d of (7)) and $\boldsymbol{v}_j$ the within-family idiosyncratic part. Equation (24) is a decomposition, not a restriction; the restriction is Assumption 1 below, on the deviations $\boldsymbol{v}_j$. Family-indexed objects ($\bar{\boldsymbol{\psi}}_c$, d_c , and below $\bar{g}_c$, $\bar{\phi}_c$) are constant within a family; proxy-indexed objects carry j . Written proxy by proxy, the moments are

$$g_j = \phi_j b_p + d_c + b_{-p} \boldsymbol{v}_j, \quad j \in c, \quad (25)$$

and (25) has the structure of a panel regression with group fixed effects. Each proxy is a unit and each family a group; the family's common contamination d_c is the group fixed effect; relevance ϕ_j is the regressor; the impact vector b_p is the coefficient; and the idiosyncratic contamination $b_{-p} \boldsymbol{v}_j$ is the error.

The analogy says what identification requires. A group fixed effect is removed by demeaning within groups, so the common contamination d_c can be eliminated without assuming that any proxy is clean. After demeaning, the coefficient is identified if the regressor varies within groups and the error is mean-independent of the regressor. Here these conditions read: relevance must differ across the members of at least one family, and a proxy's idiosyncratic contamination must be

unrelated to its relevance. Demeaning also uses one degree of freedom per group, which is why the panel must contain at least one more proxy than families (the order condition (42) of Section 5.5).

The analogy departs from a standard panel regression in one respect, and it is the reason the estimator is an eigenvector rather than a regression coefficient: the regressor is not observed. Each moment g_j is the product of an unknown scalar ϕ_j and the unknown vector b_p , so the demeaned moments are not regressed on relevance but factored into relevance and a common direction. The identifying restriction therefore concerns the idiosyncratic contamination and its relation to relevance, and it is the following.

Assumption 1 (Random-effects block structure). *Proxies are grouped into C families. Within family c , with $\boldsymbol{\phi}_c := \{\phi_k\}_{k \in c}$ the relevances of its members and conditionally on the partition,*

$$\boldsymbol{\psi}_j = \bar{\boldsymbol{\psi}}_c + \boldsymbol{v}_j, \quad \mathbb{E}[\boldsymbol{v}_j \mid \boldsymbol{\phi}_c, c] = \mathbf{0}, \quad d_c := b_{-p} \bar{\boldsymbol{\psi}}_c, \quad j \in c. \quad (26)$$

Exact block structure is the special case $\boldsymbol{v}_j \equiv \mathbf{0}$.

Assumption 1 says that the members of a family share the *mean* of their contamination, not the contamination itself. Every proxy may be contaminated, and the idiosyncratic part $\boldsymbol{v}_j$ may be large; what is required is that, within a family, a proxy's idiosyncratic contamination has mean zero whatever the relevances of the family's members. It is the random-effects model of meta-analysis, with proxies as studies and families as strata, and exact block structure is its fixed-effect special case. Two features of the condition matter. First, it conditions on relevance. It implies that the deviations average to zero within the family, $\mathbb{E}[\bar{\boldsymbol{v}}_c \mid c] = \mathbf{0}$ with $\bar{\boldsymbol{v}}_c := |c|^{-1} \sum_{j \in c} \boldsymbol{v}_j$, but it is stronger: it also rules out a family in which the more relevant proxies are systematically more exposed to some non-policy shock. Proposition 3 below shows that the stronger form is needed. Second, it conditions on the relevances of all members of the family rather than on proxy j 's own, because the estimator uses the gap between a proxy's relevance and its family's average, $\phi_j - \bar{\phi}_c$, which depends on all of them. The assumption says nothing about the *dispersion* of the deviations. Their spread matters later, for whether the target's signal dominates the within-family scatter (Proposition 4) and for the precision of the estimate (Appendix E.6), and the moment conditions those results need (independence across proxies, a common conditional variance, finite fourth moments) are collected in Assumption 2.

Demeaning the moments within families, $\tilde{g}_j := g_j - \bar{g}_c$ with $\bar{g}_c = |c|^{-1} \sum_{k \in c} g_k$ for $j \in c$ (and likewise $\tilde{\phi}_j, \tilde{\boldsymbol{v}}_j$),

$$\tilde{g}_j = \tilde{\phi}_j b_p + b_{-p} \tilde{\boldsymbol{v}}_j, \quad (27)$$

because d_c is constant within a family. The family-common contamination cancels *exactly*, without a clean instrument and for every value of $\boldsymbol{v}_j$. What remains is the target, carried by the within-family *dispersion* of relevance, plus the idiosyncratic contamination. That dispersion is the quantity on which the rest of the section turns, and it is denoted

$$\sigma_\phi^2 := \sum_{c=1}^C \sum_{j \in c} \tilde{\phi}_j^2 = \sum_c \sum_{j \in c} (\phi_j - \bar{\phi}_c)^2, \quad (28)$$

the total within-family variation of relevance across the panel: the sum, over every proxy, of the squared gap between its relevance and its family's average. It is zero when every family's members are equally relevant, in which case the within transform has removed everything, and it grows

with the number of families that contain proxies of unequal relevance and with how unequal they are. It plays for the direction the role the first stage plays for the scale: it is the signal the target leaves in the within-demeaned moments. Section 5.3 states when that signal identifies the target.

5.3 Identification

Identification asks two questions. The first is whether the model identifies b_p at all: which restriction on the contamination rules out every alternative target that generates the same moments. Proposition 3 answers it, and the answer is Assumption 1 together with within-family variation in relevance. The second is whether the object the estimator of Section 5.4 computes, the leading eigenvector of the within-demeaned moments, recovers that target. Proposition 4 answers it. Under exact block structure the answer is yes whenever relevance varies within a family; under random effects it requires in addition an ordering condition, which the proposition derives as necessary and sufficient in large panels. Throughout, the partition is taken as given; Section 5.5 states what it must satisfy. Both propositions use one further assumption, on the relevances and on the regularity of the idiosyncratic deviations.

Assumption 2 (Relevance heterogeneity and regularity). (i) $\sigma_\phi^2 > 0$, with σ_ϕ^2 the within-family dispersion of relevance (28). (ii) Under Assumption 1, conditionally on the relevances the v_j are independent across proxies with a common conditional variance $\Sigma_v := \text{Var}(v_j \mid \phi_c, c)$, the same in every family, and finite fourth moments.

Part (i) is the rank condition: relevance varies within at least one family, without which the within transform removes the target along with the contamination. Part (ii) is regularity for the random-effects case: the tilt formula of Proposition 4(ii) holds for any realised deviations, while its mean and order, the large-panel limit of part (iii) and inference use the independence of the deviations and their common variance Σ_v .

Which restriction identifies the target. The within transform removes the family-common contamination for any deviations v_j ; Proposition 3 states what else is needed to identify the target.⁸

Proposition 3 (Identification). (i) Under the restriction that the deviations average to zero within each family, $\mathbb{E}[\bar{v}_c \mid c] = \mathbf{0}$, alone, b_p is not identified, at any m and in population. (ii) Under Assumptions 1 and 2(i) the shear is excluded and b_p is identified up to scale; and if instead $v_j = \alpha_0 \tilde{\phi}_j + \eta_j$ with $\alpha_0 \in \mathbb{R}^{n-1}$ fixed and η_j satisfying (26), every estimator identifies $b_p + b_{-p}\alpha_0$.

The derivations are in Appendix E.4. The two parts explain why Assumption 1 conditions on relevance. The shear moves the target by $b_{-p}\alpha$ and each loading by $-\phi_j\alpha$. Writing $\phi_j = \bar{\phi}_c + \tilde{\phi}_j$, the family mean absorbs $-\bar{\phi}_c\alpha$, a change in d_c that the within transform removes, and each deviation moves by $-\tilde{\phi}_j\alpha$, with $\tilde{\phi}_j$ the proxy's relevance net of the family mean. If the only restriction is that the deviations average to zero within the family, the shear goes undetected: the $\tilde{\phi}_j$ sum to zero

⁸The obstacle to identification is a *shear*. By (23), for any $\alpha \in \mathbb{R}^{n-1}$, $(b_p + b_{-p}\alpha)\phi' + b_{-p}(\Psi - \phi\alpha')' = b_p\phi' + b_{-p}\Psi' = G$: moving the target by $b_{-p}\alpha$ inside the space of the non-policy impact responses, and removing $\phi_j\alpha$ from the loading of each proxy, leaves every moment unchanged (Proposition E.1, Appendix E.2). The data therefore identify b_p only under a restriction on the loadings Ψ that the sheared loadings $\Psi - \phi\alpha'$ violate for every $\alpha \neq \mathbf{0}$. Because the relevance contrasts sum to zero within a family, a restriction on within-family averages of the deviations survives the shear; this is part (i) of the proposition, and it is why Assumption 1 conditions on the relevances.

within the family by construction, so the sheared deviations still average to zero for every α , and b_p is not identified (part (i)). Under (26) the sheared deviation has conditional mean $-\tilde{\phi}_j\alpha$, which is nonzero whenever relevance varies within the family; the restriction then holds only at $\alpha = 0$, and the target is pinned down (part (ii)). What breaks identification is therefore contamination correlated with relevance in a direction the family does not share, not idiosyncratic contamination, however large.

The last clause of part (ii) shows that the restriction cannot be weakened. A relevance-correlated component common to every family, α_0 , generates the moments of the shifted target $b_p + b_{-p}\alpha_0$, and no estimator can distinguish the two. A relevance-correlated component that *differs across families* is detectable, because each family then identifies its own direction $b_p + b_{-p}\alpha_c$ and the family-by-family directions disagree; the between-family part of the overidentification statistic is that test (Appendix E.7), and it is reported with every estimate in Section 6.

Proposition 3 identifies b_p through a restriction on a conditional mean, which holds on average over the draw of the idiosyncratic deviations v_j . Whether the object the estimator computes attains that target in the panel is the question of Proposition 4 below: it does so exactly under exact block structure, up to a mean-zero error in a finite panel under random effects, and in the limit of many proxies when the ordering condition holds.

The metric. The second question is answered in the metric of the reduced-form covariance. Because $\Sigma_u = BB'$, $B'\Sigma_u^{-1}B = I_n$, so $b'_p\Sigma_u^{-1}b_{-p} = 0'$: in the Σ_u^{-1} metric the target is orthogonal to the contamination space. Equivalently, for any square root, $\mathcal{Q} := \Sigma_u^{-1/2}B = [q_p \ \mathcal{Q}_{-p}]$ is an orthogonal matrix and the whitened moments are

$$g_j^w := \Sigma_u^{-1/2}g_j = \phi_j q_p + \mathcal{Q}_{-p}\psi_j, \quad q'_p \mathcal{Q}_{-p} = 0', \quad \|q_p\| = 1. \quad (29)$$

The metric matters only under random effects. Under exact block structure the within scatter has rank one and its eigenvector is b_p in any metric (Proposition 4(i)). Under random effects, in the units of the data the idiosyncratic scatter $b_{-p}Nb'_{-p}$ is not orthogonal to b_p , so it tilts the leading eigenvector by an amount whose mean is not zero: a bias that does not vanish as proxies are added, so the unwhitened eigenvector is inconsistent. In the Σ_u^{-1} metric the idiosyncratic scatter lies in the orthogonal complement of the target and can only compete with it as a rival axis, which is what Proposition 4(ii)–(iii) exploit. Whitening costs nothing, since the reduced form supplies Σ_u . The metric serves identification, not efficiency against sampling error, which is handled separately (Proposition E.2(iv)).

When the eigenvector recovers the target. The object the estimator computes is the scatter of the whitened within-demeaned moments at the population moments, $S_0 := \tilde{G}^w \tilde{G}^{w'}$, with $\tilde{g}_j^w = \tilde{\phi}_j q_p + \mathcal{Q}_{-p}\tilde{v}_j$ and $\tilde{G}^w = [\tilde{g}_1^w, \dots, \tilde{g}_m^w]$. It is what the estimator would compute with unlimited data on the proxies at hand: the sampling error in the estimated moments, which vanishes as T grows, is removed, but the realised deviations v_j of the m proxies are kept. Two realised quantities describe those deviations: the cross-moment between relevance and idiosyncratic contamination, $\zeta := \sum_j \tilde{\phi}_j v_j$, and the idiosyncratic scatter, $N := \sum_j \tilde{v}_j \tilde{v}_j'$. Under exact block structure both are zero and S_0 is a single axis along the target, with weight σ_ϕ^2 . Under random effects ζ tilts that axis and N adds rival axes in the contamination space. Proposition 4 asks whether the leading eigenvector of

S_0 is still q_p , the whitened target.

Proposition 4 (Recovery by the leading eigenvector). *Let $S_0 := \tilde{G}^w \tilde{G}^{w'}$ at the population moments.*

(i) (Exact block structure.) *If $v_j \equiv \mathbf{0}$, the population within scatter is $\tilde{G}\tilde{G}' = \sigma_\phi^2 b_p b_p'$, of rank one, so whenever $\sigma_\phi^2 > 0$ the leading eigenvector of $\tilde{G}\tilde{G}'W$ is b_p for any $W \succ 0$; in the Σ_u^{-1} metric $S_0 = \sigma_\phi^2 q_p q_p'$, q_p is its unique leading eigenvector, and $b_p = \Sigma_u^{1/2} q_p$.*

(ii) (Random effects, finite m .) *For any realised $\{v_j\}$, identically,*

$$S_0 = \sigma_\phi^2 q_p q_p' + q_p (\mathcal{Q}_{-p} \zeta)' + (\mathcal{Q}_{-p} \zeta) q_p' + \mathcal{Q}_{-p} N \mathcal{Q}_{-p}', \quad (30)$$

and, if $\sigma_\phi^2 > \lambda_{\max}(N)$ and $\|\zeta\|$ is small relative to the gap ς below, its leading eigenvector is

$$q_p + \mathcal{Q}_{-p} (\sigma_\phi^2 I - N)^{-1} \zeta + O(\|\zeta\|^2 / \varsigma^2), \quad \varsigma := \sigma_\phi^2 - \lambda_{\max}(N) > 0, \quad (31)$$

the target plus a tilt that vanishes when $\zeta = \mathbf{0}$. Under Assumptions 1 and 2(ii), the tilt has mean zero over the draw of the $\{v_j\}$ to first order (exactly if their distribution is symmetric), and is of order $\|\Sigma_v\|^{1/2} / \sigma_\phi$, hence $O(m^{-1/2})$ when σ_ϕ^2 grows in proportion to m ; when the idiosyncratic scatter is small relative to σ_ϕ^2 the tilt is $\sigma_\phi^{-2} \mathcal{Q}_{-p} \zeta$.

(iii) (Random effects, $m \rightarrow \infty$.) *Under Assumptions 1 and 2(ii), as $m \rightarrow \infty$ with $\sigma_\phi^2 / m \rightarrow \bar{\sigma}_\phi^2$ and $m^{-1} \sum_c (|c| - 1) \rightarrow \bar{\kappa} \in (0, 1]$, the share of the proxies that survives within-family demeaning (one is lost per family), $m^{-1} S_0 \rightarrow \bar{\sigma}_\phi^2 q_p q_p' + \bar{\kappa} \mathcal{Q}_{-p} \Sigma_v \mathcal{Q}_{-p}'$, and q_p is its unique leading eigenvector iff*

$$\bar{\sigma}_\phi^2 > \bar{\kappa} \lambda_{\max}(\Sigma_v) \quad (\text{the ordering condition}), \quad (32)$$

in which case the leading eigenvector of S_0 converges to q_p .

The derivations are in Appendix E.4. Part (i) is the benchmark: parts (ii) and (iii) relax it. With no idiosyncratic contamination, every demeaned moment lies on the line through b_p , S_0 has rank one, and its only non-zero eigenvector is the target, whenever $\sigma_\phi^2 > 0$. Part (i) is the special case of part (ii) with $v_j \equiv \mathbf{0}$, and the only case in which the metric does not matter, which is why estimation error in $\hat{\Sigma}_u$ moves the estimate only at second order under exact block structure (Section 5.4). Identification fails only when no family contains two proxies of unequal relevance, in which case the within transform removes the target along with the contamination.

Part (ii) is the case of the application, a finite panel under random effects. It does not say that the leading eigenvector equals the target; it says that the error is a variance rather than a bias. The cross-moment ζ is zero in expectation by Assumption 1 but not in any realised panel, and it tilts the leading eigenvector by approximately $\sigma_\phi^{-2} \mathcal{Q}_{-p} \zeta$, an amount with mean zero that shrinks at rate $m^{-1/2}$ as relevance-heterogeneous proxies are added. No estimator removes it, since it is the realised draw of a restriction that holds only in expectation; it enters the uncertainty of the direction as a second variance component, as the heterogeneity term does in a random-effects meta-analysis (Appendix E.6). The idiosyncratic scatter N does not tilt the target axis, but it creates rival axes in the contamination space, and the target is the leading one only if $\sigma_\phi^2 > \lambda_{\max}(N)$.

Part (iii) delivers two results. As proxies are added, $\zeta / m \rightarrow \mathbf{0}$ while $N / m \rightarrow \bar{\kappa} \Sigma_v$, so the tilt of part (ii) vanishes and q_p is an exact eigenvector of the limit scatter: the target of the estimator

converges to the true b_p . And q_p is the leading eigenvector, and a unique one, if and only if the ordering condition (32) holds: within families, the dispersion of relevance must exceed the dispersion of idiosyncratic contamination. The ordering condition is therefore not an additional assumption but the exact requirement for the eigenvector to recover the target; when it fails, the estimator returns a contamination direction, whatever T . Both forms of the condition can be computed from the within residuals of the fit: the finite-panel condition $\hat{\sigma}_\phi^2 > \lambda_{\max}(\hat{N})$ of part (ii), which governs the panel at hand, and its large-panel counterpart, $\hat{\sigma}_\phi^2/m$ against $\hat{\kappa} \lambda_{\max}(\hat{\Sigma}_v)$ with $\hat{\kappa} = (m - C)/m$ (Appendix E.7).

No proxy is assumed clean, and no anchor instrument is used, but three conditions have replaced the exclusion restriction of the single-proxy SVAR. Contamination has random-effects block structure (Assumption 1), relevance varies within families (Assumption 2(i)), and the ordering condition (32) holds. Each has a failure mode. If the more relevant proxies of a family are more contaminated in a direction the family does not share, the shear is live and every estimator is biased by a fixed amount that no sample size reduces. If relevance is homogeneous within a family, that family does not contribute to identification. If idiosyncratic contamination is large relative to the dispersion of relevance, the ordering condition fails, and the leading eigenvector is a contamination direction. A fourth failure, a family that combines two contamination channels, is a failure of the partition, taken up in Section 5.5. Proposition 8 extends the result to the pooled instrument of Section 5.6, so that the shock series and the first stage of Section 6 inherit it.

5.4 The estimator

Section 5.3 identifies the target as the leading eigenvector of the whitened within scatter. The estimator is its sample counterpart, and it is derived here directly in the whitened coordinates of that section. Write $\hat{g}_j^w := \hat{\Sigma}_u^{-1/2} \hat{g}_j$ for the whitened estimated moments, so that, by (29), $\hat{g}_j^w = \phi_j q_p + d_c^w + e_j^w$ for $j \in c$, with $d_c^w := Q_{-p} \bar{\psi}_c$ the family's common contamination and e_j^w collecting the idiosyncratic contamination $Q_{-p} v_j$ and the $O_p(T^{-1/2})$ sampling error. The estimator fits this panel regression by least squares, treating the family effects, the relevances and the direction as unknowns:

$$Q(q, \{\phi_j\}, \{d_c^w\}) = \sum_{c=1}^C \sum_{j \in c} \|\hat{g}_j^w - \phi_j q - d_c^w\|^2. \quad (33)$$

In the units of the data this is the least-squares criterion with weight $\hat{\Sigma}_u^{-1}$; Appendix E.3 gives the derivation for a general weight. The direction that minimises (33) is called the within estimator. Because the direction multiplies the unknown relevances, the fit is a factorisation rather than a regression. The criterion is quadratic in $\{d_c^w\}$ and in $\{\phi_j\}$ for fixed q , so it is minimised by profiling: the linear parameters are concentrated out one group at a time, each step has a closed form, and the last step is an eigenproblem.

Step 1: the family contamination, for given $(q, \{\phi_j\})$. Only the terms of family c involve d_c^w ; with $\bar{g}_c^w$ and $\bar{\phi}_c$ the family means of $\hat{g}_j^w$ and ϕ_j , the first-order condition gives

$$\hat{d}_c^w(q, \phi) = \bar{g}_c^w - \bar{\phi}_c q, \quad \text{so that} \quad \hat{g}_j^w - \phi_j q - \hat{d}_c^w = \tilde{g}_j^w - \tilde{\phi}_j q, \quad (34)$$

with $\tilde{g}_j^w := \hat{g}_j^w - \bar{g}_c^w$. This is the within transformation of Section 5.2, and it has two consequences. The family-common contamination is profiled out exactly, whatever its value, because the family

effects are estimated rather than assumed away. And the family means of relevance disappear with it: only the contrasts $\tilde{\phi}_j = \phi_j - \bar{\phi}_c$ survive, because the levels determine only the sums $\bar{\phi}_c q + d_c^w$ and cannot separate the two, which is the confounding of the levels (Section 5.6) seen from inside the criterion.

Step 2: the relevance contrasts, for given q . Substituting (34) into (33) gives the within criterion,

$$Q^w(q, \{\tilde{\phi}_j\}) := \min_{\{d_c^w\}} Q = \sum_j \|\tilde{g}_j^w - \tilde{\phi}_j q\|^2, \quad (35)$$

which separates across j . For fixed q each term is a scalar least-squares problem, solved by the projection coefficient of the demeaned moment on the candidate direction,

$$\hat{\phi}_j(q) = \frac{q' \tilde{g}_j^w}{q' q}. \quad (36)$$

The minimised value of term j is the squared length of the part of $\tilde{g}_j^w$ orthogonal to q , and summing over j gives the concentrated criterion, a function of the direction alone,

$$Q^c(q) := \min_{\{\tilde{\phi}_j\}} Q^w = \sum_j \|\tilde{g}_j^w\|^2 - \frac{q' \tilde{G}^w \tilde{G}^{w'} q}{q' q}, \quad \tilde{G}^w := [\tilde{g}_1^w, \dots, \tilde{g}_m^w]. \quad (37)$$

The first term does not depend on q ; the second is a Rayleigh quotient, invariant to the scale and sign of q ; the estimator therefore delivers a direction, which the normalisation $\beta = b_p / (s_p' b_p)$ of (3) then scales.

Step 3: the direction. Minimising (37) is maximising the Rayleigh quotient

$$f(q) := \frac{q' S q}{q' q}, \quad S := \tilde{G}^w \tilde{G}^{w'}, \quad (38)$$

over $q \neq 0$. S is symmetric and positive semidefinite, so at a unit vector the quotient is a weighted average of its eigenvalues, with weights that sum to one; its maximum is the largest eigenvalue, attained at the corresponding eigenvector, which is unique up to sign when that eigenvalue is simple (Appendix E.3). Hence

$$\hat{q} = \arg \min_q Q^c(q) \iff \tilde{G}^w \tilde{G}^{w'} \hat{q} = \lambda_{\max} \hat{q}, \quad Q^c(\hat{q}) = \sum_j \|\tilde{g}_j^w\|^2 - \lambda_{\max} : \quad (39)$$

the estimate of the whitened direction is the first principal component of the whitened within-demeaned moments. The minimised value is the within scatter less its leading eigenvalue, and it is the quantity the overidentification test and the robust set of Appendix E.6 are built on. Substituting $\hat{q}$ back into (36) and (34) returns the fitted contrasts $\hat{\phi}_j$ and family effects $\hat{d}_c^w$, so the three steps deliver every parameter of (33). Under exact block structure the population criterion is zero at q_p and positive at every other direction (Proposition 4(i)).

Figure 2 illustrates the three steps in a two-dimensional space of whitened moments. In that space each family is a cloud of points on its own line: the points are $\hat{g}_j^w$, the line runs in the direction q_p because every member's moment is $\phi_j q_p$ plus the family's common offset d_c^w , and the offset is

where the line sits. Step 1 moves every cloud to the origin, which removes the offset together with the family's mean relevance; what survives is each family's spread *along* q_p , the relevance contrasts $\tilde{\phi}_j$, plus the idiosyncratic and sampling scatter around the line. Steps 2 and 3 then take the axis of the pooled cloud, the direction along which the demeaned moments are most spread out, which is their first principal component. The second eigenvalue measures the scatter off that axis, and the ratio of the two, the eigengap λ_1/λ_2 reported with every estimate, measures how well the axis is determined.

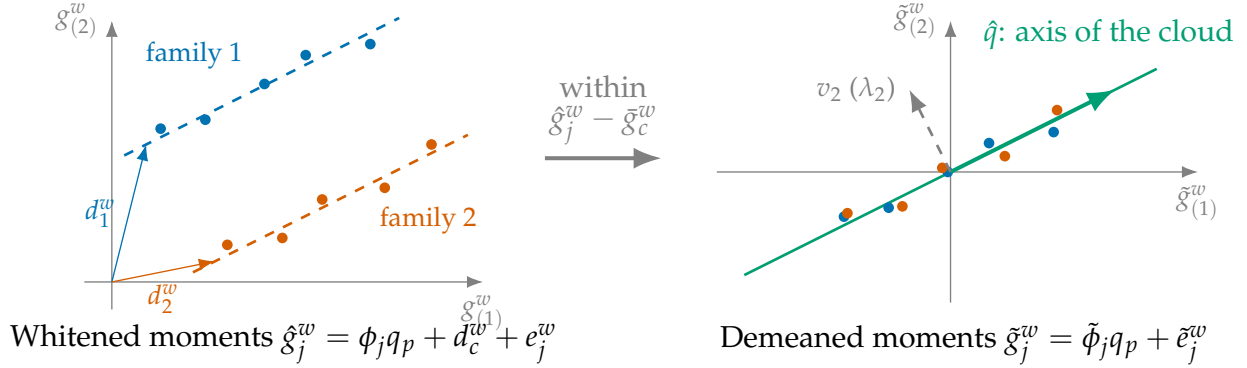


Figure 2: The geometry of the estimator: demean within, then take the axis of the cloud. Two families of five proxies in a two-dimensional space of whitened moments. Left: each family's moments lie on a line parallel to the target, displaced by the family's common contamination. Right: demeaning within families moves each line to the origin, and the leading eigenvector of the pooled scatter, its first principal component, is the common direction. The second eigenvalue λ_2 measures the scatter off the axis.

The eigenvector characterisation holds in general. Under exact block structure the target also has explicit instrumental-variable forms, which show what the eigenvector is doing. At the population moments $\tilde{g}_j = \tilde{\phi}_j b_{pp}$, so every within-demeaned moment is a multiple of the target and, after normalisation,

$$\beta = \frac{\tilde{g}_j}{s_p' \tilde{g}_j} \quad \text{for any } j \text{ with } \tilde{\phi}_j \neq 0, \quad \text{and} \quad \beta = \frac{\tilde{G} \tilde{G}' s_p}{s_p' \tilde{G} \tilde{G}' s_p} = \frac{\sum_j (s_p' \tilde{g}_j) \tilde{g}_j}{\sum_j (s_p' \tilde{g}_j)^2}, \quad (40)$$

since $s_p' \tilde{g}_j = \tilde{\phi}_j b_{pp}$. The first is the proxy-SVAR estimand of the *contrast instrument* $z_{jt} - |c|^{-1} \sum_{k \in c} z_{kt}$, $j \in c$, a single proxy net of its family mean: differencing within the family removes d_c , and the contrast is a valid instrument with first stage $\tilde{\phi}_j b_{pp}$. The second is the estimand of the pooled contrast instrument $\sum_c \sum_{j \in c} \tilde{\phi}_j (z_{jt} - |c|^{-1} \sum_{k \in c} z_{kt})$, whose weights sum to zero within every family, so that the family-common contamination is annihilated across families, and which pools the contrasts within families with weights proportional to their relevance, so that whatever is idiosyncratic to a proxy is averaged. That is “pool within, annihilate across” as an instrument, and (40) is its impact vector. The instrument is infeasible, because its weights $\tilde{\phi}_j$ are unknown; the eigenvector of (39) combines the same contrasts without requiring them.

Back to the units of the data. The impact vector is recovered by unwhitening and normalising,

$$\hat{b}_p = \hat{\Sigma}_u^{1/2} \hat{q}, \quad \hat{\beta} = \frac{\hat{b}_p}{s'_p \hat{b}_p}, \quad \text{equivalently} \quad \tilde{G} \tilde{G}' \hat{\Sigma}_u^{-1} \hat{b}_p = \lambda_{\max} \hat{b}_p, \quad (41)$$

with $\tilde{G}$ the within-demeaned moments in the units of the data. Any square root of $\hat{\Sigma}_u$ serves, the symmetric root or the Cholesky factor, because $\hat{b}_p$ solves the right-hand eigenproblem in (41), which involves no root; the whitened eigenvector $\hat{q}$ depends on the root, $\hat{b}_p$ does not.⁹ The estimation error in $\hat{\Sigma}_u$ moves $\hat{b}_p$ only at second order under exact block structure, where the within scatter has rank one and its eigenvector is b_p in every metric, but at first order under random effects, by an amount proportional to the heterogeneity term of (65).¹⁰

5.5 The family partition

The estimator takes the partition into families as given; it is the one input the researcher supplies. Two counting questions decide what a panel can deliver. The first is whether there are enough proxies inside the families to identify the direction: each family spends one proxy on its common contamination, so only the remaining $m - C$ proxies carry information about the direction. The second is whether there are enough families for a clean combination of the proxies to use the *level* of relevance, and not only its within-family contrasts: a combination of family levels is clean only if it annihilates every family's common contamination, which requires more families than the number of distinct directions those contaminations occupy. There are only $n - 1$ non-policy shocks, so there are at most $n - 1$ such directions.

Proposition 5 (The two order conditions). (O1) Enough proxies inside families. *Identification of the direction requires*

$$m - C \geq 1, \quad (42)$$

and the rank condition of Assumption 2(i): at least one family contains two proxies of unequal relevance. (O2) Enough families to use the levels. *Let*

$$r := \text{rank}[\bar{\psi}_1, \dots, \bar{\psi}_C] \leq \min(C, n - 1) \quad (43)$$

be the number of distinct directions of the family contaminations. A clean combination with nonzero family totals exists iff $C > r$; since $r \leq n - 1$, a sufficient condition, whatever the realised rank, is

$$C > n - 1. \quad (44)$$

Condition (O2) is $C > r$, and (44) guarantees it. When (42) fails nothing is identified; when (O2) fails the direction is still identified, but the level component of clean strength vanishes and clean strength collapses to its contrast-only value σ_ϕ^2 .

The derivations are in Appendix E.4, and Proposition 6 in Section 5.6 makes the strength statement precise. The two conditions both ask for more, more proxies and more families, but they answer

⁹A generalised eigenvalue routine called on $(\tilde{G}\tilde{G}', \Sigma_u)$ returns x with $\tilde{G}\tilde{G}'x = \lambda\Sigma_u x$, which is $\Sigma_u^{-1}b$ and not b .

¹⁰Both the sampling error of $\hat{\Sigma}_u$ and the window on which it is estimated matter under random effects. The application estimates Σ_u on the VAR window; the moment-window estimate is the natural check, and Section 6.1 reports it.

different questions. (O1) concerns the direction: without two proxies of unequal relevance in some family, demeaning removes the target along with the contamination. (O2) concerns strength: with too few families, the only clean combinations are within-family contrasts, which carry the target only through the dispersion of relevance, and the level of relevance, usually the larger part, is lost. (O2) is also the condition that becomes harder to meet as the VAR grows: each added variable adds a non-policy shock, and hence a direction the family contaminations may occupy, while the number of families is fixed by the proxy menu.

One point matters when (O2) is checked in the data. The rank r is that of the C family centroids, the estimated common contaminations, and not that of the proxy-level residuals. The residuals also contain each proxy's idiosyncratic contamination, fill all $n - 1$ directions whatever C is, and would declare (O2) violated where it holds. And since a failure of (O2) costs strength but not identification, a low surviving relevance (Section 5.6) in such a panel reflects the loss of the level component, not a shock that has not been cleaned.

The two conditions govern the design and the reading of the application. (O1) is why a family of near-copies of one series contributes nothing to the direction, however many members it has (Section 6.3); (O2) is why the identified response changes when the family count first exceeds the number of non-policy shocks (Section 6.3), and why the variable set is enlarged one variable at a time rather than all at once (Section 7.4).

Splitting and merging. The partition can err in two directions, and the two errors are not symmetric. Splitting a true family in two preserves block structure: if the members of a family share a mean contamination, so do the members of each part. The only cost is precision, because the contrasts across the two parts are lost; clean strength is unchanged. Merging two families with different mean contaminations breaks block structure: the within-family contrasts of the merged family no longer cancel the contamination but carry the difference $b_{-p}(\bar{\psi}_c - \bar{\psi}_{c'})$ between the two means, and the estimate is biased by an amount that no sample size removes. Splitting costs precision; merging costs validity. When in doubt, split, keeping at least two proxies of unequal relevance in each family. The same asymmetry qualifies Assumption 2(ii), the independence of the deviations: two proxies that share an idiosyncratic component, overlapping contracts for instance, should form their own family, since otherwise their shared component does not average out.

Assigning proxies to families. The assignment follows from what a family is: a set of proxies whose contamination shares a mean, that is, proxies that see the non-policy shocks through the same channel. For high-frequency surprises the channel is the contract and the construction. Proxies built from the same contract, over the same window and purged of the same predictable component, pick up news about the economy in the same way, and differ in relevance because they differ in maturity or horizon; they form a family. Series built from the policy record or from a model that share a source and a construction form a family in the same sense. Proxies whose construction differs, in the asset class, the window or the purge, are not grouped however correlated they are: correlation through the policy shock is relevance, not shared contamination, and relevance is what the within transform must keep. Each family needs at least two proxies of unequal relevance, and three where the menu allows, both for the family-size floor of the audit below and for the residuals that the heterogeneity estimator of Appendix E.6 uses. A family that could reach three members only by taking in a proxy of a different construction is better left with two, by the asymmetry above. The assignment is an economic statement about the proxies, made before the moments are formed.

The number of families. The data can check the number of families, but not the assignment of each proxy. Counting the families requires only that the residuals, once the target is removed, form

C clusters; assigning a proxy requires its residual to be clearly closer to its own family's centroid than to any other, a separation that the sampling noise of monthly moments does not deliver. The economic partition is therefore audited rather than replaced (Appendix E.7). For each candidate number of families, a k -means search assigns the proxies, and a penalised criterion selects the count. The selected count must lie in a window set by the order conditions, above the contamination rank by (O2) and at most $m - 1$ by (O1). What matters in a disagreement is its direction. A selected count below the economic one means that the economic partition is finer than the data can resolve, which costs precision and not validity; a count above it means that the data would split an economic family, the one disagreement that signals a merger and hence a biased direction.

5.6 Strength: from pooling to a clean instrument

Combining the proxies into a single instrument is what adds strength, but how they are combined decides whether the combination is also clean. This subsection contrasts standard pooling, which averages the proxies and leaves their common contamination in place, with constrained pooling, which chooses the weights to remove it.

Standard pooling. The natural way to combine proxies is to average them. A combined proxy $w'z_t$ has moment

$$\sum_j w_j g_j = (w'\phi) b_p + b_{-p}(\Psi'w), \quad (45)$$

so it identifies b_p tilted by $b_{-p}(\Psi'w)/(w'\phi)$, and it is clean iff its weights annihilate the loadings, $\Psi'w = \mathbf{0}$. Split each loading into a component common to the whole panel and an idiosyncratic one, $\psi_j = \bar{\psi} + \xi_j$, with $\bar{\psi}$ the panel-wide mean loading and the ξ_j independent, mean zero and of variance Σ_ψ . The contamination of any combination is then

$$\Psi'w = \left(\sum_j w_j \right) \bar{\psi} + \sum_j w_j \xi_j. \quad (46)$$

For the equal-weight average the second term is $O_p(m^{-1/2})$, while the first is $\bar{\psi}$ exactly, since the weights of any average sum to one (Appendix E.1). Averaging removes the idiosyncratic contamination and nothing else: the averaged proxy identifies the direction of $b_p + b_{-p}\bar{\psi}/\bar{\phi}$, $\bar{\phi}$ the mean relevance, a tilt that no number of proxies reduces. The common component is removed only by weights that sum to zero, which is what the within estimator does. But the within estimator discards the family levels $\bar{g}_c = \bar{\phi}_c b_p + d_c$, which carry information on b_p confounded by d_c ; they can be used only through weights that cancel the d_c . The direction is already delivered by Section 5.4, so what the levels add is strength: the first stage of a single pooled instrument, which produces the shock series and the scale of the responses. This subsection derives that instrument, shows that it identifies the impulse responses of the within estimator, and says what its first stage measures.

Constrained pooling. A combined instrument $w'z_t = \sum_j w_j z_{jt}$ inherits the three parts of its members, (22), each combined with the same weights:

$$w'z_t = \underbrace{(w'\phi)}_{\text{relevance}} \varepsilon_t^p + \underbrace{(\Psi'w)'}_{\text{contamination}} \varepsilon_t^{-p} + \underbrace{w'\omega_t}_{\text{noise}}, \quad \text{Var}(w'\omega_t) = w'\Sigma_\omega w. \quad (47)$$

Its concentration parameter, the population counterpart of the first-stage F of a single instrument, is $\mu^2(w) = T(w'\phi)^2 / (w'\Sigma_\omega w)$: relevance squared over noise. The strongest clean instrument therefore solves a single program: maximise the concentration subject to zero contamination,

$$w^* = \arg \max_w \frac{(w'\phi)^2}{w'\Sigma_\omega w} \quad \text{s.t.} \quad \Psi'w = \mathbf{0}. \quad (48)$$

With homoskedastic independent noise, $\Sigma_\omega = \sigma_\omega^2 I$, the solution is

$$w^* \propto P_\perp \phi, \quad P_\perp := \text{the orthogonal projector onto } \ker(\Psi'), \quad \mu^2(w^*) = T \|P_\perp \phi\|^2 / \sigma_\omega^2. \quad (49)$$

The solution is exact. Appendix E.5 derives it and gives the exact solution for general noise, $w^* \propto \Sigma_\omega^{-1} \phi^\perp$, with $\phi^\perp$ the residual of ϕ on the columns of Ψ in the Σ_ω^{-1} metric. The optimal clean instrument is the relevance vector with its component along the contamination loadings removed: $P_\perp$ is the *annihilator* of the contamination, and clean strength is the squared length of what survives it. Relative to the unconstrained pool, whose weights are ϕ itself, what is lost is $\|\phi\|^2 - \|P_\perp \phi\|^2$, the part of relevance collinear with the contamination: a proxy is useful to the pool to the extent that its relevance is not explained by its loadings.

Pool within, annihilate across. The constraint in (48) says that a clean combination must cancel the contamination, but not which combinations do. Under block structure they can be described explicitly, and the description separates the two operations. With $a_c := \sum_{j \in c} w_j$ the family totals of the weights,

$$\Psi'w = \sum_c a_c \bar{\psi}_c + \sum_j w_j v_j, \quad (50)$$

and under exact block structure the second sum is absent. *Within* a family the weights are then unconstrained, and the program spends them on strength, in proportion to relevance; *across* families the totals must make the family contaminations cancel, which is possible only if there are more families than independent contamination directions, the condition (O2) of Proposition 5. Proposition 6 makes the decomposition exact.

Proposition 6 (Clean strength: contrast and level components). *Under exact block structure the clean weight vectors, $\ker(\Psi') = \{w : \Psi'w = 0\}$, are the within-family contrasts (family totals $a_c = 0$), of dimension $m - C$, plus the family-constant vectors whose totals annihilate the centroids ($\sum_c a_c \bar{\psi}_c = 0$), a space $\mathcal{K}$ of dimension $C - r$. Hence $\dim \ker(\Psi') = m - r$, and the clean strength $\|P_\perp \phi\|^2$ of (49) decomposes as*

$$\|P_\perp \phi\|^2 = \underbrace{\sigma_\phi^2}_{\text{contrasts: available for every } C} + \underbrace{\|P_{\mathcal{K}} \phi\|^2}_{\text{levels: generically nonzero iff } C > r}. \quad (51)$$

The derivations are in Appendix E.4. The contrasts are clean whatever the partition and carry σ_ϕ^2 , the same quantity that identifies the direction; they are the pooled contrast instrument of (40). The levels add strength only when there are more families than contamination directions. This is why a family can add strength without adding identifying variation. A family whose members are near-copies of one series contributes almost nothing to σ_ϕ^2 , and hence to the direction, but its level still enters the second term when (O2) holds; in the application this is the role of the Romer–Romer

family (Section 6.3). The share of relevance that survives the annihilator, $\|P_{\perp}\phi\|^2/\|\phi\|^2$, is the measure Section 6.3 reports as families are added. Proposition 7 shows why pooling raises strength.

Proposition 7 (Pooling adds concentration). *Let $\mu^2(w) = T(w'\phi)^2/(w'\Sigma_{\omega}w)$ be the concentration parameter of the combined proxy $w'z_t$, and $\mu_j^2 = T\phi_j^2/\sigma_{\omega j}^2$ that of proxy j alone. (i) With independent measurement noise, the unconstrained pool $w^{\text{GLS}} = \Sigma_{\omega}^{-1}\phi$ attains $\mu^2(w^{\text{GLS}}) = \sum_j \mu_j^2$: concentrations add, and the pool is at least as strong as its strongest proxy. (ii) Under exact block structure with homoskedastic independent noise σ_{ω}^2 , the pooled contrast instrument $\sum_c \sum_{j \in c} \tilde{\phi}_j(z_{jt} - \bar{z}_{ct})$ of (40) has concentration $T\sigma_{\phi}^2/\sigma_{\omega}^2 = \sum_j T\tilde{\phi}_j^2/\sigma_{\omega}^2$, which is at least that of any single contrast $z_{jt} - \bar{z}_{ct}$; and the constrained-optimal instrument (49) attains $T\|P_{\perp}\phi\|^2/\sigma_{\omega}^2 \geq T\sigma_{\phi}^2/\sigma_{\omega}^2$, the maximum over all clean combinations.*

The derivations are in Appendix E.5. Independent noise averages out in the combination while relevance accumulates, so the concentration of the pool is the sum of the individual concentrations, and the annihilator removes from that sum only the part of relevance collinear with the contamination. A panel of individually weak proxies can therefore deliver a pooled instrument well above conventional thresholds without any of them being strong on its own.

The pooled instrument. In the data Ψ is unobserved, and the fit of Section 5.4 supplies its stand-in. Split each whitened level moment into the part along the fitted direction and the rest,

$$\hat{G}^w = \hat{q}\hat{\phi}' + \hat{R}, \quad \hat{\phi}_j := \hat{q}'\hat{g}_j^w, \quad \hat{R} := (I - \hat{q}\hat{q}')\hat{G}^w, \quad (52)$$

so that $\hat{\phi}$ collects the level relevances and $\hat{R}$ the residual loadings, what the moments contain beyond the target direction. At the population moments $R = Q_{-p}\Psi'$ by (29), the contamination alone, and $\ker R = \ker \Psi'$; so $\ker \hat{R}$ estimates the clean combinations, and the feasible version of (49) is¹¹

$$\hat{w} := P_{\ker \hat{R}} \hat{\phi}, \quad \text{scaled so that } \hat{w}'\hat{\phi} = 1, \quad z_t^* := \sum_j \hat{w}_j z_{jt}. \quad (53)$$

Two features of $\hat{R}$ matter. Its rank is at most $n - 1$ by construction, so $\ker \hat{R}$ has dimension at least $m - (n - 1)$ whatever the partition, and no estimate of the contamination rank enters the instrument. And it is built from the fitted direction: whatever error $\hat{q}$ carries, $\hat{R}$ carries with it, which is the content of the next result.

Equivalence with the within estimator. The pooled instrument is a proxy-SVAR instrument like any other, and its estimand is the moment $\text{Cov}(u_t, z_t^*) = \hat{G}\hat{w}$ normalised on the policy coordinate, (6). Proposition 8 shows that this estimand coincides with the direction of the within estimator: in population under block structure, and in the sample as an algebraic identity.

¹¹On unequal supports z_t^* is formed with the support weights of step 5 of Algorithm 1, so that its moment is $\hat{G}\hat{w}$ exactly. The precision-weighted form, with the proxies' purged sampling variances $\hat{\sigma}_j$ in the role of the noise variances, replaces the Euclidean projection by the Σ_{ω}^{-1} one of Appendix E.5, which makes the pooling weights, though not the direction (footnote 12), invariant to rescaling any proxy. In the units of the data, $\hat{\phi}_j = \hat{b}_p' \hat{\Sigma}_u^{-1} \hat{g}_j$ with $\hat{b}_p = \hat{L}\hat{q}$ the direction of unit $\hat{\Sigma}_u^{-1}$ -norm before normalisation.

Proposition 8 (The pooled instrument identifies the impulse responses of the within estimator).

(i) Population. Under exact block structure, let w be any clean combination, $\Psi'w = 0$, with $w'\phi \neq 0$; in particular $w = w^*$. Then

$$Gw = (w'\phi) b_p, \quad \text{so} \quad \frac{Gw}{s_p' Gw} = \frac{b_p}{s_p' b_p} \quad \text{and} \quad \Phi_h \frac{Gw}{s_p' Gw} = \Phi_h \frac{b_p}{s_p' b_p} = \Phi_h \beta \quad \text{for every } h, \quad (54)$$

the impact vector and impulse responses that the criterion (33) identifies (Proposition 4(i)). (ii) Sample. For any $\hat{w} \in \ker \hat{R}$ with $\hat{\phi}'\hat{w} \neq 0$, and in particular for (53),

$$\hat{G}^w \hat{w} = (\hat{\phi}'\hat{w}) \hat{q} \quad \Longleftrightarrow \quad \hat{G}\hat{w} = (\hat{\phi}'\hat{w}) \hat{L}\hat{q}, \quad (55)$$

exactly, with $\hat{L}$ the root used in the whitening: the proxy-SVAR impact vector built from z_t^* is $\hat{b}_p = \hat{L}\hat{q}$ rescaled, and after normalisation the two routes deliver the same impact vector $\hat{\beta} = \hat{b}_p / (s_p' \hat{b}_p)$, the same $\hat{\Theta}_h = \hat{\Phi}_h \hat{\beta}$ at every horizon, and the same value in every bootstrap draw, provided both use the same $\hat{\Sigma}_u$ and the same moments.

Both parts follow in one line. For (i), $Gw = b_p(\phi'w) + b_{-p}(\Psi'w)$ by (23), and $\Psi'w = 0$. For (ii), $\hat{G}^w \hat{w} = \hat{q}(\hat{\phi}'\hat{w}) + \hat{R}\hat{w}$ by (52), and $\hat{R}\hat{w} = 0$; unwhitening multiplies both sides by $\hat{L}$. In both cases the normalisation on the policy coordinate removes the scalar. Two consequences follow. First, the pooled instrument is *not* a second estimate of the direction: it reproduces $\hat{b}_p$ because the annihilator that makes it clean is built from $\hat{b}_p$, and the family levels enter only through the weights. What it adds is the shock series z_t^* and the strength of the scale step. Second, the equivalence is an identity in the sample and does not depend on block structure; block structure is what makes the common object the *true* b_p (part (i)). Under random effects the within direction is b_p up to the heterogeneity error of Proposition 4(ii), and the pooled instrument inherits exactly that error. For the empirical section this means that one set of impulse responses, one bootstrap band and one first stage describe the identification.

Testing strength. Because the pooled instrument reproduces the direction, its first stage is a statement about the scale step alone, and it is tested as the first stage of any single proxy is. The scaling $\hat{w}'\hat{\phi} = 1$ gives the pooled proxy a unit loading on the policy shock; when it is clean it is a proxy of the form (4) with $\phi = 1$ and $\psi = 0$, and its first stage sits at the floor b_{pp} of Proposition 1(ii). Its first-stage F is the heteroskedasticity-robust squared t -ratio of $\hat{u}_t^p$ on z_t^* over the moment window, as in Table 1, read against the same thresholds as a single proxy's, since the proxy-SVAR remains just-identified with one instrument. The concentration μ^2 of Proposition 7 is reported alongside; the two agree to first order when the noise in z_t^* is the only thing separating it from the policy innovation. The F measures the strength of the scale step, not the precision of the direction, which is governed by σ_ϕ^2 and can be modest when F is large; the precision of the direction is measured by the eigengap and the level set of Appendix E.6.

5.7 The algorithm

Sections 5.3 to 5.6 are population arguments: the criterion, the profiling steps and Proposition 4 say what is identified and by which object, the order conditions say what the partition must satisfy,

and the constrained program says what strength it allows. The algorithm below is the sequence of computations the empirical section runs on the data, with sample quantities in place of population ones at every step. Steps 1–3 are the estimator of the direction; steps 4–5 build the annihilator instrument of Section 5.6 on top of it; steps 6–7 are the outputs and their uncertainty.

Algorithm 1 (Implementation of the within estimator). Inputs: the VAR residuals $\hat{u}_t$ and $\hat{\Sigma}_u = \hat{L}\hat{L}'$ (estimated on the VAR window); proxies z_{jt} on their supports S_j , in the units in which block structure is asserted;¹² the partition $c(\cdot)$ of Section 5.5.

1. *Moments.* $\hat{g}_j = T_j^{-1} \sum_{t \in S_j} \hat{u}_t z_{jt}$, $j = 1, \dots, m$.
2. *Whiten and demean.* $\hat{g}_j^w = \hat{L}^{-1} \hat{g}_j$; with weights $\omega_j > 0$, $\tilde{g}_j^w = \hat{g}_j^w - \sum_{k \in c} \omega_k \hat{g}_k^w / \sum_{k \in c} \omega_k$ for $j \in c$ (any weights remove d_c exactly); $\tilde{G}^w = [\tilde{g}_1^w, \dots, \tilde{g}_m^w]$.
3. *Direction.* $\hat{q} = \arg \max_{\|q\|=1} q' \tilde{G}^w \tilde{G}^{w'} q$, $\omega = \text{diag}(\omega_j)$, sign fixed by $s_p' \hat{L} \hat{q} > 0$; $\hat{b}_p = \hat{L} \hat{q}$, and $\hat{\beta} = \hat{b}_p / (s_p' \hat{b}_p)$. Equal weights give the estimator of Section 5.4; the application uses precision weights $\omega_j = 1/\hat{v}_j$, $\hat{v}_j$ the sampling variance of $\hat{g}_j^w$ under the purged score (Appendix E.6), the efficient version of Proposition E.2(iv). Precision weights also make the direction invariant to rescaling a family; with equal weights, the families published in the largest units would dominate the scatter by scale alone.
4. *Loadings and residuals.* $\hat{\phi}_j = \hat{q}' \tilde{g}_j^w$ (contrast relevances), $\hat{\phi}_j = \hat{q}' \hat{g}_j^w$ (level relevances),¹³ whitened within residuals $\hat{r}_j = \tilde{g}_j^w - \hat{\phi}_j \hat{q}$, residual loadings $\hat{R} = \hat{G}^w - \hat{q} \hat{\phi}'$ as in (52), family centroids $\hat{d}_c^w$ from the level residuals.
5. *Pooled instrument.* $\hat{w} = P_{\ker \hat{R}} \hat{\phi}$, (53) (or its precision-weighted form (63)), scaled to $\hat{w}' \hat{\phi} = 1$; $z_t^* = \sum_j \hat{w}_j (T/T_j) \tilde{z}_{jt} \mathbf{1}\{t \in S_j\}$, with $\tilde{z}_{jt}$ proxy j demeaned on its support and T the length of the moment window, so that $T^{-1} \sum_t \hat{u}_t z_t^* = \hat{G} \hat{w}$ exactly and Proposition 8 holds for the series itself;¹⁴ its first-stage F describes the scale step.
6. *Responses.* $\hat{\Theta}_h = \hat{\Phi}_h \hat{\beta}$; the shock series from the proxy-SVAR step with z_t^* .
7. *Uncertainty.* The two-component variance and the level set for the direction (below); the moving-block bootstrap re-running steps 1–6, the weights included, on each draw, with the partition held fixed.

The algorithm is not iterative. The pooled instrument of step 5 reproduces the direction of step 3 exactly (Proposition 8), so it requires no rank selection and does not alter the direction; the identity $\hat{G}^w \hat{w} = (\hat{\phi}' \hat{w}) \hat{q}$ is checked numerically at every fit. The partition is the only input beyond the data,

¹²Exact block structure is a statement in the proxies' own units. Rescaling proxy j by $1/s_j$ divides its loading by s_j , so when the s_j differ within a family its common loading is no longer common and the within transform no longer removes it. Standardising the proxies by their own standard deviations would therefore bias the estimator. The precision weights of step 3 make $\hat{b}_p$ invariant to rescaling a whole family, not proxies within one. Contamination that scales with relevance along the family's direction, $\psi_j = s_j \tilde{\psi}_c$, is a different model, scale heterogeneity, which the within-family part of the overidentification statistic detects (Appendix E.7).

¹³Both are whitened inner products, $\hat{\phi}_j = \hat{b}_p' \hat{\Sigma}_u^{-1} \hat{g}_j$, not $\hat{b}_p' \hat{g}_j$; the Euclidean inner product reintroduces the metric error that step 2 removes.

¹⁴With equal supports this is $\hat{w}' z_t$. Restricting to the common support instead discards the months in which some proxy is missing, and the covariance of the resulting series with $\hat{u}_t$ is no longer $\hat{G} \hat{w}$: its impact vector is not $\hat{b}_p$ rescaled.

and the one object the bootstrap of step 7 holds fixed. The empirical section refers to the pair formed by the direction of step 3 and the pooled instrument of step 5 as the block-hybrid identification.

Uncertainty. Step 7 has two ingredients. The direction’s error has two components: a sampling component, driven by the purged-proxy scores, and a between-proxy heterogeneity component of order $m^{-1/2}$. They are uncorrelated, and both are estimable from the residuals of the fit (Proposition E.2, Appendix E.6). The moving-block bootstrap over t that re-runs steps 1–6, the bootstrap shown to be valid for proxy-SVARs by Jentsch and Lunsford (2022), is consistent for the sampling component; it does not capture the heterogeneity component, so the reported bands reflect sampling uncertainty. The weak-identification-robust set for the direction is the level set of the concentrated criterion (70), which uses no instrument (Appendix E.6). The diagnostics reported with every fit are in Appendix E.7.

6 Empirical application to high-frequency monetary surprises

The preceding sections developed the estimator and the conditions under which it identifies; this section takes it to a panel of U.S. monetary proxies and asks the motivating question directly: can a menu of individually weak, contaminated instruments be combined into a strong and credible identification of the monetary policy shock? The families are formed economically, one construction and one purging per family, so that the proxies of a family are pooled for strength and their common contamination is annihilated across families. On that design the answer is yes, with two qualifications that the diagnostics make precise. The pooled instrument has a first stage of $F = 23.9$, higher than that of any external proxy in the panel but one, and it describes a contraction in which output falls by 5.3% at a fourteen-month trough, the headline CPI falls at every horizon, core prices respond later, and the credit spread widens. The qualifications are that the identifying variation comes mostly from two of the five families, the Swanson (2021) factors above all, and that the direction is identified but not sharply. The result survives the policy indicator, the lag length, the price measure, the addition of variables one at a time, and the removal of any one of the three families other than the Swanson and Cholesky families; the pre-2008 sample keeps the impact responses and the first stage but gives a deeper, more persistent contraction (Section 7).

6.1 Data, the proxy panel, and the partition

The system is a monthly VAR in (IP, CPI, EBP, Fed Funds), with industrial production and the CPI in log levels times 100, the excess bond premium of Gilchrist and Zakrajšek (2012), six lags and a constant, estimated on 1973m1–2019m12. The identifying moments $\hat{g}_j$ are formed on each proxy’s own support inside 1988m1–2019m12, the window over which the Bauer and Swanson (2023b) controls, and hence every purged proxy, exist; the supports run from 311 to 383 months (Appendix Table 3). The companion matrix is thus estimated on the longer window and the identification uses the shorter one, the standard proxy-SVAR arrangement applied one level up: each proxy contributes on its own support without any change to the estimator. Estimating Σ_u on the moment window instead of the VAR window, the check suggested in Section 5.4, moves the direction by 0.1° to 2.9° across the specifications reported below. Responses are to a +1 percentage-point move in the policy indicator, over 36 months. Bands are 68% median-recentred percentile intervals, (69), from a residual moving-block bootstrap (Jentsch and Lunsford, 2022) with 1,000 draws, in which

the VAR, Σ_u , the moments (on the moment window only) and the weights are re-estimated in every draw and only the partition is held fixed.

The estimator is Algorithm 1 in its efficient form. The family means and the within scatter are precision-weighted, each proxy weighted by $1/\hat{v}_j$, where $\hat{v}_j$ is the sampling variance of $\hat{g}_j^w$ under the purged score, which is the version that attains the efficiency bound of Proposition E.2(iv); unlike the equal-weight version, its direction does not depend on the units in which each family happens to be published.¹⁵

The sixteen proxies (Appendix Table 3) are the purged versions of five instrument families: high-frequency futures surprises (MP1, FF4 and ED4 from the Acosta et al. (2025) database, and the Bauer and Swanson (2023b) surprise), the three Swanson (2021) factors, the information-robust surprise of Miranda-Agrippino and Ricco (2021), the extended Romer and Romer (2004) shock of Section 4.1, and the recursive (Cholesky) innovations of Section 4.2 for three policy rates. “Purged” means orthogonalised on the six Bauer and Swanson (2023b) controls, at the announcement level and then summed within the month (PTA), jointly with the instrument’s own first stage (PTA-FWL), or after monthly aggregation (ATP). The moments are formed in the units in which each family is published, which is the unit in which block structure is asserted (footnote 12): the futures surprises and the Bauer and Swanson (2023b) surprise in percentage points of the futures-implied rate, the Swanson (2021) factors in the unit-variance normalisation in which they are published, the information-robust surprise in percentage points, the extended Romer–Romer shock in percentage points of the intended target, and the recursive innovations in percentage points of the policy rate concerned. Before the purge, three of the four instruments of Appendix Table 2, the Bauer–Swanson surprise, the Romer–Romer shock and the recursive innovation, are predictable from the controls; the fourth, the information-robust surprise, which is already purged of the Greenbook forecasts, is not.

Raw and purged versions of the same instrument are kept in separate families. In a family holding both, the within transform would leave only the moment of the component the purge removed, which is contaminated by construction; Appendix F.2 shows that pooling such pairs rotates the direction and is rejected decisively by the overidentification test.

Appendix Table 4 compares the two partitions that satisfy the rule: purged families (P2) and raw families (P3), with and without the Swanson family, unweighted and precision-weighted. The purged, precision-weighted partition with the Swanson family is the headline. It is chosen on two grounds stated before any response is looked at: the assignment rule, and the fact that precision weighting is the efficient version of the estimator and the only one that does not depend on the units in which each family is published. Without precision weights the estimate is effectively the Swanson family’s, whose unit-variance factors take 99% of the identifying variation; Appendix Figure 14 compares the two estimates, whose directions are 12° apart. That the headline is also the only row with an output contraction, a price decline at every horizon and a moderate credit response is a finding, not the reason for the choice. The data-driven audit of Section 5.5 agrees on the count: the precision-weighted penalised criterion selects $\hat{C} = 5$ at every penalty up to a tenth of the noise floor, the upper end of the admissible window (71), which here is $4 \leq \hat{C} \leq 5$, while the fit at $\hat{C} = 5$ reproduces the economic assignment only in part (adjusted Rand index 0.54), as expected

¹⁵The overidentification statistic (67) uses the full sampling covariance of the moments, not only its diagonal, since proxies built from the same announcements or the same VAR are correlated up to 0.99. It has $(n-1)(m-C-1)$ degrees of freedom and splits into a within-family part on $(n-1)\sum_c(|c|-2)$ and a between-family part on $(n-1)(C-1)$, which asks whether the families agree on the direction (Appendix E.7).

at this sample size. The silhouette would select $\hat{C} = 2$, which is why it is read as a diagnostic of separation and not as a selector.

6.2 The headline impulse responses

Figure 3 is the headline. A one-point funds-rate tightening lowers industrial production by 1.3% on impact and by 5.3% at the trough, fourteen months out (68% band $[-8.7, -3.3]$); consumer prices fall at every horizon, by 0.1% on impact, 1.0% after two years and 1.4% after three, with no price puzzle in the point estimate; the excess bond premium rises by 0.9 points on impact (68% band $[0.2, 1.7]$) and returns to zero within eighteen months; and the funds rate rises for a few months and then turns negative from the eighth month, a reversal discussed below. The bands exclude zero for output at every horizon and for the credit spread over the first sixteen months; the price response is imprecise, with a band of $[-2.5, +0.1]$ at two years. The impact response of output is small relative to the trough, a quarter of it, and core prices respond later than the headline index (Section 7.6).

Appendix Table 5 summarises the estimate and its diagnostics. The pooled instrument has a first stage of $F = 23.9$ ($\mu^2 = 22.4$). Exact block structure is rejected at 5% ($J_T = 46.0$ on 30 degrees of freedom, $p = 0.031$), and the rejection is entirely between families: the within-family part is 17.8 on 18 degrees of freedom ($p = 0.47$) and the between-family part 28.2 on 12 ($p = 0.005$). The proxies of a family therefore share their contamination as the model assumes, while the families disagree somewhat on the direction. The ordering condition (32) holds with a margin of about five ($\hat{\sigma}_\phi^2/m = 1.5 \times 10^{-3}$ against a largest idiosyncratic eigenvalue of 3.2×10^{-4}). The eigengap of the weighted within scatter is 3.1, and the level set of Appendix E.6 extends 18.6° from the estimate: the direction is identified, but not sharply.

The reversal of the funds rate. The funds-rate response overshoots: after the unit tightening it falls below zero from the eighth month, to a trough of -0.57 points at nineteen months. Because the response path $\Phi_h \hat{\beta}$ is linear in the impact vector, it splits into the part set off by the funds rate itself, a persistent tightening common to every identification, and the parts set off by the impact responses of the other variables. The overshooting comes almost entirely through the excess bond premium: a rise in the premium is followed by systematic easing, as documented by Gilchrist and Zakrajšek (2012) and estimated in the policy rule by Caldara and Herbst (2019), who also find the funds rate turning negative after a tightening. The larger the premium’s impact response, the earlier and deeper the reversal. The strong single instruments raise the premium by 0.05 to 0.26 points per point of funds rate and keep the funds rate positive for five years; the pooled estimate raises it by 0.93, and the orthogonalised surprise by 1.97, with a reversal to -1.5 points (Figure 5). The overshooting is therefore a measure of how much credit-market disturbance the identified shock carries, largest for the weakest of these instruments, as the magnification of Section 3.3 predicts. On the 1985–2008 VAR even the recursive benchmark turns negative after about two years, so part of the reversal is a property of the VAR’s dynamics (Section 7.3).

6.3 What identifies the shock

Figure 4 and Appendix Table 6 decompose the identification. Individually the proxies are weak to moderate: MP1 and the Swanson target factor have first stages of 14, the other high-frequency surprises between 2 and 7, the forward-guidance and asset-purchase factors 4 and 0.1, the information-

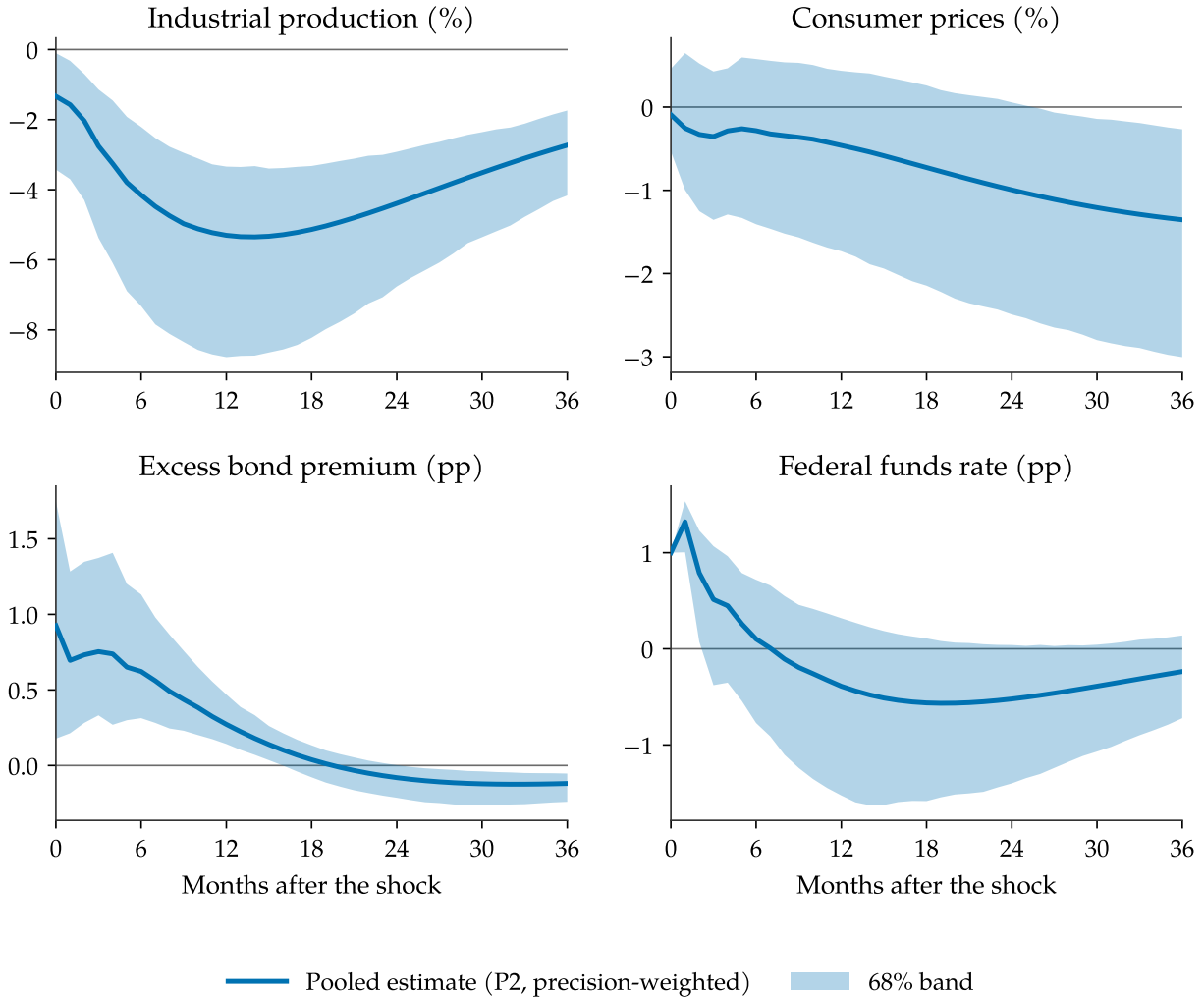


Figure 3: The pooled monetary policy shock. Responses to a +1 percentage-point shock to the federal funds rate. Pool-within/annihilate-across estimator on sixteen purged proxies in five families, precision-weighted, in the Σ_u^{-1} metric. VAR in (IP, CPI, EBP, Fed Funds) with six lags, 1973m1–2019m12; moments on each proxy’s support within 1988m1–2019m12. Shaded: 68% median-recentred percentile bands from the moving-block bootstrap, 1,000 draws, with the VAR, the moments and the weights re-estimated in each draw.

robust variants about 6, and the Romer–Romer variants 18 to 30. The Cholesky innovations are strong (F of 30 to 534) because they are functions of the VAR’s own innovations; in the pool they are one family whose common contamination is annihilated like any other’s, and what they contribute to the direction is their within-family contrast. The pooled instrument’s $F = 23.9$ is higher than that of every external proxy except one Romer–Romer variant (PTA-FWL, $F = 29.6$).

The right panel of Figure 4 shows where the identifying variation comes from: 79% from the Swanson factors and 19% from the Cholesky family, with the Swanson family’s own direction 10° from the pooled one and the Cholesky family’s 37° . The high-frequency, information-robust and Romer–Romer families contribute almost nothing to the direction. For the last two the reason is

mechanical: their three members are variants of one series purged in three ways (correlations 0.98 to 0.995), so there is almost no relevance dispersion within the family, and their own directions, 77° and 89° from the pooled one, are noise. They still enter the *level* of the pooled instrument, and hence its first stage.

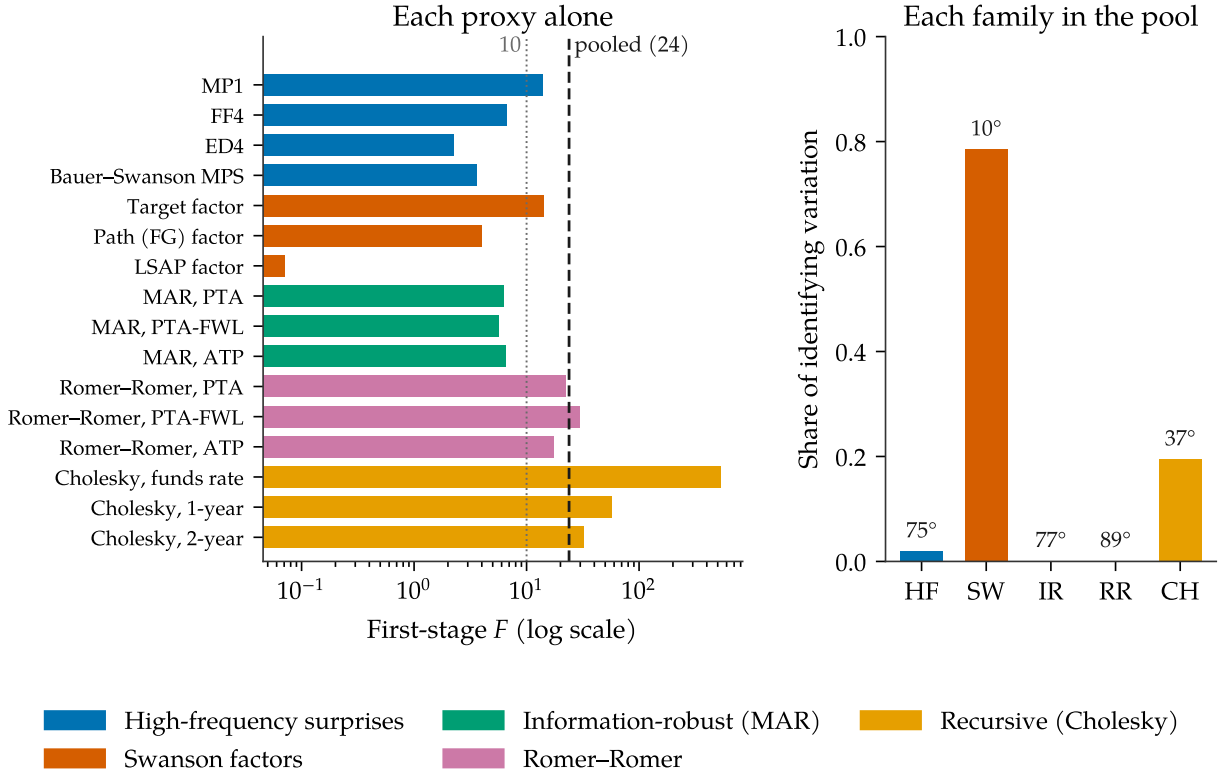


Figure 4: What identifies the shock. Left: first-stage F of each proxy entered alone against the funds-rate innovation on the moment window (log scale); the dotted line marks $F = 10$, the dashed line the pooled instrument. Right: each family’s share of the identifying variation $\sum_{j \in c} \hat{\phi}_j^2 / \hat{\sigma}_j$ in the precision-weighted within scatter; the label is the angle between the direction the family identifies on its own and the pooled direction.

The Swanson family. The three factors form one family under the assignment rule: they come from the same announcements, the same intraday window and the same purge, and differ in the dimension of policy they load on, which is the relevance heterogeneity the within transform needs. The premise that they share their contamination is testable inside the family, and the test is borderline: $J_c = 7.7$ on three degrees of freedom ($p = 0.052$), against $p \geq 0.35$ for every other family. This is where the Fed information effect (Nakamura and Steinsson, 2018; see Bauer and Swanson, 2023a for the contrary reading), which the literature locates in the forward-guidance factor, would show up. The estimate is sensitive to how the family is cut. With the target factor on its own and the forward-guidance and asset-purchase factors as a two-member family, the direction moves by 27° , the first stage falls to 12.5, exact block structure is no longer rejected ($p = 0.13$), and

output falls by 0.5% on impact and 4.6% at the trough, prices by 1.1% at two years. Dropping the asset-purchase factor rotates the direction by 37° and brings back a price puzzle (+0.7% at two years); dropping the forward-guidance factor moves it by 11° and leaves the responses close to the headline. The headline direction therefore rests on the contrast between the asset-purchase factor and the other two, and without the family the estimate falls back to the recursive benchmark (Section 7.1).

Near-duplicate proxies. The information-robust and Romer–Romer families each hold three purgings of one series. They add proxies and degrees of freedom without adding information about the direction, and they make the between-family test harder to read: a family that cannot identify a direction on its own has an own direction that is noise, and its disagreement with the pooled direction inflates J_{between} . Keeping one purging per series, the Bauer–Swanson convention, turns both into single-proxy families, which contribute to the level of the pooled instrument but not to the direction: the direction moves by 0.2° and the first stage is 23.2. Dropping the information-robust family altogether removes the rejection of exact block structure (Section 7.1).

How many families the pool needs. Appendix Table 9 adds the families one at a time, and it puts condition (O2) of Proposition 5 on display. With the high-frequency and Cholesky families alone, or with the Romer–Romer family added, the pool is strong ($F \approx 50$) but the response is a small, slow contraction with prices rising for two years, close to the recursive benchmark: with two or three families (O2) fails in its conservative form ($C \leq n - 1 = 3$), and the within contrasts that remain come almost entirely from the Cholesky family. The response changes when the Swanson family enters, which also brings the count to four so that (O2) is met; Section 7.1 shows that it is the Swanson family, not the count, that matters: output falls by 1.3% on impact and 5.3% at the trough, and prices fall. Adding the information-robust family changes nothing at the reported precision, because its three variants carry no relevance dispersion: a family of near-copies adds nothing to the direction, and little to the level because its members are weak.

The surviving-relevance column measures the same thing directly. It is the share of the relevance vector that survives the annihilator, computed in the precision-weighted metric,

$$\|P_{\ker(\hat{R}D)}D\hat{\phi}\|^2/\|D\hat{\phi}\|^2,$$

with $D = \text{diag}(\hat{\sigma}_j^{-1/2})$, so that each proxy’s relevance is measured in units of its own sampling noise. With two or three families it is 0.12: (O2) fails, the level component of Proposition 6 is lost, and what survives is the within-family contrasts alone. It jumps to 0.43 when the Swanson family enters and (O2) is met, and is 0.39 in the headline.

Units and the first stage. The strength numbers need one qualification. The headline’s pooled instrument uses the annihilator of Algorithm 1, $\hat{w} = P_{\ker \hat{R}}\hat{\phi}$, a Euclidean projection, so its F and μ^2 depend on the units in which the proxies are published: multiplying the high-frequency proxies by 100 moves F from 23.9 to 17.1 and μ^2 from 22.4 to 19.0. The precision-weighted form of the constrained-optimal instrument, (63), with noise variance proportional to $\hat{\sigma}_j$, is invariant to rescaling any proxy and gives $F = 20.9$ and $\mu^2 = 19.2$. The two instruments give identical impulse responses,

because both annihilate the fitted contamination and so reproduce $\hat{b}_p$ exactly (Proposition 8); they differ only in the shock series (correlation 0.89) and in the strength they report.

6.4 What pooling adds

The purge alone does not solve the problem it addresses: on the headline VAR it weakens the [Bauer and Swanson \(2023b\)](#) surprise from $F = 14.5$ to $F = 3.6$ and enlarges its responses, as Section 4.4 showed (Appendix Figure 13). Figure 5 compares the pooled estimate with the orthogonalised surprise alone, the most aggressively cleaned and individually one of the weakest proxies in the panel. Pooling raises the first stage from 3.6 to 23.9, narrows the bands at almost every horizon, halves the credit response on impact (+0.9 against +2.0), cuts the price response at two years by two thirds (−1.0% against −3.0%), and moderates the output trough (−5.3% against −6.6%).

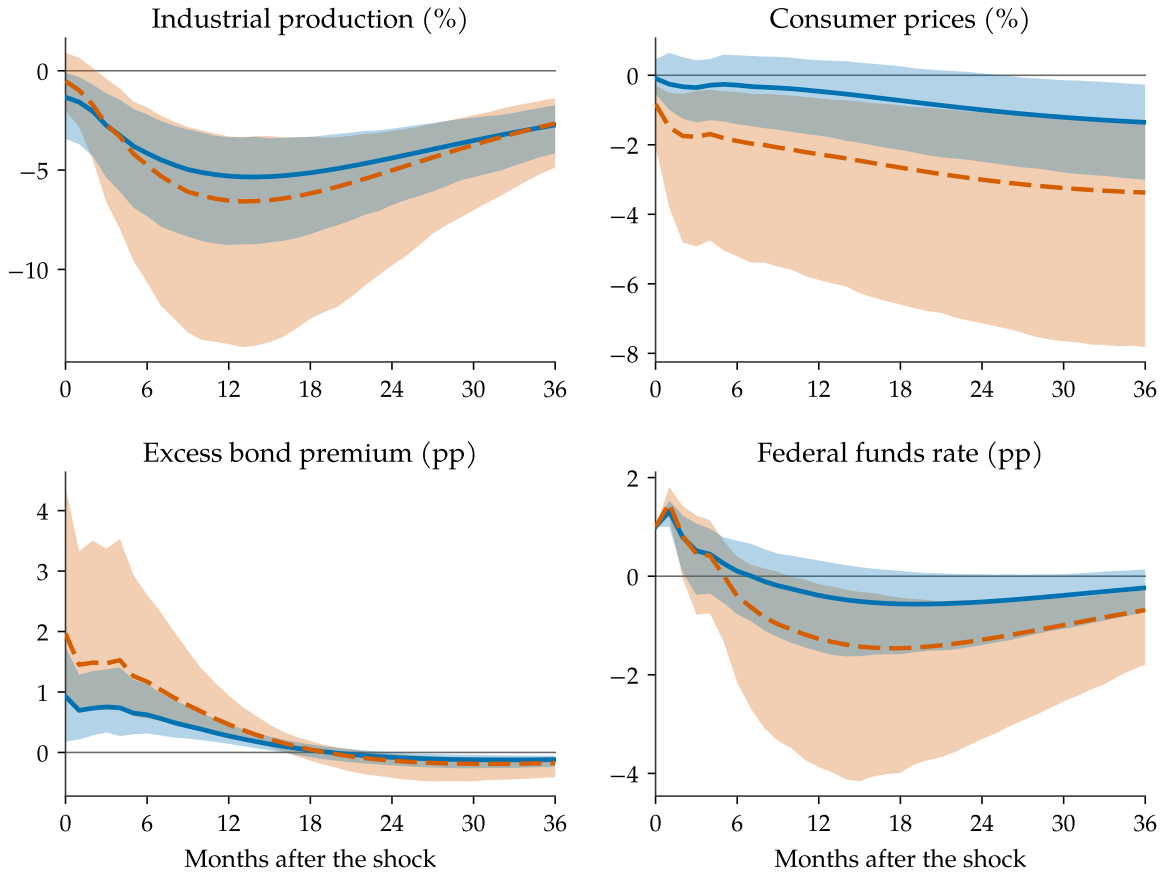
Appendix Table 8 places the pooled estimate among the single proxies and two alternative pools. The largest single-instrument magnitudes come from the weakest instruments, the orthogonalised surprise ($F = 3.6$, trough −6.6%) and the information-robust surprise ($F = 6.3$, −8.4%), and the smallest from the two strongest, the Romer–Romer shock ($F = 22.3$, −1.7%) and the recursive innovation ($F = 534$, −1.4%); the Swanson target factor ($F = 14.2$, −6.8%) is the exception. Two pools matter. Pooling all sixteen proxies *without* the within transform gives a stronger first stage ($F = 32.8$) but an output response about half as large on impact (−0.7%), because the family-common contamination is not removed, and the strong Cholesky family dominates the level. The high-frequency and Swanson families alone give a weak pool ($F = 7.2$) with a larger response; adding the other families is what makes the pool strong.

What the annihilator removes. The within transform discards each family’s mean moment, and with it the family’s common contamination $d_c = b_{-p}\bar{\psi}_c$. Figure 6 shows what d_c would have done to the responses had it been kept. For each family take the precision-weighted mean of its members’ moments, $\bar{g}_c = \sum_{j \in c} (\hat{g}_j / \hat{v}_j) / \sum_{j \in c} (1 / \hat{v}_j) = \bar{\phi}_c b_p + d_c$, the moment of the family’s average proxy used alone. Writing $\bar{\phi}_c^p := \bar{\phi}_c b_{pp}$ for the family mean’s relevance in units of the policy rate, so that $\bar{g}_c = \bar{\phi}_c^p \beta + d_c$, and $d_{c,p}$ for the funds-rate coordinate of d_c , the family mean’s normalised impact vector differs from the pooled one by

$$\frac{\bar{g}_c}{\bar{\phi}_c^p + d_{c,p}} - \beta = \frac{d_c - d_{c,p} \beta}{\bar{\phi}_c^p + d_{c,p}}, \quad (56)$$

up to the family mean of the idiosyncratic and sampling errors, and its responses at horizon h by Φ_h times the same vector. Each bar is this difference, on impact and after twelve months. It is therefore d_c mapped into responses, with two qualifications for reading it: only the part of d_c off the target counts, because the part along β cannot be told apart from relevance, which is the shear of Appendix E.2; and it is divided by the family mean’s own loading on the funds rate, so that a family with little relevance can show a large bar from a small d_c , the magnification of Section 3.3 at the level of a family. The whiskers are 68% intervals from the headline bootstrap. Appendix Table 7 adds a unit-free measure of the same object, the contamination share $\|P_{\hat{q}}^\perp \bar{g}_c^w\|^2 / \|\bar{g}_c^w\|^2$, the fraction of the whitened family mean that lies off the pooled direction.

The two wide-window families, whose proxies are monthly innovations rather than surprises on the day of an announcement, point toward a *milder* contraction. The Cholesky and Romer–Romer



— Pooled estimate ($F = 23.9$) - - - Orthogonalised Bauer-Swanson alone ($F = 3.6$) 68% bands

Figure 5: The value added of pooling. Pooled estimate (solid, $F = 23.9$) against the orthogonalised Bauer-Swanson surprise alone (dashed, $F = 3.6$), each with its 68% band as in Figure 3.

family means would have raised the twelve-month output response by 4.7 and 3.9 points (68% intervals $[2.6, 8.1]$ and $[1.9, 7.5]$), put prices 0.6 to 0.8 points higher, and lowered the credit-spread response on impact by 0.7 to 1.0 points. Their means are also the most contaminated: 81% and 43% of each lies off the pooled direction. The announcement-based families lean the other way, toward a *stronger* contraction: the high-frequency and information-robust means would have lowered prices by 1.1 to 1.4 points at twelve months and raised the credit response on impact by about 0.8 points, with intervals that exclude zero; the information-robust mean would also have deepened the twelve-month output fall by 3.2 points. The Swanson family's mean is the least biased, with every interval covering zero. The contamination share and the size of the response bias need not line up: a family mean almost aligned with $\hat{q}$ (information-robust, 5%) can still move the responses a lot if its small off-direction component loads on the slow-moving variables.

The two groups of biases have opposite signs, which is why the pooled estimate sits between the single proxies of Appendix Table 8, and why pooling *without* the within transform, where

the strong Cholesky family dominates the level, halves the output response on impact. The bars are deviations from the pooled estimate, not from the truth: reading them as biases requires the block-structure assumption, which Section 6.3 tests but cannot prove. What they show without that assumption is how much the partition matters for the answer: the within transform moves the twelve-month output response by up to five points, in opposite directions for different families.

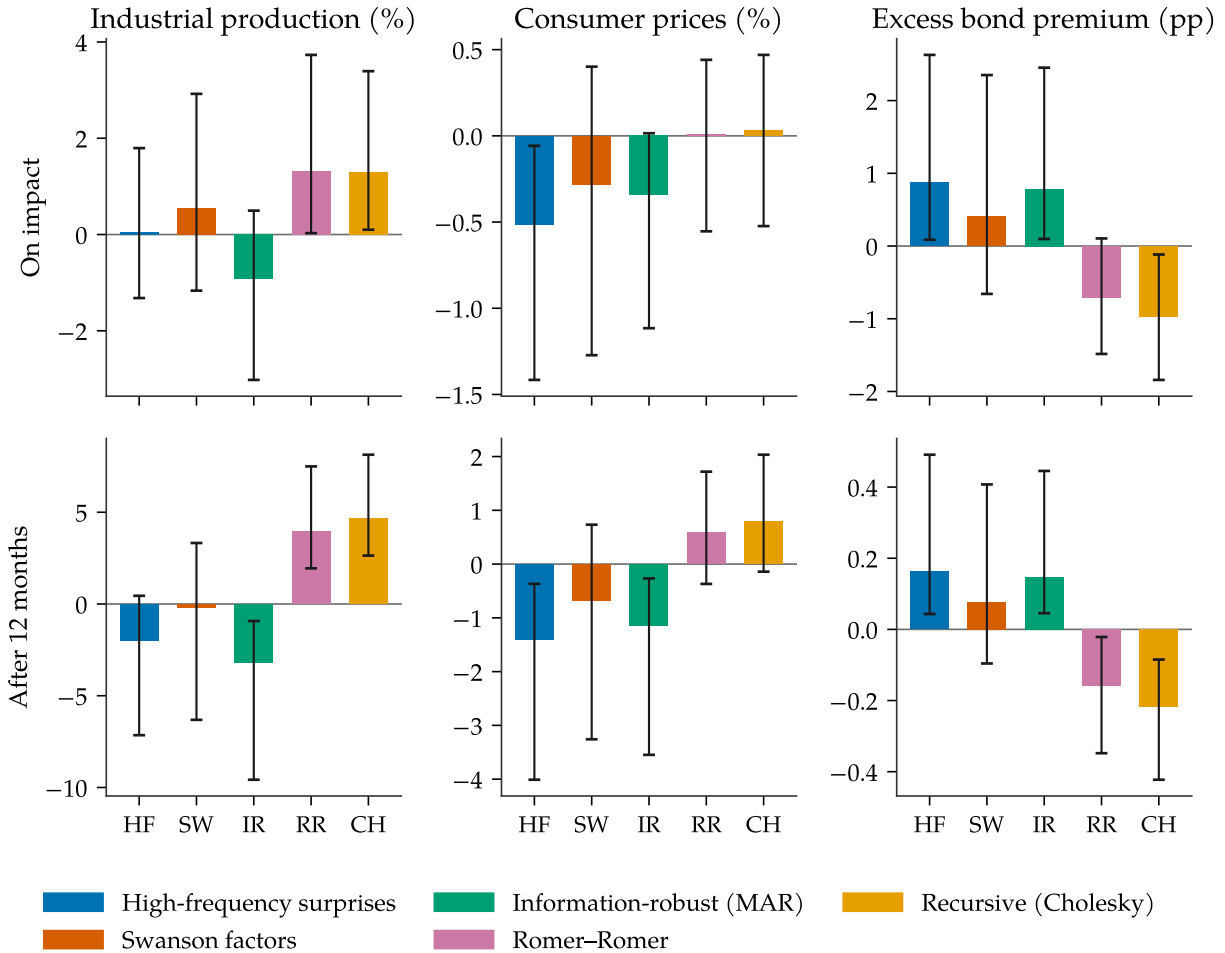


Figure 6: The family-common component removed by the annihilator. For each family, the response to a +1 percentage-point funds-rate shock identified by the family’s precision-weighted mean moment $\bar{g}_c$ (its average proxy used alone), minus the pooled response, (56). Top row: on impact; bottom row: after twelve months. Whiskers: 68% median-recentred percentile intervals, 1,000 draws, with the VAR, the moments, the weights and the pooled direction re-estimated in each draw. HF: high-frequency surprises; SW: Swanson factors; IR: information-robust; RR: Romer-Romer; CH: recursive (Cholesky).

6.5 All database proxies

The headline is not a product of the curation. Figure 7 re-estimates the pooled shock on every proxy in the instrument database with at least 60 observations in the moment window, 88 series in 36

families (each family again one source and one purging), raw and purged alike, and including the [Jarociński and Karadi \(2020\)](#), [Bu et al. \(2021\)](#) and [Aruoba and Drechsel \(2024b\)](#) series that are not in the headline panel. The response paths correlate at 0.995 with the headline's and the first stage is essentially the same ($F = 23.4$); the output response is somewhat smaller and the price response flatter. The recursive benchmark, which orders the funds rate after output and prices and the credit spread after the funds rate, imposes zero impact responses of output and prices and delivers a small, slow contraction with a price puzzle.

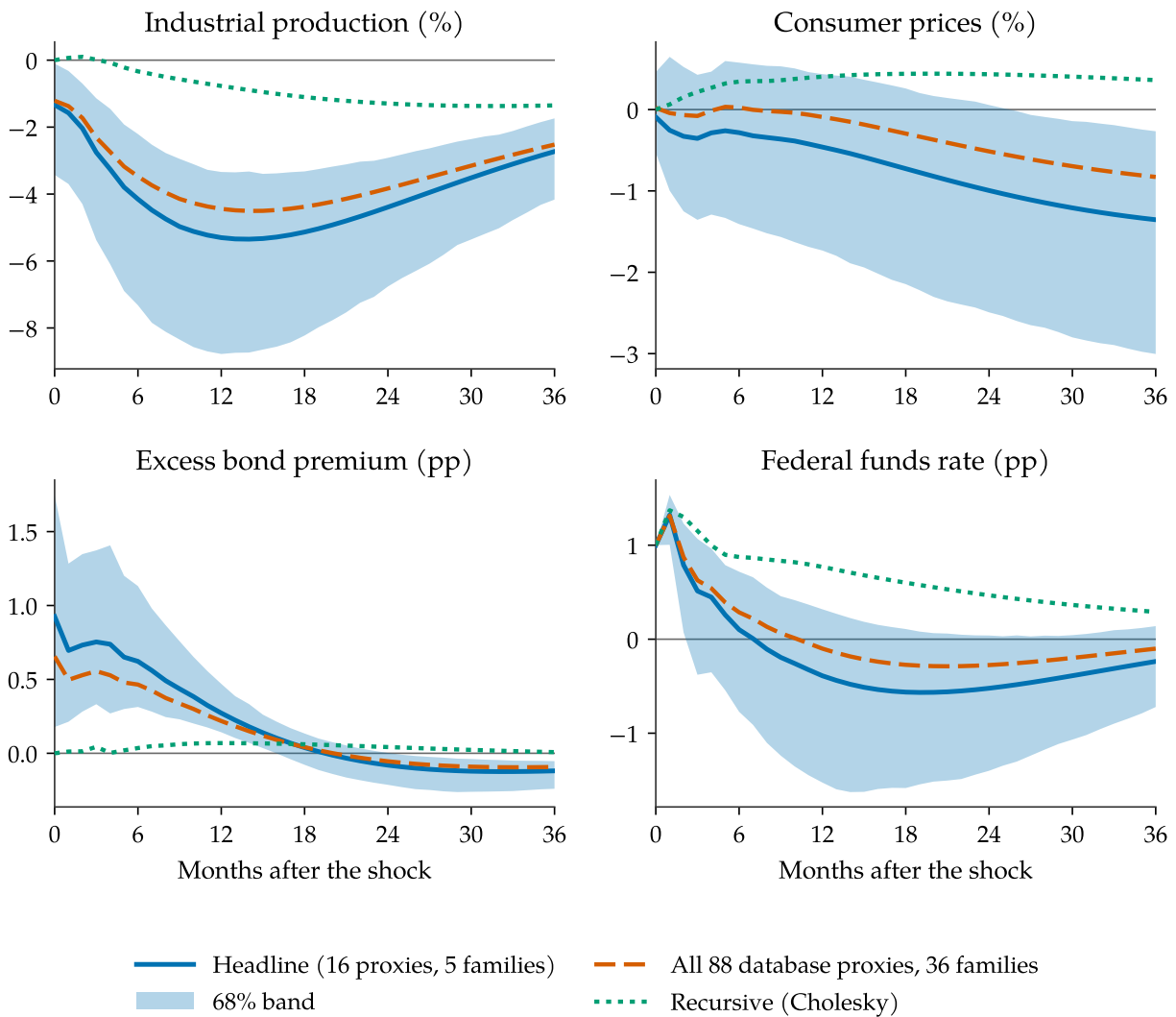


Figure 7: All database proxies and the recursive benchmark. Headline (solid, with 68% band); the same estimator on all 88 proxies of the database with at least 60 observations in the moment window, grouped into 36 families by source and purging (dashed); recursive identification with the funds rate ordered third (dotted).

7 Robustness

The identification is stress-tested along six dimensions: the families that enter it, the policy indicator, the sample, the variable set, the lag length and the measure of prices. Every exercise holds the headline families, weights and windows fixed unless stated otherwise.

7.1 Leaving one family out

Appendix Table 10 and Appendix Figure 15 drop each family in turn and re-estimate with the remaining four ($C = 4 > n - 1 = 3$, so (O2) still holds). Three of the five families can be dropped without consequence. Without the high-frequency, information-robust or Romer–Romer family the direction moves by at most 3.4° and the responses are unchanged at the reported precision. Dropping the information-robust family also removes the rejection of exact block structure ($p = 0.40$), in line with Section 6.3: its three near-copies inflate J_T without informing the direction.

The other two cannot. Without the Swanson family the direction rotates by 45° and the estimate falls back to the small, slow contraction of the two- and three-family rows of Appendix Table 9: output troughs at -2.2% after nineteen months, prices rise, and the credit response is a third of the headline's. Its first stage is high ($F = 50.6$) because the Cholesky proxies dominate the level, but its concentration halves ($\mu^2 = 10.8$). The headline direction is therefore the Swanson family's, as the identification shares of Appendix Table 6 already said. Without the Cholesky family, whose proxies are functions of the VAR's own innovations, the direction moves by 14° and the responses are larger: output falls by 2.6% on impact and 8.4% at the trough, prices by 2.0% at two years, and the credit spread rises by 1.4 points on impact. The first stage drops below ten ($F = 9.8$, $\mu^2 = 18.4$). The internal proxies thus do not create the result; they strengthen the pool and pull the responses toward the recursive benchmark.

7.2 The policy indicator: the shadow rate and the Treasury yields

Over 2009–2015 the funds rate is at its floor, so the headline identifies from the months in which the rate moves. Replacing the funds rate with the Wu and Xia (2016) shadow rate, equal to the funds rate away from the zero lower bound and to the Wu–Xia estimate at it, carries identification through those years, where the forward-guidance and asset-purchase factors move. Appendix Figure 16 shows that the result is essentially unchanged: $F = 18.1$, output falls by 1.6% on impact and 6.4% at the trough, prices fall at every horizon, and the credit spread rises by 1.1 points on impact. The shadow-rate responses are slightly larger than the funds-rate ones, and exact block structure is not rejected at 5% ($p = 0.055$).

Appendix Table 11 varies the indicator further. Normalised to the one-year Treasury yield the pool collapses ($F = 0.8$) and the responses explode: the monthly innovation in a one-year yield is dominated by term-premium and expectations news that the instruments do not explain, and no amount of pooling manufactures strength the maturity does not carry. Normalised to the two-year yield the first stage is high ($F = 43.5$) but the responses are not credible (output rises on impact, prices rise at two years); that strength comes from the Cholesky family, whose two-year member is a function of the two-year innovation itself, a mechanical identity rather than external identification. The funds rate and the shadow rate are the indicators the panel identifies.

7.3 A pre-zero-lower-bound VAR

Estimating the VAR on 1985m1–2008m12, with moments on 1988m1–2008m12, keeps the impact responses almost exactly (output -1.2% , prices 0.0% , credit spread $+0.9$) and the first stage ($F = 25.4$), but produces a more persistent contraction: the output trough is -8.1% at 29 months (Appendix Figure 17). The direction is better determined on this window (eigengap 8.7) and exact block structure is rejected more clearly ($p = 0.011$). The difference in persistence appears to be a property of the Great-Moderation dynamics rather than of the identification: on the same VAR the recursive benchmark's funds-rate response also turns negative after about two years.

7.4 Larger systems: one added variable at a time

Enlarging the system to seven variables at once violates (O2) with five families ($C = 5 \leq n - 1 = 6$): the contamination can occupy six directions and five families cannot span an annihilator over them, so the identification falls back on within-family contrasts and the surviving relevance collapses. Adding one variable at a time keeps (O2) ($C = 5 > n - 1 = 4$) and asks the question the enlargement is meant to answer: how do other variables respond to the *same* shock? Each panel of Figure 8 is a separate five-variable VAR, the headline four plus the variable shown, estimated with the headline families, weights and windows; only the added variable's response is plotted. The additions are the unemployment rate, a commodity-price index, the S&P 500 and housing starts. The two-year Treasury yield is not among them: it is close to a second policy indicator, and with a yield in the system the pooled direction no longer separates a funds-rate shock from a move in the whole curve, so exact block structure is rejected decisively ($p < 0.001$).

A one-point tightening raises the unemployment rate by 1.1 points at a sixteen-month peak, lowers commodity prices by about 7% over the same horizon, lowers the S&P 500 by about 23% at six months, and lowers housing starts by 9.5% on impact and 18% at a four-month trough, with a band that excludes zero from the first month to about the twenty-second month. Housing responds first and fastest, ahead of output and unemployment, as the interest-rate channel predicts. In all four systems the responses of output and prices are close to the headline and exact block structure is not rejected at 1%; the housing system is not rejected at 5% ($p = 0.081$).

7.5 Lag length

Re-estimating the headline with twelve lags instead of six, with the same families, weights, windows and bootstrap, moves the impact vector by 4.1° and leaves the pooled instrument as strong ($F = 24.6$, $\mu^2 = 24.2$). The responses keep their shape: output falls by 1.1% on impact and 4.1% at a ten-month trough, prices by 0.8% at two years, and the credit spread rises by 0.9 points on impact (Appendix Figure 18). The price and funds-rate responses lie inside the headline's bands at every horizon, and output does over the first eighteen months; after that output recovers somewhat faster and lies just above the headline band, and the credit spread returns to zero about six months sooner. The two sets of bands overlap throughout. Exact block structure is rejected more firmly with twelve lags ($p = 0.003$) than with six ($p = 0.031$).

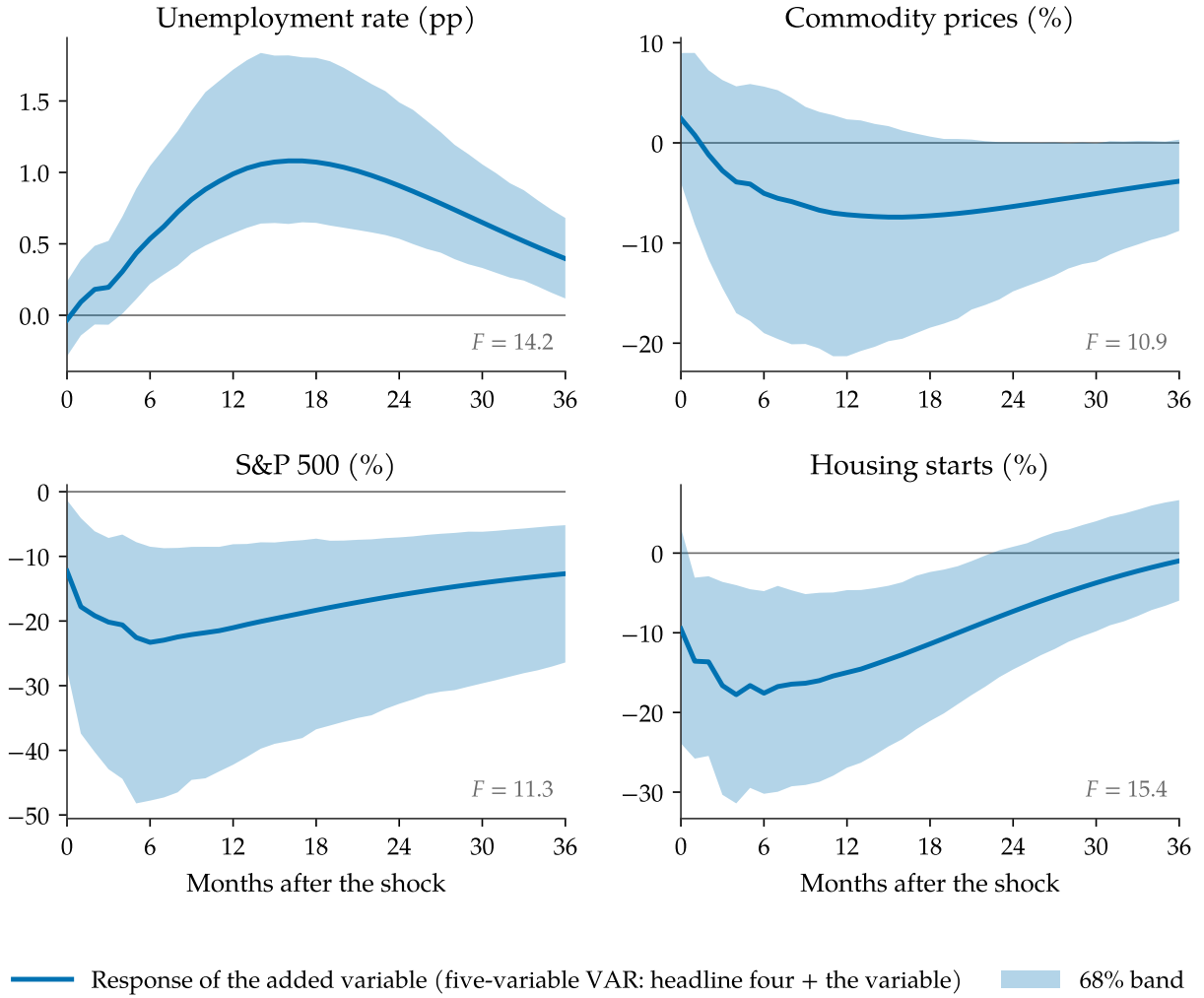


Figure 8: Larger systems, one added variable at a time. Each panel is estimated separately: a five-variable VAR in the headline four variables plus the variable shown, with the headline families, precision weights, windows and bootstrap; the panel plots only the response of the added variable to a +1 percentage-point funds-rate shock, with its 68% band. F is the first stage of the pooled instrument in that system.

7.6 The price measure: levels, inflation rates and core prices

The headline enters prices as the log level of the CPI, the convention of the proxy-SVAR literature since [Gertler and Karadi \(2015\)](#), so that the price response is cumulative and the lags absorb the trend. Two other conventions are common, year-over-year inflation, $100(\log P_t - \log P_{t-12})$, and month-over-month inflation, $100\Delta \log P_t$, with P_t the CPI, and each changes the reduced form on which the moments are formed, and so the direction. Core CPI is a further check, since the headline index moves with energy prices, which respond quickly to financial conditions. Each replacement keeps $n = 4$, so the families, the order conditions and the partition are as in the headline. Appendix Figures 19 and 20 report, for each of the six systems, the price response under the pooled estimate

and under the orthogonalised [Bauer and Swanson \(2023b\)](#) surprise alone. The responses are not on a common scale across rows: the level response is cumulative, the year-over-year response is a twelve-month moving sum of the monthly responses, and the month-over-month response is the monthly rate.

The comparison of the paper holds under every price measure. The pooled instrument is strong in all six systems, with a first stage between 19.5 and 25.4, while the orthogonalised surprise is weak in all of them, between 2.6 and 4.6, and the weak instrument reports the larger price effect in every panel. Core prices respond less and later than the headline index: under the pooled estimate the core level rises slightly in the first year and falls below zero only after about twenty months, to about -0.5% after three years, with a band that contains zero, against about -1.9% under the orthogonalised surprise. The month-over-month responses under the pooled estimate are short-lived, a fall in headline inflation of about 0.2 points on impact that is gone within three months, against about 0.9 points for the orthogonalised surprise. In year-over-year terms, pooled headline inflation falls by about 1.9 points within the first half-year and is still about 0.7 points lower after three years, against about 4 points for the orthogonalised surprise; core inflation moves by at most 0.2 points in either direction, against a trough of about 0.9 points for the orthogonalised surprise.

8 Conclusion

High-frequency monetary policy surprises are informative but not necessarily clean or strong. The purge that removes their predictability is necessary, but at meeting frequency it removes proportionally more of the policy-shock variation than of the contamination, and whatever contamination survives is divided by the smaller first stage. The purged surprise therefore yields larger responses than any conventional identification scheme on the same reduced form. [Bauer and Swanson \(2023b\)](#) read this enlargement as the removal of an attenuation bias; this paper argues that it is residual contamination magnified by a weaker first stage.

Because the bias scales with the inverse of the first stage, the productive response is to preserve strength rather than to clean further. The shock is the portion of the decision not owed to economic conditions, and the announcement window reveals only the slice of it the market had not already priced. Instruments built from the decision itself, an extended [Romer and Romer \(2004\)](#) shock and a Cholesky-based instrument, carry contamination of their own but keep their first stage under the same purge, and their responses are not enlarged.

When no single proxy is both strong and clean, the pool-within, annihilate-across estimator replaces a clean instrument with a restriction on the structure of the contamination. Pooling *within* families of proxies that share a contamination direction buys strength, annihilating *across* families buys validity, and the direction is identified by the dispersion of relevance within families rather than by its level. On sixteen purged U.S. proxies in five families, the estimator delivers a strong first stage and a standard monetary contraction: output falls, the headline CPI falls at every horizon, core prices respond later, and the credit spread widens.

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Appendix

A Proofs for Section 3

Throughout, the structural shocks are mutually orthogonal with unit variance, so that $\mathbb{E}[\varepsilon_t^{-p} \varepsilon_t^{-p'}] = I$ and $\mathbb{E}[\varepsilon_t^{-p} \varepsilon_t^p] = \mathbf{0}$, and the policy innovation is $u_t^p = b_{pp} \varepsilon_t^p + \lambda' \varepsilon_t^{-p}$.

Proof of Proposition 1 (what the purge does to the pair). Setup. Write the contamination as $\chi_t := \psi' \varepsilon_t^{-p}$ and its predictable part as $\hat{\chi}_t := P_{\mathcal{W}} \chi_t$. Because $P_{\mathcal{W}}$ is linear and $P_{\mathcal{W}} \varepsilon_t^p = 0$, the purge removes only predictable contamination:

$$\hat{z}_t = P_{\mathcal{W}} z_t = \hat{\chi}_t, \quad z_t^\perp = \phi \varepsilon_t^p + \chi_t - \hat{\chi}_t.$$

Multiplying by ε_t^{-p} and taking expectations, the policy-shock term drops out and

$$\psi^\perp = \mathbb{E}[\varepsilon_t^{-p} z_t^\perp] = \psi - a, \quad a := \mathbb{E}[\varepsilon_t^{-p} \hat{\chi}_t].$$

(i) *Validity.* Expanding the square,

$$\|\psi^\perp\|^2 = \|\psi\|^2 - 2\psi' a + \|a\|^2.$$

The middle term is $\psi' a = \mathbb{E}[\chi_t \hat{\chi}_t] = \text{Var}(\hat{\chi}_t)$, since the projection residual $\chi_t - \hat{\chi}_t$ is orthogonal to $\hat{\chi}_t$. The last term is the variance of $a' \varepsilon_t^{-p}$, the projection of $\hat{\chi}_t$ on the shocks, so $\|a\|^2 \leq \text{Var}(\hat{\chi}_t)$. Together,

$$\|\psi^\perp\|^2 \leq \|\psi\|^2 - 2\text{Var}(\hat{\chi}_t) + \text{Var}(\hat{\chi}_t) = \|\psi\|^2 - \text{Var}(\hat{\chi}_t),$$

which is the claim, strict whenever $\text{Var}(\hat{\chi}_t) > 0$.

(ii) *First stage.* Multiply $z_t^\perp$ by $u_t^p = b_{pp} \varepsilon_t^p + \lambda' \varepsilon_t^{-p}$ and take expectations. The cross terms between ε_t^p and ε_t^{-p} vanish by orthogonality of the shocks, and $\mathbb{E}[\varepsilon_t^p \hat{\chi}_t] = 0$ because $\hat{\chi}_t$ lies in the span of the controls, to which ε_t^p is orthogonal. With $\text{Var}(\varepsilon_t^p) = 1$,

$$\pi^\perp = \text{Cov}(z_t^\perp, u_t^p) = \phi b_{pp} + \lambda' \mathbb{E}[\varepsilon_t^{-p} (\chi_t - \hat{\chi}_t)] = \phi b_{pp} + \lambda' \psi^\perp.$$

(iii) *Share lost.* Covariance is additive over $z_t = \hat{z}_t + z_t^\perp$, so $\pi = \text{Cov}(\hat{z}_t, u_t^p) + \pi^\perp$. Dividing by π gives $\pi^\perp / \pi = 1 - \delta_\pi$. $\square$

Derivation of the correspondence (10) and of (11). Variances. The fitted value and the residual of the purge regression are orthogonal, so $\text{Var}(z_t) = \text{Var}(\hat{z}_t) + \text{Var}(z_t^\perp)$ and, by the definition of R^2 ,

$$\text{Var}(z_t^\perp) = (1 - R^2) \text{Var}(z_t), \quad \text{Var}(\hat{z}_t) = R^2 \text{Var}(z_t).$$

First identity. By Proposition 1(iii), $\text{Cov}(z_t^\perp, u_t^p) = (1 - \delta_\pi) \text{Cov}(z_t, u_t^p)$. Dividing by the standard deviations,

$$\rho^\perp = \frac{(1 - \delta_\pi) \text{Cov}(z_t, u_t^p)}{\sqrt{1 - R^2} \sqrt{\text{Var}(z_t) \text{Var}(u_t^p)}} = \frac{1 - \delta_\pi}{\sqrt{1 - R^2}} \rho.$$

Second identity. Writing each covariance in the definition of δ_π as a correlation times standard deviations,

$$\delta_\pi = \frac{\hat{\rho} \sqrt{\text{Var}(\hat{z}_t)}}{\rho \sqrt{\text{Var}(z_t)}} = \sqrt{R^2} \frac{\hat{\rho}}{\rho}.$$

Third identity. With the same T for both instruments and $F = T\rho^2/(1 - \rho^2) \approx T\rho^2$ when ρ is small, $F^\perp/F \approx (\rho^\perp/\rho)^2$, which is the square of the first identity.

Condition (11). Since $\rho > 0$, the first identity gives $\rho^\perp < \rho$ iff $1 - \delta_\pi < \sqrt{1 - R^2}$, that is $\delta_\pi > 1 - \sqrt{1 - R^2}$. The approximation uses $\sqrt{1 - R^2} \approx 1 - \frac{1}{2}R^2$ for small R^2 . $\square$

Proof of Proposition 2 (magnification). Since $\psi \neq \mathbf{0}$, $d \neq \mathbf{0}$ by the argument following (7), so $\|d\| > 0$. By (7), $\tilde{\beta} - \beta = d/\pi$ and $\tilde{\beta}^\perp - \beta = d^\perp/\pi^\perp$. Because $\pi, \pi^\perp > 0$, the norms factor as

$$\varrho = \frac{\|d^\perp\|/\pi^\perp}{\|d\|/\pi} = \frac{\|d^\perp\|/\|d\|}{\pi^\perp/\pi} = \frac{1 - \delta_d}{1 - \delta_\pi},$$

using the definitions of δ_d and δ_π . Positivity of $\pi^\perp/\pi$ gives $\delta_\pi < 1$, and $\|d^\perp\| \geq 0$ gives $\delta_d \leq 1$.

Condition (13). The denominator $1 - \delta_\pi$ is positive, so $\varrho > 1$ iff $1 - \delta_d > 1 - \delta_\pi$, that is $\delta_d < \delta_\pi$. If $\delta_d \geq 0$, the numerator is at most one, so $\varrho \leq 1/(1 - \delta_\pi)$, with equality iff $\delta_d = 0$. $\square$

Proof of Corollary 2 (projection on a finite menu). *Setup.* With $\mathbf{W} = A'\varepsilon_t^{-p}$ and $\mathbb{E}[\varepsilon_t^{-p}\varepsilon_t^{-p'}] = I$, the projection of $\chi_t = \psi'\varepsilon_t^{-p}$ on the controls has coefficient $(A'A)^{-1}A'\psi$, so

$$\hat{\chi}_t = (P_A\psi)'\varepsilon_t^{-p}, \quad \psi - \psi^\perp = P_A\psi, \quad \psi^\perp = (I - P_A)\psi.$$

The controls are functions of the non-policy shocks only, so $P_{\mathcal{W}}\varepsilon_t^p = 0$ holds and Proposition 1 applies.

(i) *What the purge removes.* By Proposition 1(ii), $\pi = \phi b_{pp} + \lambda'\psi$ and $\pi^\perp = \phi b_{pp} + \lambda'\psi^\perp$; by linearity, $d = D\psi$ and $d^\perp = D\psi^\perp$. Subtracting,

$$\pi - \pi^\perp = \lambda'P_A\psi, \quad d - d^\perp = DP_A\psi.$$

Dividing the first by π gives δ_π , and δ_d follows from its definition with $d^\perp = D(I - P_A)\psi$. Finally, $\psi^\perp = (I - P_A)\psi = \mathbf{0}$ iff $\psi \in \mathcal{A}$.

(ii) *The menu spans λ .* If $\lambda \in \mathcal{A}$, then $P_A\lambda = \lambda$ and

$$\lambda'\psi^\perp = \lambda'(I - P_A)\psi = 0.$$

Hence $\pi^\perp = \phi b_{pp}$ and $d^\perp = b_{-p}\psi^\perp - \beta\lambda'\psi^\perp = b_{-p}\psi^\perp$, and $\tilde{\beta}^\perp = \beta + d^\perp/\pi^\perp$ is the first expression in (15). For the raw first stage, $\lambda'\psi = \lambda'P_A\psi$, so $\pi = \phi b_{pp} + \lambda'P_A\psi$, and $1 - \delta_\pi = \pi^\perp/\pi$ is the second expression. The magnification condition is then (13).

(iii) *Adding one control.* For a unit vector $w \perp \mathcal{A}$, the projection on the enlarged menu is $P_{A \oplus w} = P_A + ww'$. Since $w'\psi = w'(I - P_A)\psi = w'\psi^\perp = k_w$, the new residual contamination is

$$(I - P_A - ww')\psi = \psi^\perp - k_w w.$$

The addition therefore removes $k_w w$ from $\psi^\perp$. By Proposition 1(ii) and linearity, it removes $\lambda'(k_w w) = \ell_w k_w$ from $\pi^\perp$ and $k_w D w$ from $d^\perp$. Applying (12)–(13) with $(\psi^\perp, \pi^\perp, d^\perp)$ as the raw triple gives (16), provided $\psi^\perp \neq 0$ and $\pi^\perp - \ell_w k_w > 0$. $\square$

B A rule of thumb for the first-stage loss and the magnification

The correspondence (10) turns three published numbers, the raw and purged first-stage F and the R^2 of the purge regression, into the first-stage loss δ_π and the magnification bound $1/(1 - \delta_\pi)$. The exercise fixes the order of magnitude; the inputs are the current estimates and the outputs move with them.

Step 1: from F to correlations. For a single instrument, $F \approx T\rho^2/(1 - \rho^2)$ (exactly for the homoskedastic statistic), so $\rho = \sqrt{F/(F + T)}$. With $T = 383$ and the funds-rate F -statistics of Table 1,

$$\rho \approx \sqrt{\frac{14.5}{14.5 + 383}} = 0.191 \text{ (raw)}, \quad \rho^\perp \approx \sqrt{\frac{3.6}{3.6 + 383}} = 0.097 \text{ (purged)}.$$

Step 2: the first-stage loss. With the $R^2 = 0.165$ of Table 2, the first identity in (10) gives

$$1 - \delta_\pi = \sqrt{1 - R^2} \frac{\rho^\perp}{\rho} = \sqrt{0.835} \times \frac{0.097}{0.191} \approx 0.46, \quad \delta_\pi \approx 0.54.$$

The sixth of the surprise's variance that is forecastable thus carries more than half of its covariance with the policy innovation. This is six times the bar $1 - \sqrt{1 - R^2} \approx 0.09$ in (11) above which the purge weakens the instrument, and the opposite of what forecastable noise would give: $\delta_\pi = 0$ and a purged F larger than the raw one.

Step 3: the magnification. By (13), the purge multiplies whatever bias it leaves behind by up to

$$\frac{1}{1 - \delta_\pi} \approx 2.2.$$

The bound is attained when the purge removes none of the contamination; in general the factor is $(1 - \delta_d)/(1 - \delta_\pi)$.

Units. In correlation units the loss is $1 - \rho^\perp/\rho \approx 0.50$. The two measures differ by the factor $\sqrt{1 - R^2}$, which accounts for the variance the purge removes from the instrument.

C The proxy zoo and the identification menu

This appendix collects the background evidence behind Sections 2.1 and 3.3. It describes the conventional schemes of Figure 1 and asks whether it is the high frequency or the purge that produces the outsized responses. The point of the exercise is not a horse race but a measurement of dispersion: how much of the disagreement in this literature is identification and how much is strength.

C.1 The conventional schemes of Figure 1

The five schemes whose range is shaded in Figure 1 are run on the reduced form of Section 6.1, and each identified shock is normalised to a +1 percentage-point funds-rate response on impact.

A candidate shock is a unit vector q with impact vector $a = \text{chol}(\Sigma_u) q$ and responses $\Phi_h a$. Three of the schemes set-identify the shock, and for these the figure plots the pointwise median of the accepted set.

Recursive. The funds rate is ordered after output and prices and before the excess bond premium, (IP, CPI, Fed Funds, EBP): the ordering of Section 4.2. This order differs from the order of y_t in Section 6.1, where the premium precedes the funds rate; the recursive identification depends on the ordering and the proxy-SVAR estimates of the paper do not.

Sign restrictions. Following Uhlig (2005), 200,000 rotations are drawn and the contractionary ones kept: $(\Phi_h a)_{\text{funds}} > 0$ and $(\Phi_h a)_{\text{CPI}} < 0$ for $h = 0, \dots, 5$, with output and the credit spread unrestricted.

Sign restrictions with a narrative filter. The accepted set is further restricted by the narrative sign restrictions of Antolín-Díaz and Rubio-Ramírez (2018): the shock is positive in October 1979 and is the overwhelming contributor to that month’s funds-rate innovation. About 2,700 of the roughly 55,000 sign-accepted draws survive.

Heteroskedasticity. Identification through a change in volatility (Lanne and Lütkepohl, 2008; Rigobon, 2003): with S_1, S_2 the residual covariances before and after the break, H solves $H' S_1 H = I$ and $H' S_2 H = \text{diag}(\tau)$, so that $e_t = H' u_t$ are shocks uncorrelated in both regimes, distinct τ_i pin the rotation down, and the impact vectors are the columns of $(H')^{-1}$. The break is placed at the Great Moderation, 1984m1, and one column is labelled the monetary shock and normalised to a unit funds-rate impact. The scheme assumes the break date, orthogonality in both regimes and a genuine change in the policy shock’s variance.

Restrictions on the systematic component. Following Arias et al. (2019), the policy row of $A_0 = B^{-1}$ is read as the rule $i_t = \gamma_y y_t + \gamma_\pi \pi_t + \gamma_f \text{EBP}_t + \varepsilon_t^p$, and the rotations kept are those with $\gamma_y > 0$ and $\gamma_\pi > 0$, the response of the premium γ_f being unrestricted; about 52,000 of 200,000 rotations are accepted. Unlike the sign restrictions, which constrain the responses to the shock, this scheme constrains only the systematic part of policy.

C.2 Is it the high frequency, or is it the purge?

Figure 1 shows the Bauer and Swanson (2023b) surprise, raw and orthogonalised, outside the range of the conventional schemes. Figure 9 asks whether this reflects high-frequency measurement or the purge, by running three further high-frequency instruments through the same system with the same normalisation: the raw Swanson (2021) target factor, the information-robust surprise of Miranda-Agrippino and Ricco (2021) and the shock of Jarociński and Karadi (2020). The last two address the information effect (Bauer and Swanson, 2023a; Nakamura and Steinsson, 2018) without projecting on pre-announcement data, by conditioning on central-bank forecasts and on the sign of the simultaneous stock-price move, respectively.

The raw Bauer–Swanson surprise is the least contractionary of the four, with an output trough of -1.9% against -2.9 to -4.7% for the others. The orthogonalised version leaves their range at almost every horizon: output reaches about -6.6% against at most -4.7% , prices -3.0% after two years against at most -0.7% , and the credit spread two percentage points on impact against at most 0.8. The two instruments that address the information effect by another route stay close to the raw surprise, with output troughs of -2.9% and -3.2% ; purged on the same controls, they too are enlarged (Appendix Table 8). The outlier is therefore produced neither by the thirty-minute window nor by the correction for central-bank information, but by the projection on pre-announcement data,

the operation that removes most of the first stage (Section 3.2). This is the second implication of the magnification reading in Section 3.3: a treatment that leaves the first stage intact has $\delta_\pi \approx 0$ in (12) and cannot magnify the contamination that survives. The comparison rests on one instrument and should be weighed accordingly.

D Instrument construction, time series, and impulse responses

This appendix collects the construction details behind Section 4, the time series of the instruments, and the impulse responses obtained under each identification.

D.1 Data and the high-frequency surprise

The high-frequency surprise mps_m is the [Bauer and Swanson \(2023b\)](#) monetary policy surprise, the first principal component of the changes in the first four quarterly Eurodollar futures in a 30-minute window around each FOMC announcement, 1988m2–2023m12, taken from their replication data. The orthogonalised surprise used in the paper is purged announcement by announcement on the six controls and then summed within the month (PTA), following [Amodeo \(2026\)](#); the orthogonalised series of [Bauer and Swanson \(2023b\)](#), which purges after monthly aggregation (ATP), is the green line of Figure 1. The MP1, FF4 and ED4 surprises of the panel of Section 6 come from the [Acosta et al. \(2025\)](#) U.S. Monetary Policy Database, 1994m2 onward. Monthly series sum the announcements within the month. Table 2 reports how predictable each raw instrument is from the pre-announcement controls of Section 2.2, the evidence referred to in Sections 2.2 and 4.

D.2 The extended Romer–Romer instrument

The projection (17) is the regression, at meeting m , of the intended target change on the pre-meeting target level, the Greenbook forecasts of output growth ($\hat{g}$) and inflation ($\hat{\pi}$), levels and revisions, from the previous quarter to two quarters ahead, and the current-quarter unemployment forecast ($\hat{u}$),

$$\Delta i_m = a + a_i i_{m-} + \sum_{h=-1}^2 (c_h^g \hat{g}_{m,h} + c_h^{g\Delta} \Delta \hat{g}_{m,h}) + \sum_{h=-1}^2 (c_h^\pi \hat{\pi}_{m,h} + c_h^{\pi\Delta} \Delta \hat{\pi}_{m,h}) + c^u \hat{u}_{m,0} + \varepsilon_m^{\text{RR}}, \quad (57)$$

where $\Delta \hat{g}_{m,h}, \Delta \hat{\pi}_{m,h}$ are revisions of each forecast relative to the previous meeting’s Greenbook; the residual is the [Romer and Romer \(2004\)](#) shock, extended here to 2019. The pre-FOMC purge (18), or equivalently its two-step form (19), yields $Z^{\text{RR},\perp}$.

D.3 Time series and impulse responses

Figure 10 shows what the purge does to the series themselves, over the moment window. The raw and purged series move together (correlations 0.84, 0.81 and 0.76), and the purge lowers their standard deviations by 13%, 19% and 25%. What it removes is concentrated in a few months. For the surprise it is the early sample (1988–1989) and the easings of late 1991 and late 2008. For the Romer–Romer shock it is the easing cycles of 2001 and 2007–2009 and several months of 2011–2012. For the recursive innovation it is, above all, the months without an FOMC announcement in the Bauer–Swanson sample, where the purged series is zero: the largest raw innovations of the window (February 2008, -1.0 ; November 2007, -0.5 ; September 2001, -0.4) are all of this kind.

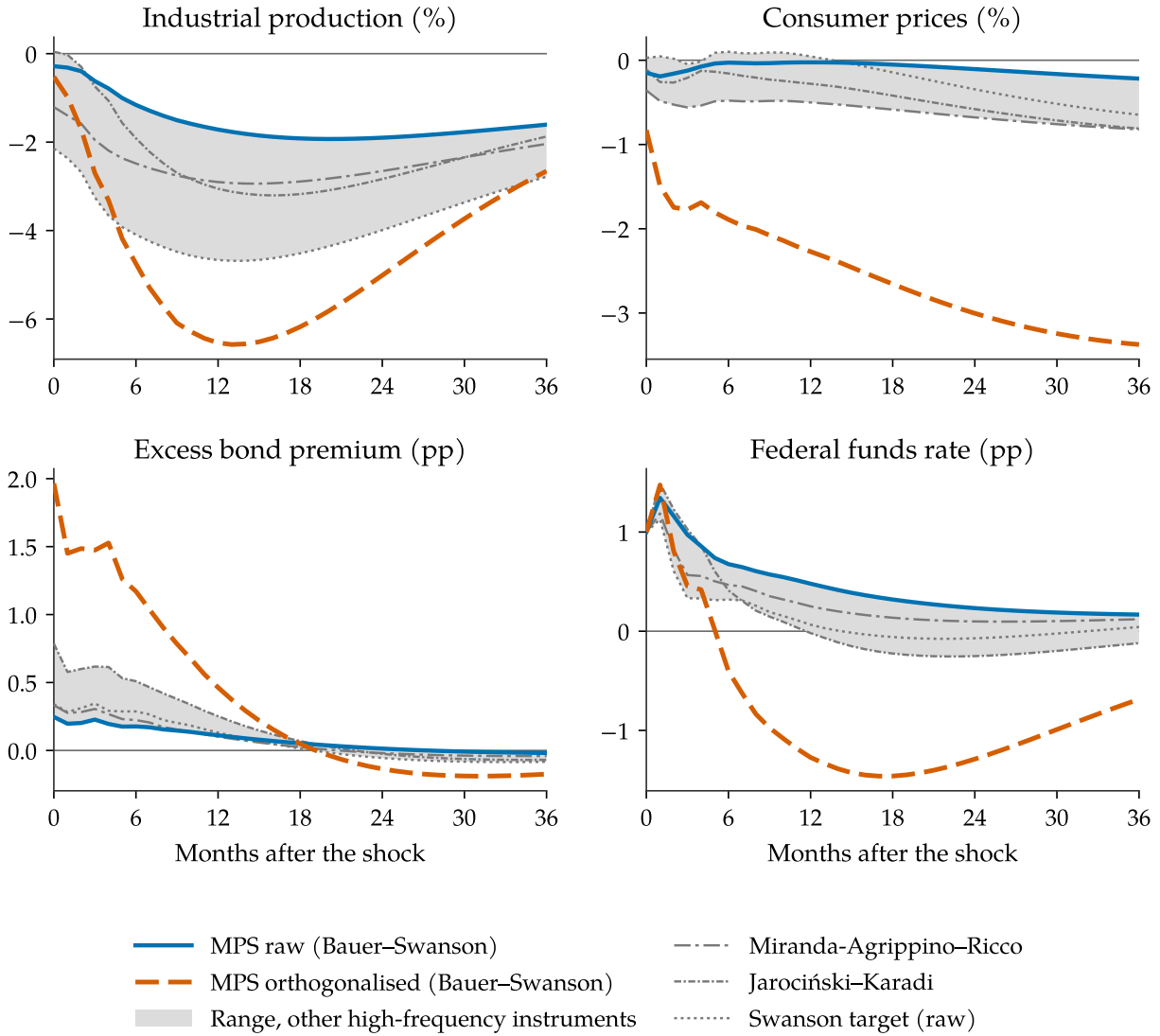


Figure 9: It is the purge, not the frequency. Single-instrument responses to a +1 percentage-point funds-rate shock on the reduced form and normalisation of Figure 1, without confidence bands. Shaded: the min-max range across the raw surprise and three further high-frequency instruments, drawn individually as grey lines, namely the raw [Swanson \(2021\)](#) target factor, the [Miranda-Agrippino and Ricco \(2021\)](#) information-robust surprise, and [Jarociński and Karadi \(2020\)](#). Blue: the raw [Bauer and Swanson \(2023b\)](#) surprise, the least contractionary of the four. Orange dashed: the same surprise orthogonalised on pre-announcement information, which lies outside the range in every panel. The last two grey instruments remove the information effect by a different route than the pre-announcement projection and deliver responses close to the raw surprise.

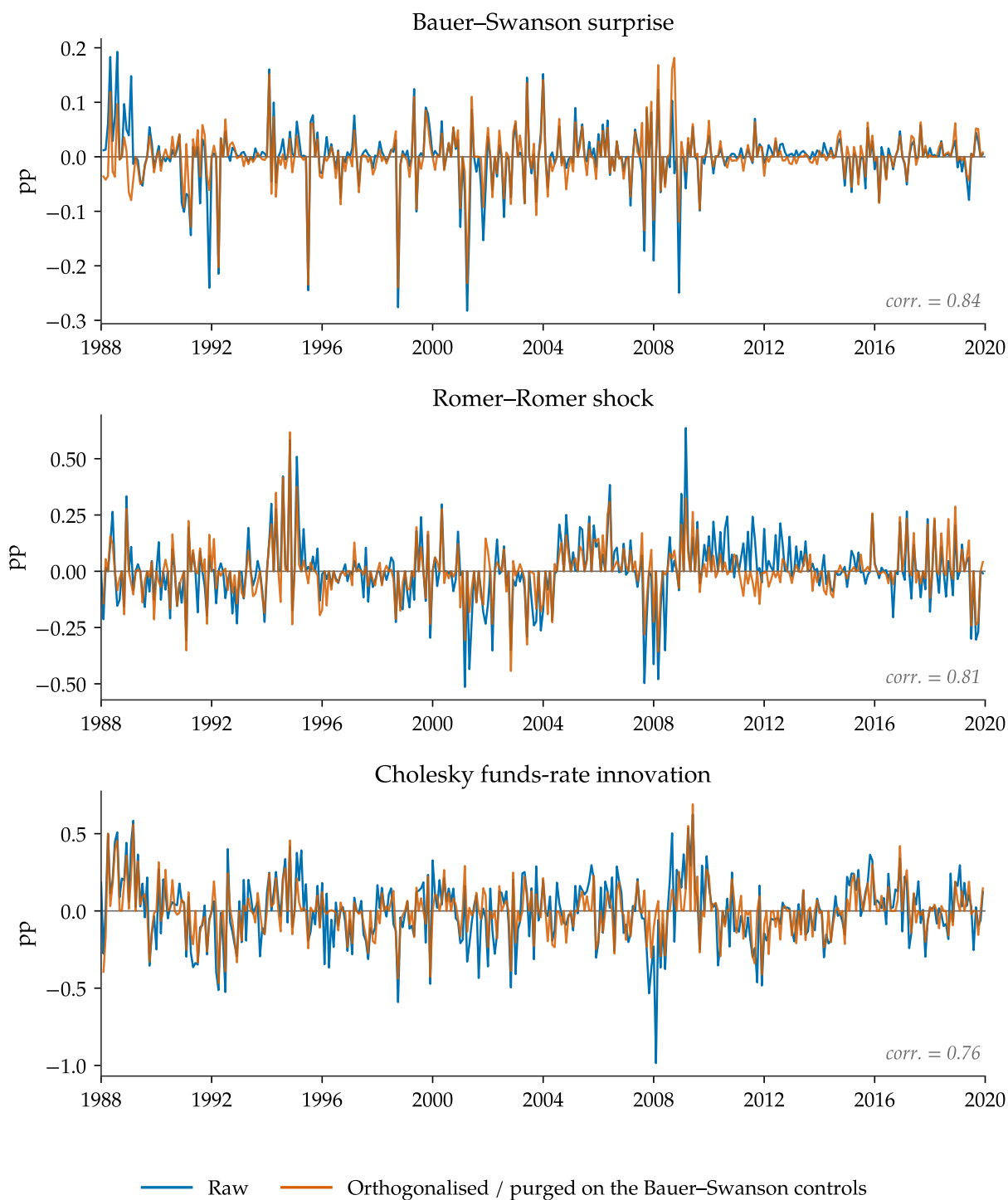


Figure 10: The raw and purged instruments, 1988m1–2019m12. Monthly series in percentage points. Top: **Bauer and Swanson (2023b)** surprise, raw and orthogonalised (PTA). Middle: extended **Romer and Romer (2004)** shock, raw and purged (PTA-FWL). Bottom: recursive (Cholesky) funds-rate innovation of the instrument database, raw and purged (ATP-FWL), which is zero in months without an FOMC announcement in the Bauer–Swanson sample. The correlation between the two series is shown in each panel.

Table 2: Predictability of the raw instruments from pre-announcement information. Each raw instrument is regressed on a constant and the six [Bauer and Swanson \(2023b\)](#) controls of Section 2.2; HC1 standard errors in parentheses; *, **, ***: significant at 10, 5 and 1%. Monetary policy surprises: the [Bauer and Swanson \(2023b\)](#) surprise mps_m ; Romer–Romer: the extended shock ϵ^{RR} of Section 4.1 before the pre-FOMC purge; MAR: the information-robust surprise of [Miranda-Agrippino and Ricco \(2021\)](#); Cholesky: the recursive funds-rate innovation of Section 4.2. The first three at meeting frequency with the meeting controls, the Cholesky innovation at monthly frequency with the monthly controls (months with an FOMC decision). NFP surprise in millions of jobs, NFP 12M in percent. Joint F : HC1 test that all six coefficients are zero.

	Monetary policy surprises	Romer–Romer	MAR	Cholesky
NFP surprise	0.093** (0.039)	0.113 (0.101)	0.101** (0.044)	0.276** (0.128)
NFP 12M	0.005** (0.002)	−0.002 (0.009)	−0.000 (0.002)	0.001 (0.012)
S&P 500 3M	0.079 (0.059)	−0.156 (0.184)	0.033 (0.056)	−0.056 (0.209)
Slope 3M	−0.011 (0.007)	−0.097*** (0.028)	−0.006 (0.009)	−0.076*** (0.025)
BCOM 3M	0.122** (0.052)	0.022 (0.147)	0.034 (0.050)	−0.041 (0.179)
Treasury skew	0.032*** (0.011)	0.117*** (0.038)	0.011 (0.010)	0.078* (0.041)
R^2	0.165	0.105	0.066	0.060
Joint F (p)	5.22 (0.000)	3.59 (0.002)	1.11 (0.356)	2.64 (0.017)
N	316 meetings	256 meetings	219 meetings	267 months
Sample	1988m2–2019m12	1988m2–2019m12	1994m2–2019m12	1988m2–2019m12

Figures 11 and 12 report the responses to the two wide-window instruments, raw and purged; Section 4.4 discusses them.

E The pool-within-annihilate-across estimator: derivations

This appendix collects the derivations behind Section 5 that are not given in the text: what averaging removes, the shear, the eigenvector step with a general weight, the proofs of the propositions of the section, the constrained-optimal instrument, inference (including the robust set for the direction), and the diagnostics and the audit of the partition. Notation is that of Section 5.

E.1 What averaging removes, and what it leaves

This subsection proves the two claims made about standard pooling in Section 5.6: averaging shrinks the idiosyncratic contamination at rate $m^{-1/2}$, and it leaves the common component untouched.

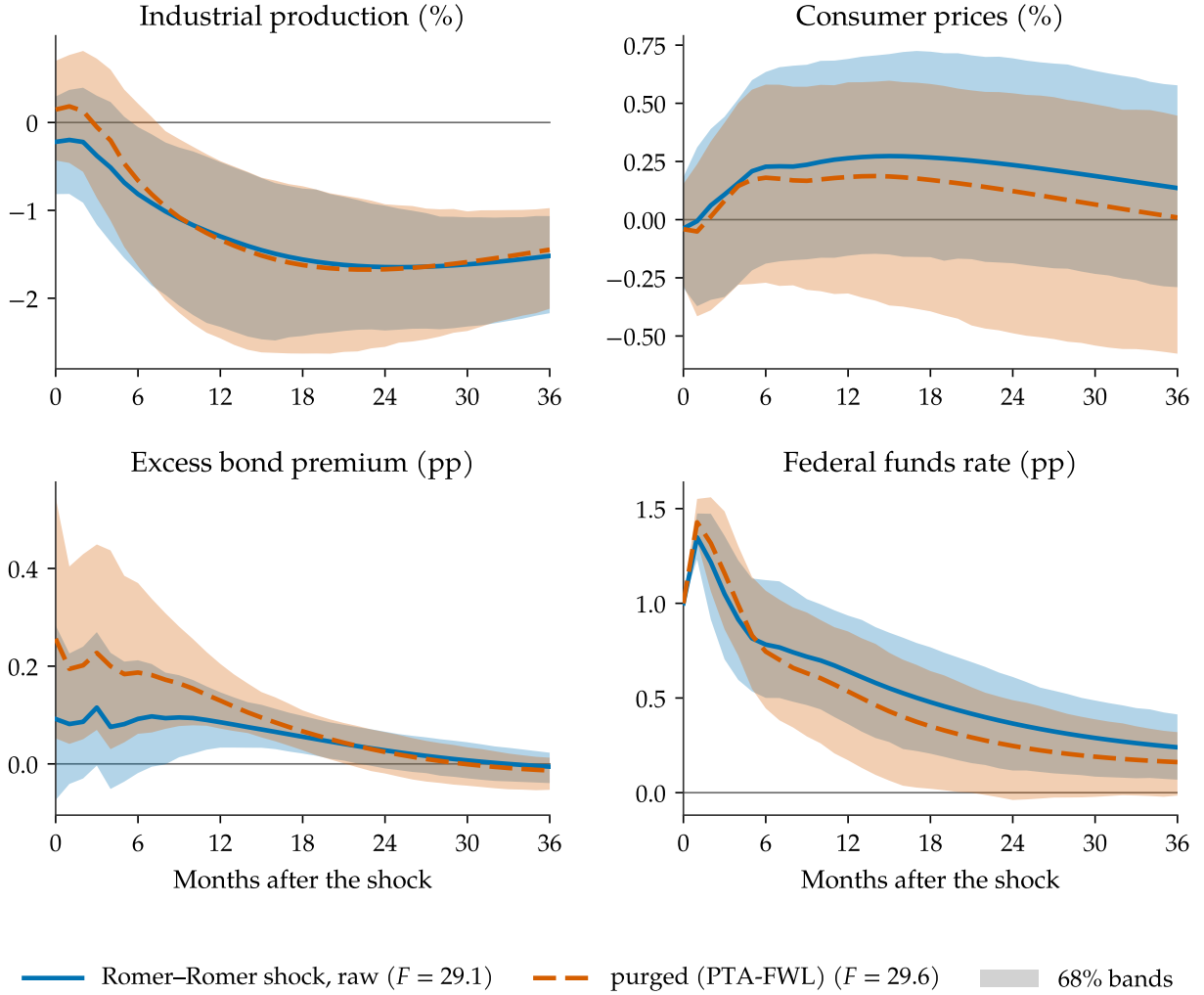


Figure 11: Extended Romer–Romer shock: raw versus purged. Single-instrument responses to the extended [Romer and Romer \(2004\)](#) shock, raw (solid, $F = 29.1$) and purged on the six Bauer–Swanson controls at the announcement level, jointly with its first stage (PTA-FWL; dashed, $F = 29.6$), each with its 68% moving-block-bootstrap band. The purge leaves the first stage and the output and price responses essentially unchanged.

The idiosyncratic component shrinks. Write $\psi_j = \bar{\psi} + \xi_j$ as in (46), with the ξ_j independent, mean zero and of variance Σ_ψ . The equal-weight average, $w_j = 1/m$, has contamination

$$\Psi'w = \bar{\psi} + \bar{\xi}_m, \quad \bar{\xi}_m := \frac{1}{m} \sum_j \xi_j,$$

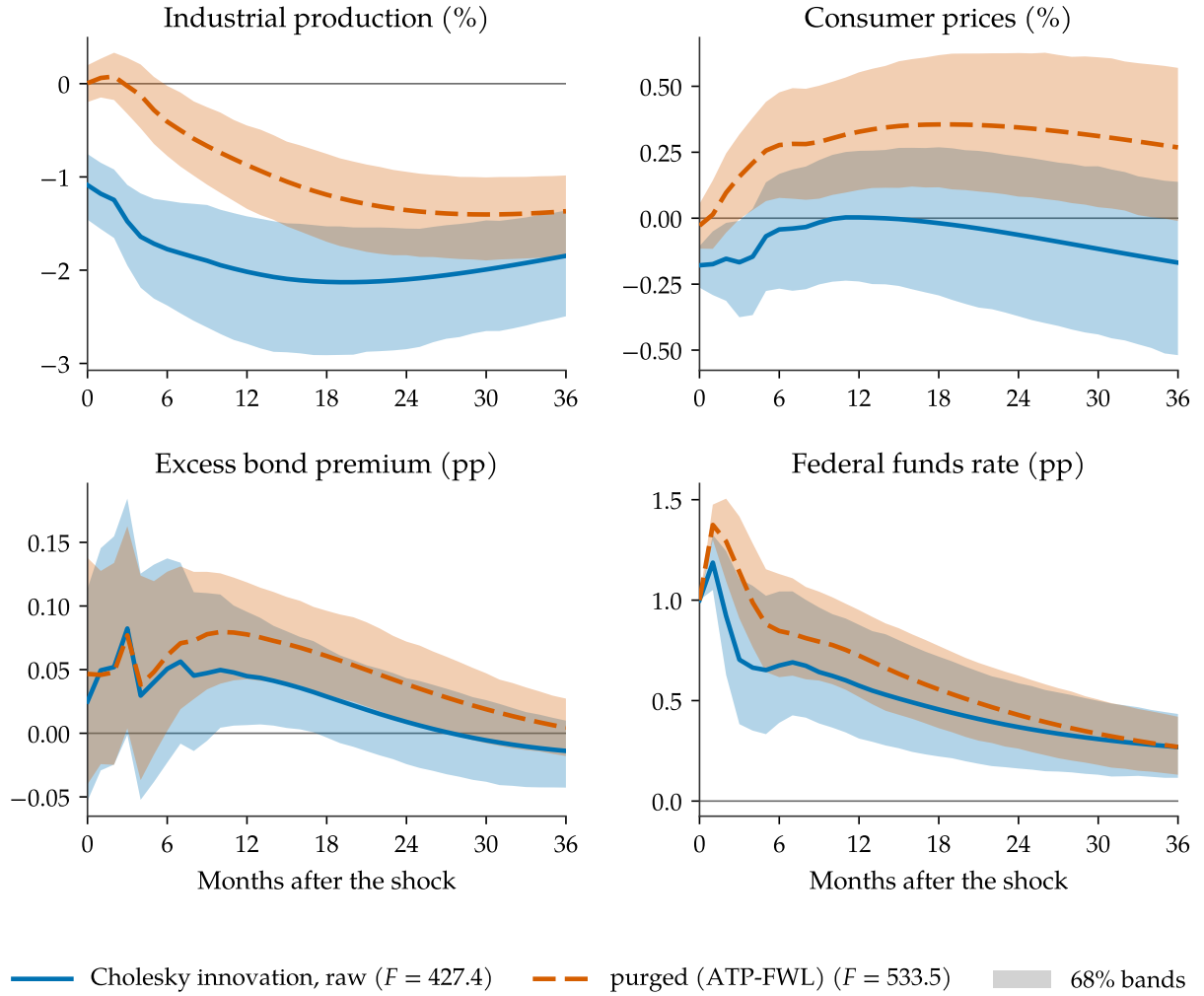


Figure 12: Cholesky-based instrument: raw versus purged. Single-instrument responses to the recursive funds-rate innovation of the instrument database, raw (solid, $F = 427$) and purged on the six Bauer–Swanson controls after monthly aggregation, jointly with its first stage (ATP-FWL; dashed, $F = 534$), each with its 68% moving-block-bootstrap band. The purge raises the first stage and shrinks the responses, the mirror image of the high-frequency surprise. The raw innovation comes from the database’s own recursive VAR, not the headline VAR, and so moves output on impact.

and by independence

$$\mathbb{E}[\bar{\xi}_m] = \mathbf{0}, \quad \text{Var}(\bar{\xi}_m) = \frac{1}{m^2} \sum_j \text{Var}(\xi_j) = \frac{\Sigma_\psi}{m}. \quad (58)$$

The average of m independent deviations has a $1/m$ share of the variance of one of them. Chebyshev's inequality turns this into a rate: for every $M > 0$,

$$\Pr(\sqrt{m} \|\bar{\xi}_m\| > M) \leq \frac{m \mathbb{E} \|\bar{\xi}_m\|^2}{M^2} = \frac{m \operatorname{tr} \operatorname{Var}(\bar{\xi}_m)}{M^2} = \frac{\operatorname{tr} \Sigma_\psi}{M^2}.$$

The bound does not depend on m , so $\bar{\xi}_m = O_p(m^{-1/2})$. Only a finite second moment is needed.

The common component does not. With $\bar{\phi} := m^{-1} \sum_j \phi_j$ the mean relevance, the moment of the averaged proxy is

$$\frac{1}{m} \sum_j g_j = \bar{\phi} b_p + b_{-p} \bar{\psi} + b_{-p} \bar{\xi}_m = \bar{\phi} \left(b_p + \frac{b_{-p} \bar{\psi}}{\bar{\phi}} \right) + O_p(m^{-1/2}). \quad (59)$$

Provided $\bar{\phi}$ stays bounded away from zero, the proxy-SVAR estimand of the averaged proxy converges to the normalised $b_p + b_{-p} \bar{\psi} / \bar{\phi}$. This is the contaminated impact vector (6) of a single proxy carrying the panel's mean loading: adding proxies measures the wrong target more precisely.

General weights. By (46), the common component enters any combination with weight $\sum_j w_j$, and the idiosyncratic component with variance $(\sum_j w_j^2) \Sigma_\psi$. Removing the first requires weights that sum to zero, a contrast; making the second small requires weights spread over many proxies. The two requirements are distinct: the equal-weight average meets the second but not the first. The estimator of Section 5.4 meets both, with weights that sum to zero within every family and are spread across its members.

E.2 The shear

Contamination can mimic the target. The lemma below shows that every alternative generating the same moments is a shift of the target inside the span of the non-policy impact responses, offset by a change in the loadings, up to scale. The proposition turns this into a condition for identification.

Lemma E.1 (Observational equivalence is a rescaled shear). *Hold b_{-p} fixed, let $B = [b_p \ b_{-p}]$ be nonsingular and $\phi \neq \mathbf{0}$. If $(\tilde{b}_p, \tilde{\phi}, \tilde{\Psi})$ generates the same G as (b_p, ϕ, Ψ) in (23), then for some $\kappa \neq 0$ and $\alpha \in \mathbb{R}^{n-1}$*

$$\tilde{b}_p = \kappa (b_p + b_{-p} \alpha), \quad \tilde{\phi} = \phi / \kappa, \quad \tilde{\Psi} = \Psi - \phi \alpha'.$$

Proof. Because B is nonsingular, $\tilde{b}_p = \kappa b_p + b_{-p} a$ for unique $\kappa \in \mathbb{R}$ and $a \in \mathbb{R}^{n-1}$. Equating the two expressions for G ,

$$b_p (\kappa \tilde{\phi} - \phi)' + b_{-p} (a \tilde{\phi}' + \tilde{\Psi}' - \Psi') = \mathbf{0},$$

that is, B times a stacked matrix equals zero, so both brackets vanish: $\phi = \kappa \tilde{\phi}$ and $\tilde{\Psi} = \Psi - \tilde{\phi} a'$. Since $\phi \neq \mathbf{0}$, $\kappa \neq 0$. Setting $\alpha := a / \kappa$ gives $\tilde{b}_p = \kappa (b_p + b_{-p} \alpha)$ and $\tilde{\Psi} = \Psi - \phi \alpha'$. $\square$

Proposition E.1 (The identification obstacle). *For every $\alpha \in \mathbb{R}^{n-1}$ the pair $(b_p + b_{-p} \alpha, \Psi - \phi \alpha')$ generates the same G as (b_p, Ψ) . Hence b_p is identified up to scale from G if and only if the maintained restriction on the contamination excludes $\Psi - \phi \alpha'$ for every $\alpha \neq \mathbf{0}$; with the contamination unrestricted it is never identified.*

Proof. Substituting the sheared pair into (23),

$$(b_p + b_{-p}\alpha)\phi' + b_{-p}(\Psi - \phi\alpha')' = b_p\phi' + b_{-p}\alpha\phi' + b_{-p}\Psi' - b_{-p}\alpha\phi' = b_p\phi' + b_{-p}\Psi' = G.$$

The two pairs are therefore observationally equivalent. If Ψ is unrestricted, every sheared $\Psi - \phi\alpha'$ is admissible, so b_p can be moved anywhere in $b_p + \text{col}(b_{-p})$ without changing G .

Conversely, suppose the restriction excludes $\Psi - \phi\alpha'$ for every $\alpha \neq \mathbf{0}$. By Lemma E.1, any admissible alternative that generates G has loadings $\Psi - \phi\alpha'$, so $\alpha = \mathbf{0}$ and $\tilde{b}_p = \kappa b_p$: the target is identified up to scale. $\square$

The shear moves the target by $b_{-p}\alpha$ and compensates by removing $\phi_j\alpha$ from the loading of proxy j : the more relevant the proxy, the more of the shift its loading absorbs. A restriction on within-family averages of the deviations is left intact by this change, because the relevance contrasts sum to zero within a family, which is why Assumption 1 conditions on the relevances.

E.3 The eigenvector step with a general weight

Section 5.4 derives the estimator in whitened coordinates. This subsection shows that the same derivation holds for any positive definite weight W , with the whitened form as the case $W = \Sigma_u^{-1}$, and proves the eigenvector result used in Step 3.

The criterion. In the units of the data the fit is

$$\sum_c \sum_{j \in c} (\hat{g}_j - \phi_j b_p - d_c)' W (\hat{g}_j - \phi_j b_p - d_c), \quad W \succ 0.$$

Steps 1 and 2 go through with W -inner products in place of Euclidean ones. The family effects, the relevance contrasts and the concentrated criterion are

$$\hat{d}_c = \bar{g}_c - \bar{\phi}_c b_p, \quad \hat{\phi}_j = \frac{b_p' W \tilde{g}_j}{b_p' W b_p}, \quad \sum_j \tilde{g}_j' W \tilde{g}_j - \frac{b_p' W \tilde{G} \tilde{G}' W b_p}{b_p' W b_p}.$$

The family effects do not depend on W . Step 3 maximises the second term,

$$f(b) = \frac{b' W \tilde{G} \tilde{G}' W b}{b' W b}, \quad b \neq \mathbf{0}.$$

Reduction to a Rayleigh quotient. Change variables to $v := W^{1/2}b$. Then $b' W b = v' v$ and

$$f = \frac{v' S v}{v' v}, \quad S := W^{1/2} \tilde{G} \tilde{G}' W^{1/2},$$

a symmetric positive semidefinite matrix.

The maximiser. Let $\lambda_1 \geq \dots \geq \lambda_n \geq 0$ be the eigenvalues of S and $v_1, \dots, v_n$ orthonormal eigenvectors. For a unit vector v ,

$$f = v'Sv = \sum_{k=1}^n \lambda_k (v'v_k)^2, \quad \sum_{k=1}^n (v'v_k)^2 = 1.$$

The quotient is a weighted average of the eigenvalues, with weights that sum to one. A weighted average is at most its largest term, with equality iff all the weight is on that term, so $f \leq \lambda_1$, with equality iff $v = \pm v_1$ when λ_1 is simple.

Back to b . The maximiser is $\hat{b} \propto W^{-1/2}v_1$. Premultiplying $Sv_1 = \lambda_1 v_1$ by $W^{-1/2}$ gives the equivalent form

$$\tilde{G}\tilde{G}'W\hat{b} = \lambda_1 \hat{b},$$

and the minimised criterion is $\sum_j \tilde{g}_j'W\tilde{g}_j - \lambda_1$. With $W = \Sigma_u^{-1}$, $v = q$ and $S = \tilde{G}^w\tilde{G}^{w'}$: the argument is Step 3 of Section 5.4, the eigenproblem is (39), and its unwhitened form is (41).

E.4 Proofs of the propositions of Section 5

Proof of Proposition 3 (identification). The shear in the family model. By Proposition E.1, the shear $b_p \mapsto b_p + b_{-p}\alpha$, $\psi_j \mapsto \psi_j - \phi_j\alpha$ leaves G unchanged. Writing $\phi_j = \bar{\phi}_c + \tilde{\phi}_j$, it splits into a change in the family mean and a change in the deviation:

$$\bar{\psi}_c \mapsto \bar{\psi}_c - \bar{\phi}_c\alpha, \quad \nu_j \mapsto \nu_j - \tilde{\phi}_j\alpha.$$

The first is absorbed by d_c . Whether b_p is identified depends on whether the restriction on the deviations rules out the second.

(i) The family average of the sheared deviations is

$$\frac{1}{|c|} \sum_{j \in c} (\nu_j - \tilde{\phi}_j\alpha) = \bar{\nu}_c,$$

because the contrasts $\tilde{\phi}_j$ sum to zero within the family. The restriction $\mathbb{E}[\bar{\nu}_c | c] = \mathbf{0}$ therefore holds for every α and does not pin down the target.

(ii) Since $\tilde{\phi}_j$ is a function of ϕ_c , under (26)

$$\mathbb{E}[\nu_j - \tilde{\phi}_j\alpha | \phi_c, c] = -\tilde{\phi}_j\alpha.$$

By Assumption 2(i) some $\tilde{\phi}_j \neq 0$, so the sheared deviations satisfy (26) only at $\alpha = \mathbf{0}$. By Lemma E.1, every alternative that generates G is such a shear up to scale; rescaling ϕ leaves the loadings, and hence the restriction, unchanged. The only admissible alternatives are therefore κb_p , and b_p is identified up to scale.

For the last statement, let $\nu_j = \alpha_0\tilde{\phi}_j + \eta_j$ and apply the shear with $\alpha = \alpha_0$. The sheared loadings are

$$\psi_j - \phi_j\alpha_0 = (\bar{\psi}_c - \bar{\phi}_c\alpha_0) + \eta_j,$$

a family mean plus a deviation η_j that satisfies (26). The pair $(b_p + b_{-p}\alpha_0, \{\psi_j - \phi_j\alpha_0\})$ thus satisfies Assumption 1 and generates the same G ; by the first part applied to this pair, $b_p + b_{-p}\alpha_0$ is the only target, up to scale, consistent with Assumption 1, and no estimator can distinguish it from b_p . $\square$

Proof of Proposition 4 (recovery by the leading eigenvector). *Setup.* Work in the orthonormal basis $\mathcal{Q} = [q_p \ \mathcal{Q}_{-p}]$ of (29), in which the target is the first axis and the contamination space the remaining $n - 1$. The whitened demeaned moments are $\tilde{g}_j^w = \tilde{\phi}_j q_p + \mathcal{Q}_{-p} \tilde{v}_j$. Two sums appear:

$$\sum_j \tilde{\phi}_j^2 = \sigma_\phi^2, \quad \sum_j \tilde{\phi}_j \tilde{v}_j = \sum_j \tilde{\phi}_j v_j = \zeta,$$

where the second equality holds because the $\tilde{\phi}_j$ sum to zero within each family.

(i) *Exact block structure.* With $v_j \equiv \mathbf{0}$, $\tilde{g}_j = \tilde{\phi}_j b_p$, so $\tilde{G}\tilde{G}' = \sigma_\phi^2 b_p b_p'$. For any $W \succ 0$, $\tilde{G}\tilde{G}'Wb = \sigma_\phi^2 (b_p' W b) b_p$, whose only eigenvector with a nonzero eigenvalue is b_p . In whitened coordinates $S_0 = \sigma_\phi^2 q_p q_p'$.

(ii) *Random effects, finite m .* Expanding $S_0 = \sum_j \tilde{g}_j^w \tilde{g}_j^{w'}$ with the two sums above gives (30). In the basis $\mathcal{Q}$ it is a bordered matrix,

$$\mathcal{Q}' S_0 \mathcal{Q} = \begin{pmatrix} \sigma_\phi^2 & \zeta' \\ \zeta & N \end{pmatrix} :$$

the target axis carries σ_ϕ^2 , the contamination space carries N , and ζ couples them. In particular q_p is an eigenvector iff $\zeta = \mathbf{0}$.

Let λ_1 be the largest eigenvalue. Evaluating the Rayleigh quotient at the first axis gives $\lambda_1 \geq \sigma_\phi^2 > \lambda_{\max}(N)$. An eigenvector (x_0, x) with eigenvalue λ_1 must have $x_0 \neq 0$: otherwise $Nx = \lambda_1 x$, which is impossible since $\lambda_1 > \lambda_{\max}(N)$. Normalising $x_0 = 1$, the second block row reads $\zeta + Nx = \lambda_1 x$, so

$$x = (\lambda_1 I - N)^{-1} \zeta, \quad \lambda_1 = \sigma_\phi^2 + \zeta' (\lambda_1 I - N)^{-1} \zeta.$$

The solution is unique, so λ_1 is simple, and the leading eigenvector is, exactly,

$$q_p + \mathcal{Q}_{-p} (\lambda_1 I - N)^{-1} \zeta.$$

The second equation gives $0 \leq \lambda_1 - \sigma_\phi^2 \leq \|\zeta\|^2 / \varsigma$. Replacing λ_1 by σ_ϕ^2 , and normalising the eigenvector to unit length, changes it by $O(\|\zeta\|^2 / \varsigma^2)$, which gives (31).

Mean and order of the tilt. Expanding $(\sigma_\phi^2 I - N)^{-1} = \sigma_\phi^{-2} I + O(\|N\| / \sigma_\phi^4)$, the tilt is $\sigma_\phi^{-2} \mathcal{Q}_{-p} \zeta$ to first order in the deviations, and $\mathbb{E}[\zeta \mid \{\phi_j\}] = \mathbf{0}$ by (26). If the distribution of the v_j is symmetric, the mean is exactly zero: $v \mapsto -v$ flips ζ , leaves N unchanged, and so flips the tilt. By independence and the common variance, $\text{Var}(\zeta \mid \{\phi_j\}) = \sum_j \tilde{\phi}_j^2 \Sigma_v = \sigma_\phi^2 \Sigma_v$, so $\|\zeta\| = O_p(\sigma_\phi \|\Sigma_v\|^{1/2})$. With $\|(\sigma_\phi^2 I - N)^{-1}\| \leq \varsigma^{-1}$ and ς of the order of σ_ϕ^2 , the tilt is $O_p(\|\Sigma_v\|^{1/2} / \sigma_\phi)$, which is $O(m^{-1/2})$ when σ_ϕ^2 grows in proportion to m .

(iii) *Random effects, $m \rightarrow \infty$.* Divide the bordered matrix by m . The corner converges to $\bar{\sigma}_\phi^2$. The border vanishes: ζ has mean zero and variance $\sigma_\phi^2 \Sigma_v$, so $\zeta / m = O_p(m^{-1/2})$. For the lower block,

$\mathbb{E}[\tilde{v}_j \tilde{v}_j' \mid c] = (1 - |c|^{-1})\Sigma_v$, and summing over j ,

$$\frac{1}{m} \mathbb{E}[N] = \frac{1}{m} \sum_c (|c| - 1) \Sigma_v \longrightarrow \bar{\kappa} \Sigma_v,$$

with N/m converging to the same limit by the law of large numbers across proxies. Hence

$$\frac{1}{m} \mathcal{Q}' S_0 \mathcal{Q} \longrightarrow \begin{pmatrix} \bar{\sigma}_\phi^2 & \mathbf{0}' \\ \mathbf{0} & \bar{\kappa} \Sigma_v \end{pmatrix}.$$

The limit is block diagonal, with eigenvalue $\bar{\sigma}_\phi^2$ on the target axis and $\bar{\kappa} \lambda_k(\Sigma_v)$ in the contamination space. The target is its unique leading eigenvector iff $\bar{\sigma}_\phi^2 > \bar{\kappa} \lambda_{\max}(\Sigma_v)$, which is (32). The leading eigenvector of $m^{-1} S_0$ then converges to q_p , because an eigenvector is continuous in the matrix at a simple eigenvalue. $\square$

Proof of Proposition 5 (the two order conditions). (O1) Under exact block structure, G supplies nm moments. The unknowns are the direction ($n - 1$ after normalisation), the m relevances, and the C family contaminations $d_c \in \text{range}(b_{-p})$, with $n - 1$ entries each. The number of overidentifying restrictions is

$$nm - (n - 1) - m - (n - 1)C = (n - 1)(m - C - 1), \quad (60)$$

which is nonnegative iff $m - C \geq 1$; it is also the degrees of freedom of the overidentification test (Appendix E.6). The count is necessary but not sufficient. The within transform leaves $m - C$ contrasts, and they carry the target only if $\sigma_\phi^2 > 0$ (Proposition 4(i)): some family must contain two proxies of unequal relevance.

(O2) A family-constant combination with family totals $a = (a_1, \dots, a_C)'$ has contamination $\sum_c a_c \bar{\psi}_c$ by (50). It is clean iff $a \in \ker[\bar{\psi}_1, \dots, \bar{\psi}_C]$, a space of dimension $C - r$, which contains a nonzero vector iff $C > r$. Since $r \leq n - 1$, $C > n - 1$ is sufficient. The strength statement is Proposition 6. $\square$

Proof of Proposition 6 (clean strength). Split any weight vector into a family-constant part with the same family totals and a remainder,

$$w = w_L + w_0, \quad (w_L)_j = a_c / |c| \text{ for } j \in c,$$

so that w_0 sums to zero within every family. The two parts are orthogonal. Under exact block structure, $\Psi' w = \sum_c a_c \bar{\psi}_c$ by (50), which depends on w only through the totals. Every contrast w_0 is therefore clean, and w_L is clean iff its totals lie in $\ker[\bar{\psi}_1, \dots, \bar{\psi}_C]$, which defines $\mathcal{K}$, of dimension $C - r$. Hence $\ker(\Psi')$ is the orthogonal sum of the contrasts and $\mathcal{K}$, and

$$\|P_\perp \phi\|^2 = \|P_{\text{contrasts}} \phi\|^2 + \|P_{\mathcal{K}} \phi\|^2.$$

The projection of ϕ on the contrasts has entries $\phi_j - \bar{\phi}_c$, so its squared norm is σ_ϕ^2 . The level term is nonzero for generic ϕ iff $\dim \mathcal{K} = C - r > 0$. $\square$

E.5 The constrained-optimal instrument

This subsection solves the program (48), normalises its solution, proves Proposition 7, and gives the feasible weight used in the application.

The solution. Change variables to

$$v := \Sigma_\omega^{1/2} w, \quad a := \Sigma_\omega^{-1/2} \phi, \quad M := \Sigma_\omega^{-1/2} \Psi,$$

so that $w' \phi = v' a$, $w' \Sigma_\omega w = v' v$ and $\Psi' w = M' v$. The program becomes

$$\max_v \frac{(v' a)^2}{v' v} \quad \text{s.t.} \quad M' v = \mathbf{0}.$$

Let $P := I - M(M' M)^+ M'$ be the orthogonal projector onto the vectors orthogonal to the columns of M . For any feasible v , $v' a = v' P a$, and Cauchy–Schwarz gives

$$(v' a)^2 = (v' P a)^2 \leq \|v\|^2 \|P a\|^2,$$

with equality iff $v \propto P a$. The maximiser is therefore $v^* \propto P a$, and returning to w ,

$$w^* \propto \Sigma_\omega^{-1} \phi^\perp, \quad \phi^\perp := \phi - \Psi(\Psi' \Sigma_\omega^{-1} \Psi)^+ \Psi' \Sigma_\omega^{-1} \phi, \quad \mu^2(w^*) = T \|P a\|^2 = T \phi^{\perp'} \Sigma_\omega^{-1} \phi^\perp. \quad (61)$$

Here $\phi^\perp$ is the residual of ϕ on the columns of Ψ in the Σ_ω^{-1} metric, and the solution is unique up to scale iff $\phi^\perp \neq \mathbf{0}$. The pseudo-inverse makes no rank assumption on Ψ . Since $\|P a\|^2 = \|a\|^2 - \|(I - P)a\|^2$, the attainable strength is that of the unconstrained pool, $T \phi' \Sigma_\omega^{-1} \phi$, less the part of relevance collinear with the contamination. With $\Sigma_\omega = \sigma_\omega^2 I$, P is the projector $P_\perp$ onto $\ker(\Psi')$, and (61) reduces to (49).

Normalisation. Scaling w^* so that the pooled proxy has a unit loading on the policy shock, and using $\phi' \Sigma_\omega^{-1} \phi^\perp = \phi^{\perp'} \Sigma_\omega^{-1} \phi^\perp$,

$$w^* = \frac{\Sigma_\omega^{-1} \phi^\perp}{\phi^{\perp'} \Sigma_\omega^{-1} \phi^\perp}, \quad w^{*'} \phi = 1, \quad \Psi' w^* = \mathbf{0}. \quad (62)$$

The pooled proxy is then $\varepsilon_t^p + w^{*'} \omega_t$: a clean proxy of the form (4) with $\phi = 1$, first stage b_{pp} , and noise variance $1/(\phi^{\perp'} \Sigma_\omega^{-1} \phi^\perp)$.

Proof of Proposition 7 (pooling adds concentration). (i) Without the constraint ($\Psi = \mathbf{0}$ above, so $P = I$), the maximum is $T \|a\|^2 = T \phi' \Sigma_\omega^{-1} \phi$, attained at $w \propto \Sigma_\omega^{-1} \phi = w^{\text{GLS}}$. With independent noise Σ_ω is diagonal, so

$$\mu^2(w^{\text{GLS}}) = T \sum_j \frac{\phi_j^2}{\sigma_{\omega j}^2} = \sum_j \mu_j^2.$$

Since each single proxy is the combination $w = e_j$, the pool is at least as strong as the strongest.

(ii) Because the $\tilde{\phi}_j$ sum to zero within each family, $\sum_{j \in c} \tilde{\phi}_j (z_{jt} - \bar{z}_{ct}) = \sum_{j \in c} \tilde{\phi}_j z_{jt}$, so the pooled contrast instrument is the combination $w = \tilde{\phi}$. It is clean under exact block structure, since its

family totals are zero, and

$$w'\phi = \sum_j \tilde{\phi}_j \phi_j = \sigma_\phi^2, \quad \text{Var}(w'\omega_t) = \sigma_\omega^2 \|\tilde{\phi}\|^2 = \sigma_\omega^2 \sigma_\phi^2, \quad \mu^2(\tilde{\phi}) = \frac{T\sigma_\phi^4}{\sigma_\omega^2 \sigma_\phi^2} = \frac{T\sigma_\phi^2}{\sigma_\omega^2}.$$

It is also the best contrast. Applying the solution above with the constraint that the family totals be zero, the optimal weight is the projection of ϕ on the contrasts, which is $\tilde{\phi}$. Any single contrast $z_{jt} - \bar{z}_{ct}$ is itself a contrast, so its concentration is at most $T\sigma_\phi^2/\sigma_\omega^2$. Finally, w^* maximises the concentration over all clean combinations, a set that contains $\tilde{\phi}$, and its value $T\|P_\perp \phi\|^2/\sigma_\omega^2$ is at least $T\sigma_\phi^2/\sigma_\omega^2$, with the difference given by Proposition 6. $\square$

The feasible instrument. In the data Ψ is replaced by the residual loadings $\hat{R}$ of (52), which satisfy $\ker R = \ker \Psi'$ at the population moments. The feasible weight is (61) with $\hat{R}'$ in the role of Ψ' ,

$$\hat{w} \propto \Sigma_\omega^{-1} \hat{\phi}^\perp, \quad \hat{\phi}^\perp := \hat{\phi} - \hat{R}'(\hat{R}\Sigma_\omega^{-1}\hat{R}')^+ \hat{R}\Sigma_\omega^{-1} \hat{\phi}, \quad \text{scaled to } \hat{w}'\hat{\phi} = 1. \quad (63)$$

With $\Sigma_\omega \propto I$ this is the Euclidean projection (53); with $\Sigma_\omega \propto \text{diag}(\hat{v}_j)$, the purged sampling variances, it is the precision-weighted form of the application. Both satisfy $\hat{R}\hat{w} = \mathbf{0}$, so by Proposition 8(ii) they deliver the same impact vector and impulse responses. They differ only in the shock series and in the strength they report, and only the precision-weighted form makes the reported strength invariant to rescaling a proxy.

E.6 Inference

The error in the estimated direction has two uncorrelated components: sampling error in the estimated moments, which vanishes as T grows, and heterogeneity across proxies, which vanishes only as m grows. This subsection derives the two components, states the large-sample properties, estimates both components from the residuals of the fit, and describes the bootstrap bands and the robust set for the direction.

The score. Let x_{t-1} collect the VAR regressors and let $\tilde{z}_{jt}$ be the residual from regressing proxy j on them, the purged proxy. The VAR residuals $\hat{u}_t$ are orthogonal to the regressors, and so is $\tilde{z}_{jt}$. Hence, with $\hat{u}_t = u_t - (\hat{\Gamma} - \Pi)'x_{t-1}$, with Π the stacked VAR coefficients,

$$\hat{g}_j = \frac{1}{T} \sum_t \hat{u}_t z_{jt} = \frac{1}{T} \sum_t \hat{u}_t \tilde{z}_{jt} = \frac{1}{T} \sum_t u_t \tilde{z}_{jt}, \quad (64)$$

with sums over the moment sample. The identity is exact when the VAR and the moments share a sample; when the moment window lies inside the VAR window, as in the application, the middle equality holds up to a term of the order of the sampling error. Two consequences follow. Estimating the VAR adds no sampling error to the moments at first order. And the variance of $\hat{g}_j$ is governed by the purged proxy $\tilde{z}_j$, not the raw one: the part of a proxy that the VAR's information set predicts cannot move the moment, and must not be counted in its standard error.

Assumption E.1 (Data). (y_t, z_t) is stationary and weakly dependent (α -mixing), with enough finite moments and fast enough decay of dependence for a central limit theorem; each proxy's share of the sample converges, $T_j/T \rightarrow \kappa_j \in (0, 1]$.

The two components. Write $\hat{q} = q_p + \delta$ with $\delta \perp q_p$, and $P_{q_p}^\perp := I - q_p q_p'$. Linearising the leading eigenvector of the within scatter around $\sigma_\phi^2 q_p q_p'$, as in the proof of Proposition 4, gives $\delta = \sigma_\phi^{-2} P_{q_p}^\perp \sum_j \tilde{\phi}_j \tilde{e}_j^w$, where $\tilde{e}_j^w$ is the whitened within-demeaned error of proxy j : sampling error plus the idiosyncratic term $\mathcal{Q}_{-p} \tilde{v}_j$. Using (64) for the first,

$$\delta = \underbrace{\frac{P_{q_p}^\perp}{\sigma_\phi^2} \frac{1}{T} \sum_{t=1}^T \Sigma_u^{-1/2} u_t \sum_j \tilde{\phi}_j \tilde{z}_{jt}}_{\text{sampling, } O_p(T^{-1/2})} + \underbrace{\frac{P_{q_p}^\perp \mathcal{Q}_{-p}}{\sigma_\phi^2} \sum_j \tilde{\phi}_j \tilde{v}_j}_{\text{heterogeneity, } O_p(m^{-1/2})} + o_p(T^{-1/2} + m^{-1/2}), \quad (65)$$

with $\tilde{z}_{jt}$ the purged proxy demeaned within its family (weighted by $\kappa_j^{-1} \mathbf{1}\{t \in S_j\}$ on unequal supports). The sampling term is a sample mean of a weakly dependent series. The heterogeneity term does not depend on T , has mean zero by (26), and is uncorrelated with the sampling term, which has mean zero given $\{\tilde{v}_j\}$. Three refinements: when the idiosyncratic scatter is a sizeable fraction of σ_ϕ^2 , the exact heterogeneity term replaces σ_ϕ^{-2} by $(\sigma_\phi^2 I - N)^{-1}$, as in (31); with the weights ω_j of Algorithm 1, $\tilde{\phi}_j$ becomes $\omega_j \tilde{\phi}_j$ in both sums and σ_ϕ^2 becomes $\sum_j \omega_j \tilde{\phi}_j^2$; and estimating Σ_u adds a first-order term under random effects only (Section 5.4).

Proposition E.2 (Large-sample properties). Under Assumptions 1, 2 and E.1 and the ordering condition (32), with a known partition, the estimator of Algorithm 1 satisfies the following.

- (i) Consistency. Under exact block structure, $\hat{q} \rightarrow_p q_p$, $\hat{\beta} \rightarrow_p \beta$ and $\hat{\Theta}_h \rightarrow_p \Theta_h$ as $T \rightarrow \infty$ at fixed m . Under random effects at fixed m , $\hat{q}$ converges to the leading eigenvector q_1^* of the realised S_0 in (30), which differs from q_p by the heterogeneity term of (65); $\hat{q} \rightarrow_p q_p$ as $m, T \rightarrow \infty$ with $m/T \rightarrow 0$.
- (ii) Distribution. Under exact block structure, $\sqrt{T} \delta \Rightarrow N(0, \Xi^*)$, with Ξ^* the long-run variance of the sampling score $\eta_t^* := \sigma_\phi^{-2} P_{q_p}^\perp \Sigma_u^{-1/2} u_t \sum_j \tilde{\phi}_j \tilde{z}_{jt}$, and $\hat{\beta}$ and $\hat{\Theta}_h$ are asymptotically normal by the delta method. Under random effects the same holds conditionally on $\{\tilde{v}_j\}$, around q_1^* , and unconditionally

$$\text{Var}(\delta) \approx \frac{1}{T} \Xi^* + \frac{1}{\sigma_\phi^2} P_{q_p}^\perp \mathcal{Q}_{-p} \Sigma_v \mathcal{Q}_{-p}' P_{q_p}^\perp. \quad (66)$$

- (iii) Feasible variance. Ξ^* is estimated by a heteroskedasticity- and autocorrelation-consistent (HAC) estimator applied to $\hat{\eta}_t^*$, and Σ_v by $\hat{\Sigma}_v$ in (68) below.
- (iv) Efficiency. If the sampling errors of the whitened moments are uncorrelated across proxies and coordinates, with a common variance σ^2 , then $\Xi^* = (\sigma^2 / \sigma_\phi^2) P_{q_p}^\perp$, the efficiency bound for the model $g_j = \phi_j b_p + d_c$. When the variances differ across proxies, the bound is attained by weighting each proxy by the inverse of its sampling variance, the precision-weighted version of Algorithm 1.

Sketch of proof. All parts follow from (65): a central limit theorem for weakly dependent series applies to the sampling term, the law of large numbers across proxies to the heterogeneity term,

and the delta method carries both to $\hat{\beta}$ and $\hat{\Theta}_h$. Part (iv) is the Gauss–Markov argument for the weighted least-squares fit (33). $\square$

Estimating the two components. Let $\hat{E} := [\hat{r}_1, \dots, \hat{r}_m]$ collect the whitened within residuals of Algorithm 1, $\hat{E} = \hat{G}^w \hat{A}'$, where $\hat{A}$ is the linear map from the moments to the residuals: within-family demeaning followed by removal of the fitted contrasts. It has rank $m - C - 1$, one lost per family and one to the fitted direction. Let $\hat{V}$ be the sampling covariance of the proxies' moments under the purged score, $\hat{V}_{jk} = \sum_{t \in S_j \cap S_k} \tilde{z}_{jt} \tilde{z}_{kt} / (T_j T_k)$, with diagonal $\hat{v}_j$. Its off-diagonal elements matter: proxies built from the same announcements or the same VAR are correlated up to 0.99. With $\hat{e}_i$ the residuals in the i th of $n - 1$ orthonormal coordinates orthogonal to $\hat{q}$, the overidentification statistic is

$$J_T := \sum_{i=1}^{n-1} \hat{e}_i' (\hat{A} \hat{V} \hat{A}')^+ \hat{e}_i, \quad \text{df} := (n - 1)(m - C - 1), \quad (67)$$

on the degrees of freedom of (60). Under exact block structure, and when the sampling covariance of the whitened moments is the same in every coordinate, $J_T \Rightarrow \chi_{\text{df}}^2$. It is Cochran's Q of meta-analysis (Cochran, 1954), extended to correlated studies; with independent proxies of common variance $\hat{v}$ and equal weights it is $\sum_j \|\hat{r}_j\|^2 / \hat{v}$. The sampling variances must enter here, not by standardising the proxies before the moments are formed, which destroys block structure (footnote 12).

Under random effects the residuals also carry heterogeneity, and J_T exceeds df. The heterogeneity covariance is estimated by

$$\hat{\Sigma}_v := \frac{\left[\hat{E} \hat{E}' - \text{tr}(\hat{A} \hat{V} \hat{A}') P_{\hat{q}}^\perp \right]_+}{\text{tr}(\hat{A} \hat{A}')} \quad (68)$$

the matrix form of the DerSimonian–Laird estimator (DerSimonian and Laird, 1986), where $[\cdot]_+$ sets negative eigenvalues to zero. The subtracted term is the residual scatter that sampling error alone would produce, and the denominator is the number of residual degrees of freedom ($m - C - 1$ with equal weights). $\hat{\Sigma}_v$ enters the variance (66) and the ordering condition (Appendix E.7).

Bootstrap bands. The bands come from the residual moving-block bootstrap of Jentsch and Lunsford (2022), which is consistent for the joint law of the VAR coefficients, $\hat{\Sigma}_u$ and the moments. When the leading eigenvalue is simple the estimator is a smooth function of these, so consistency carries over to the direction and the responses by the delta method. Every step after the VAR is repeated in each draw; only the economic partition is held fixed, and if C is chosen from the data, that choice is repeated too. The bootstrap resamples over time with the proxies fixed, so the bands reflect sampling uncertainty and not the heterogeneity component. The small-sample bias of OLS in the VAR coefficients shifts the draws, so a raw percentile band is centred on the bootstrap median rather than on the estimate. The reported band keeps the bootstrap dispersion and recentres it on the estimate $\hat{\theta}$, the median-recentred percentile interval

$$\left[\hat{\theta} + q_{\alpha/2}(\theta^*) - \text{med}(\theta^*), \hat{\theta} + q_{1-\alpha/2}(\theta^*) - \text{med}(\theta^*) \right], \quad (69)$$

with θ^* the bootstrap draws of a response.

The robust set for the direction. The expansion (65) is reliable only when the first eigenvalue of the within scatter is well separated from the second, the counterpart for the direction of a strong first stage. The robust set does not rely on it. It uses the criterion itself,

$$\text{LR}(q_0) = \frac{Q^c(q_0) - Q^c(\hat{q})}{Q^c(\hat{q})}, \quad Q^c(q) = \text{tr } \hat{S} - q' \hat{S} q, \quad (70)$$

the proportional rise in the criterion when the direction is held at q_0 instead of fitted, and collects the q_0 it does not reject. It needs no variance estimate. The critical values are simulated, because the χ^2 approximation is too generous at low signal-to-noise: at each q_0 , draw within-scatter data under the null, $q_0(q_0' \tilde{G}^w) + E^*$ with independent normal entries in E^* of variance $Q^c(q_0)/((n-1)(m-C))$, within-demean, recompute the statistic, and keep q_0 if $\text{LR}(q_0)$ is below the simulated $(1-\alpha)$ quantile. The set is traced along rays from $\hat{q}$. When $\hat{S}$ is close to rank one it contains every direction within a fixed angle of $\hat{q}$, and that angle summarises how sharply the direction is identified. Mapping each retained q_0 through $b_p = \Sigma_u^{1/2} q_0$ and the moving-average coefficients Φ_h gives a direction-robust band for the responses. The set has correct asymptotic coverage under strong identification; under weak identification its validity is conjectured, not proved.

E.7 Diagnostics, and the audit of the partition

Two diagnostics, both computed from the within residuals of Algorithm 1, are reported with every estimate.

The ordering condition. The finite-panel condition of Proposition 4(ii) is checked as $\hat{\sigma}_\phi^2 > \lambda_{\max}(\hat{N})$, with $\hat{N}$ the scatter of the whitened within residuals. The large-panel condition (32) is checked as $\hat{\sigma}_\phi^2/m > \hat{\kappa} \lambda_{\max}(\hat{\Sigma}_v)$ with $\hat{\kappa} = (m-C)/m$; dropping $\hat{\kappa}$ gives a conservative version. If the condition fails, the leading eigenvector is a contamination direction, not the target.

Do the families agree? J_T splits into a within-family and a between-family part. Fit each family with at least three proxies on its own, with direction $\hat{q}_c$, and let J_c be (67) for that family alone, on $(n-1)(|c|-2)$ degrees of freedom. Then

$$J_{\text{between}} := J_T - \sum_c J_c, \quad (n-1)(C-1) \text{ degrees of freedom.}$$

The between part asks whether the families agree on the direction. It detects contamination correlated with relevance that differs across families; a component common to all families cannot be detected, since it is observationally equivalent to a shifted target (Proposition 3(ii)). The within part tests exact block structure inside each family. When it rejects, residuals aligned with the family's own $\hat{d}_c$ point to differences in scale across its members (footnote 12), and residuals orthogonal to it point to idiosyncratic contamination, which $\hat{\Sigma}_v$ measures. Because the families' residuals are correlated through $\hat{V}$, the χ^2 reference is approximate for each part, though not for J_T itself. Regressing $\hat{r}_j$ on $\hat{\phi}_j$ is not a test: the coefficient is zero at the fitted direction by construction.

The audit of the partition. Section 5.5 explains why the economic partition is audited rather than replaced. The audit minimises (33) jointly over the direction and the assignment. Once the target

is removed, each proxy's residual is approximately its family's common contamination d_c , so the residuals can be clustered.

1. *Candidate directions.* Take the $\binom{m}{2}$ pairwise differences of the whitened moments, $\hat{g}_j^w - \hat{g}_k^w$.
2. *Clustering.* For each candidate q , form the residuals $P_q^\perp \hat{g}_j^w$, cluster them into C groups of at least three proxies by size-constrained k -means, and evaluate the concentrated criterion $Q^c(q; \mathcal{P})$ under that partition $\mathcal{P}$.
3. *Refinement.* Keep the pair $(q, \mathcal{P})$ with the smallest criterion, then alternate: re-cluster given the direction, re-estimate the direction given the clusters.

Searching over directions rather than partitions keeps the problem tractable, and the minimum family size of three rules out the degenerate answer of a partition into pairs.

Selecting C . The criterion always falls when a family is added, so the count is chosen by a penalised criterion, $m^{-1}Q^c(\hat{q}; \hat{\mathcal{P}}_C) + \gamma(n-1)C$ with a fixed penalty γ . The scree of the residual scatter and the silhouette are not used as selectors: both reward separation, so they merge two families that share a contamination direction at different magnitudes, which is exactly the case $C > r$ in which the level component of (51) is available. They are reported, with the adjusted Rand index, as measures of separation. By the two order conditions a selected $\hat{C}$ must satisfy

$$\hat{r}_{\text{centroid}} < \hat{C} \leq m-1, \quad \text{and, with the minimum family size, } \hat{C} \leq m/3, \quad (71)$$

the lower end being (O2) and the upper end (O1). What is reported is the penalised criterion over C , whether the fit at the economic $\hat{C}$ reproduces the economic partition, and, if not, which proxies move.

F Supplementary tables and robustness figures

This appendix collects the tables and supplementary figures of Section 6 and the tables and figures of Section 7.

F.1 Tables for the empirical application

Table 3: The proxy panel. Sixteen purged proxies in five families. Support is the span of non-missing months inside the 1988m1–2019m12 moment window; T_j its length. PTA: purge-then-aggregate; ATP: aggregate-then-purge; FWL: purged jointly with the instrument’s own first stage.

Family	Proxy	Support	T_j
High-frequency surprises	MP1, orthogonalised (PTA)	1994m02–2019m12	311
	FF4, orthogonalised (PTA)	1994m02–2019m12	311
	ED4, orthogonalised (PTA)	1994m02–2019m12	311
	Bauer–Swanson MPS, orthogonalised (PTA)	1988m02–2019m12	383
Swanson factors	Swanson federal funds factor, orthogonalised (PTA)	1994m02–2019m12	311
	Swanson forward guidance factor, orthogonalised (PTA)	1994m02–2019m12	311
	Swanson asset purchase factor, orthogonalised (PTA)	1994m02–2019m12	311
Information-robust (MAR)	Miranda–Agrippino–Ricco, orthogonalised (PTA)	1994m02–2019m12	311
	Miranda–Agrippino–Ricco, orthogonalised (PTA, FWL)	1994m02–2019m12	311
	Miranda–Agrippino–Ricco, orthogonalised (ATP)	1994m02–2019m12	311
Romer–Romer	Romer–Romer, orthogonalised (PTA)	1988m02–2019m12	383
	Romer–Romer, orthogonalised (PTA, FWL)	1988m02–2019m12	383
	Romer–Romer, orthogonalised (ATP)	1988m02–2019m12	383
Recursive (Cholesky)	Cholesky, federal funds rate, orthogonalised (ATP, FWL)	1988m02–2019m12	383
	Cholesky, 1-year Treasury, orthogonalised (ATP, FWL)	1988m02–2019m12	383
	Cholesky, 2-year Treasury, orthogonalised (ATP, FWL)	1988m02–2019m12	383

Table 4: Partitions that satisfy the assignment rule. P2: purged families; P3: raw families. Responses of output ($h = 0, 12, 24$), prices ($h = 24$) and the excess bond premium ($h = 0$) to a +1 percentage-point funds-rate shock; F of the pooled instrument; J_T with the full sampling covariance; $\angle$: angle to the unweighted P2 direction; last column: the family carrying the largest share of the identifying variation.

Partition	Swanson	Weights	Output			Prices	Spread	F	J (df)	$\angle$	Largest family share
			$h = 0$	12	24	24	0				
P2 (purged)	with	unweighted	-2.22	-8.23	-6.33	-2.25	+1.59	11.2	43.8 (30)	0°	SW 99%
P2 (purged)	with	precision	-1.33	-5.30	-4.39	-1.00	+0.93	23.9	46.0 (30)	12°	SW 79%
P2 (purged)	without	unweighted	+0.07	-1.88	-2.08	+0.16	+0.35	43.6	38.0 (24)	64°	CH 99%
P2 (purged)	without	precision	-0.13	-1.97	-2.15	+0.23	+0.28	50.6	42.2 (24)	56°	CH 93%
P3 (raw)	with	unweighted	-2.27	-4.87	-4.13	-0.42	+0.36	12.5	38.4 (21)	26°	SW 92%
P3 (raw)	with	precision	-1.52	-3.96	-3.53	-0.08	+0.33	21.6	37.0 (21)	26°	CH 52%
P3 (raw)	without	unweighted	-0.88	-3.24	-3.05	+0.14	+0.34	55.6	32.9 (15)	32°	CH 99%
P3 (raw)	without	precision	-1.05	-3.42	-3.18	+0.13	+0.32	52.2	33.3 (15)	30°	CH 91%

Table 5: The headline and its main variants. T : months in the moment window. F and μ^2 : first stage and concentration of the pooled instrument. J_T : overidentification statistic with the full sampling covariance; J_{between} : its between-family part. λ_1/λ_2 : eigengap of the weighted within scatter. Responses to a +1 percentage-point shock to the policy indicator; the output trough is the largest absolute response, with its horizon.

Specification	T	F	μ^2	J_T (df)	p	J_{between} (df)	I^2	λ_1/λ_2	IP_0	IP trough (h)	CPI_0	EBP_0
Headline (funds rate)	384	23.9	22.4	46.0 (30)	0.031	28.2 (12)	0.35	3.1	-1.33	-5.35 (14)	-0.09	+0.93
Unweighted	384	11.2	21.0	43.8 (30)	0.050	26.3 (12)	0.32	4.2	-2.22	-8.24 (13)	-0.33	+1.59
Shadow rate	384	18.1	19.8	43.3 (30)	0.055	27.1 (12)	0.31	2.8	-1.56	-6.40 (14)	-0.14	+1.14
VAR 1985–2008	252	25.4	33.5	50.7 (30)	0.011	32.2 (12)	0.41	8.7	-1.18	-8.12 (29)	-0.01	+0.91

Table 6: The families in the headline. Share of the identifying variation; angle between the family’s own direction and the pooled one; the family’s own overidentification statistic J_c against its own direction.

Family	$ c $	share of $\hat{\sigma}_\phi^2$	own direction ($^\circ$)	J_c (df)	p
High-frequency surprises	4	0.02	75	3.6 (6)	0.725
Swanson factors	3	0.79	10	7.7 (3)	0.052
Information-robust (MAR)	3	0.00	77	1.3 (3)	0.733
Romer–Romer	3	0.00	89	1.8 (3)	0.608
Recursive (Cholesky)	3	0.19	37	3.3 (3)	0.350

Table 7: The family-common component. Contamination share: fraction of the family’s mean moment (in the Σ_u^{-1} metric) orthogonal to the pooled direction. Remaining columns: response implied by the family mean minus the pooled response, on impact and at twelve months.

Family	contam. share	IP_0	CPI_0	EBP_0	IP_{12}	CPI_{12}	EBP_{12}
High-frequency surprises	0.12	+0.04	-0.51	+0.87	-1.99	-1.40	+0.16
Swanson factors	0.10	+0.54	-0.28	+0.41	-0.20	-0.68	+0.08
Information-robust (MAR)	0.05	-0.92	-0.34	+0.78	-3.21	-1.15	+0.15
Romer–Romer	0.43	+1.32	+0.01	-0.71	+3.94	+0.59	-0.16
Recursive (Cholesky)	0.81	+1.29	+0.03	-0.96	+4.68	+0.79	-0.22

Table 8: Single proxies and pools on the headline VAR and moment window. F : first stage against the funds-rate innovation. Responses to a +1 percentage-point funds-rate shock; the output trough is the largest absolute response, with its horizon.

Identification	F	IP_0	IP trough (h)	CPI_0	CPI_{24}	EBP_0
Bauer–Swanson surprise, raw	14.5	-0.28	-1.93 (20)	-0.15	-0.11	+0.25
Bauer–Swanson surprise, orthogonalised	3.6	-0.52	-6.58 (13)	-0.83	-3.00	+1.97
Swanson funds-rate factor, PTA	14.2	-2.28	-6.84 (13)	-0.11	-1.33	+1.03
Romer–Romer, PTA	22.3	-0.13	-1.69 (23)	-0.11	+0.05	+0.19
MAR information-robust, PTA	6.3	-2.20	-8.35 (13)	-0.46	-2.53	+1.69
Cholesky funds rate, ATP-FWL	533.5	+0.01	-1.40 (30)	-0.03	+0.34	+0.05
HF and Swanson families only	7.2	-2.63	-8.37 (13)	-0.22	-1.99	+1.42
All 16 proxies pooled, no within transform	32.8	-0.67	-3.53 (15)	-0.21	-0.73	+0.66
Pooled estimate (headline)	23.9	-1.33	-5.35 (14)	-0.09	-1.00	+0.93

Table 9: Adding families one at a time (precision-weighted, headline VAR and window). (O2) is met in its conservative form when $C > n - 1 = 3$. Surv. rel.: share of the precision-weighted relevance vector that survives the annihilator.

Families	m	C	$C > n - 1$	F	surv. rel.	IP_0	IP trough (h)	CPI_0	CPI_{24}	EBP_0
HF + CH	7	2	no	50.6	0.12	-0.16	-2.22 (19)	+0.10	+0.23	+0.27
HF + CH + RR	10	3	no	50.5	0.12	-0.15	-2.22 (19)	+0.10	+0.23	+0.28
HF + CH + RR + SW	13	4	yes	23.8	0.43	-1.34	-5.34 (14)	-0.09	-0.99	+0.92
HF + CH + RR + SW + IR	16	5	yes	23.9	0.39	-1.33	-5.35 (14)	-0.09	-1.00	+0.93

F.2 The assignment rule in the data

The assignment rule of Section 5.5 places a raw and a purged version of the same instrument in different families, and the data show why it must. Write the raw proxy as the purged one plus the component the purge removed, $z^{\text{raw}} = z^{\text{pur}} + \hat{z}$. In a family holding only the two, the within transform leaves $\tilde{g}^{\text{raw}} = -\tilde{g}^{\text{pur}} = \frac{1}{2} \text{Cov}(u, \hat{z})$: the contamination the two share cancels exactly, but the only variation the family feeds the estimator is the moment of $\hat{z}$, the component that is contaminated by construction. Its relevance and its contamination come from the same series, which is the relevance-correlated case that Assumption 1 rules out.

In the data 55–99% of each $\hat{z}$ contrast (in the Σ_u^{-1} metric) is orthogonal to the target direction; putting the raw/purged pairs into the panel moves the direction by 6° with the Swanson family and by $75\text{--}81^\circ$ without it, and J_T rejects exact block structure with $p < 10^{-9}$, against $p = 0.05$ for the purged-only partition. Grouping the two wide-window instruments, Romer–Romer and Cholesky, into one purged family, which is the design the window argument of Section 4 suggests, gives the same direction as separate families (0.1°) but fits no better ($p = 0.004$), so the families are kept separate.

F.3 Figures for the empirical application

F.4 Robustness

Table 10: Leaving one family out. Angle: between the estimated impact vector and the headline one, in the Σ_u^{-1} metric. J_T tests exact block structure in the four-family pool.

Family dropped	m	F	μ^2	J_T (df)	p	angle ($^\circ$)	IP_0	IP trough (h)	CPI_{24}	EBP_0
None (headline)	16	23.9	22.4	46.0 (30)	0.031	0.0	-1.33	-5.35 (14)	-1.00	+0.93
High-frequency surprises	12	24.8	22.7	39.9 (21)	0.008	3.4	-1.22	-5.29 (14)	-1.05	+0.97
Swanson factors	13	50.6	10.8	42.2 (24)	0.012	44.7	-0.13	-2.22 (19)	+0.23	+0.28
Information-robust (MAR)	13	23.8	22.4	25.1 (24)	0.401	0.2	-1.34	-5.34 (14)	-0.99	+0.92
Romer–Romer	13	22.7	22.0	36.6 (24)	0.048	0.1	-1.34	-5.36 (14)	-1.00	+0.93
Recursive (Cholesky)	13	9.8	18.4	41.4 (24)	0.015	14.3	-2.62	-8.35 (13)	-1.99	+1.42

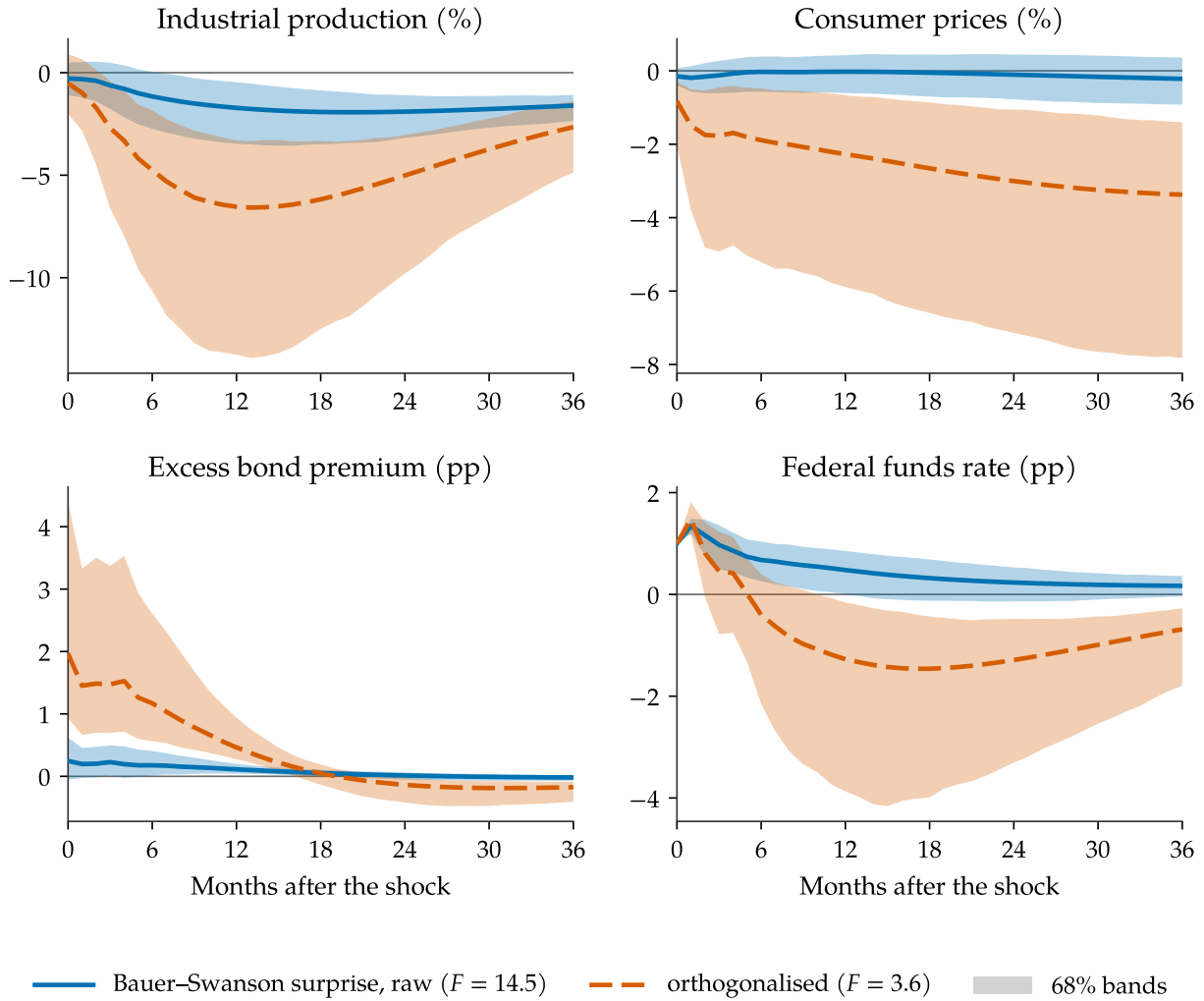


Figure 13: Purging the surprise costs it most of its first stage and enlarges its responses. Single-instrument responses to the [Bauer and Swanson \(2023b\)](#) surprise, raw (solid, $F = 14.5$) and orthogonalised on the six pre-announcement controls (dashed, $F = 3.6$), each with its 68% moving-block-bootstrap band. Same VAR and moment window as the headline. The weaker instrument reports the larger effect in every panel.

Table 11: Policy indicators (headline families, weights, VAR and window). Responses to a +1 percentage-point shock to the indicator.

Policy indicator	F	μ^2	IP_0	IP trough (h)	CPI_0	CPI_{24}	EBP_0
Federal funds rate	23.9	22.4	-1.33	-5.35 (14)	-0.09	-1.00	+0.93
Shadow rate (Wu-Xia at the ZLB)	18.1	19.8	-1.56	-6.40 (14)	-0.14	-1.39	+1.14
1-year Treasury yield	0.8	19.1	+4.69	+17.32 (12)	+0.64	+5.67	-3.76
2-year Treasury yield	43.5	38.6	+0.15	-0.99 (36)	+0.01	+0.77	-0.35

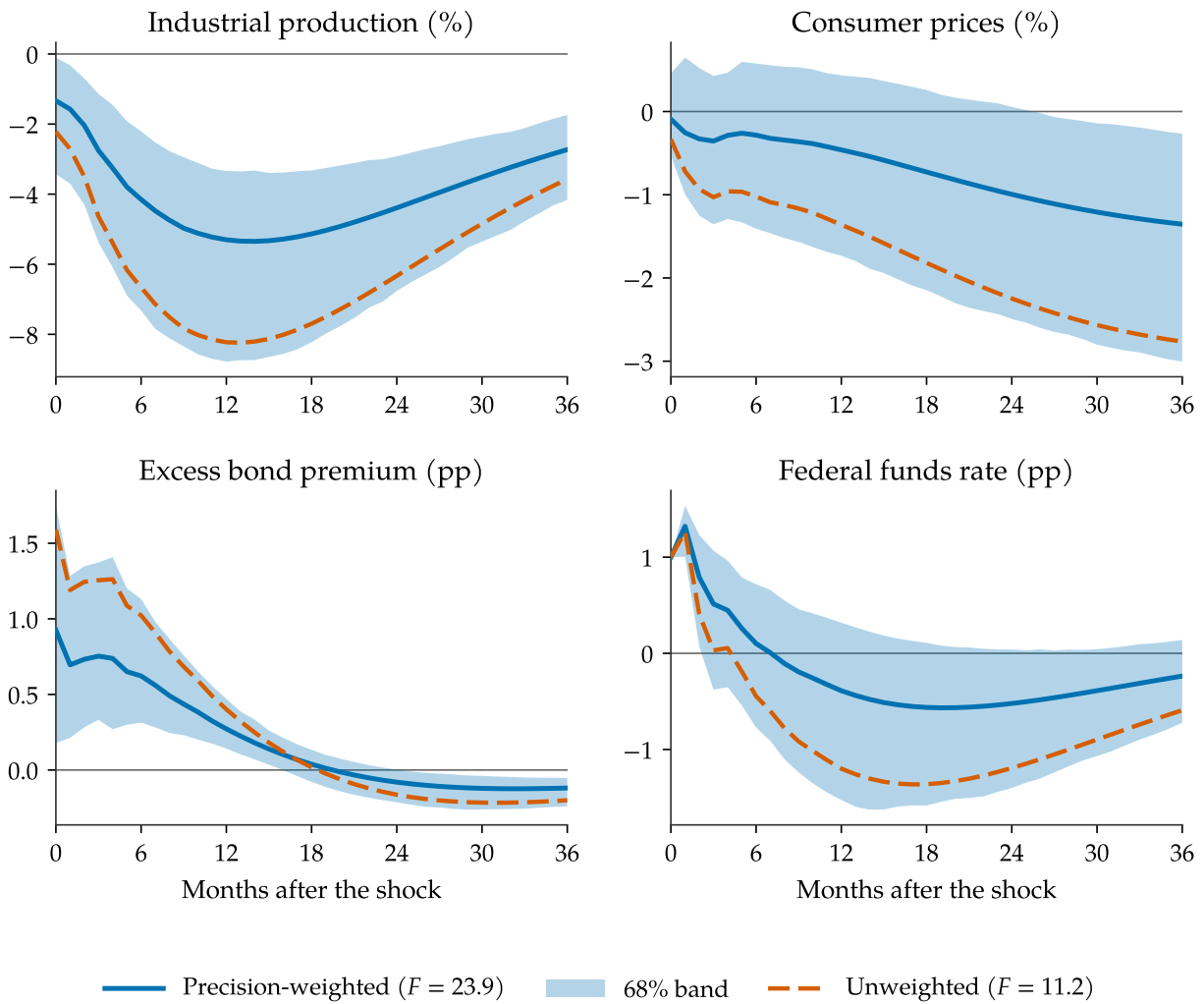


Figure 14: **Precision-weighted against unweighted.** Headline (solid, with 68% band) and the same estimator with equal weights (dashed). Same VAR, proxies and families.

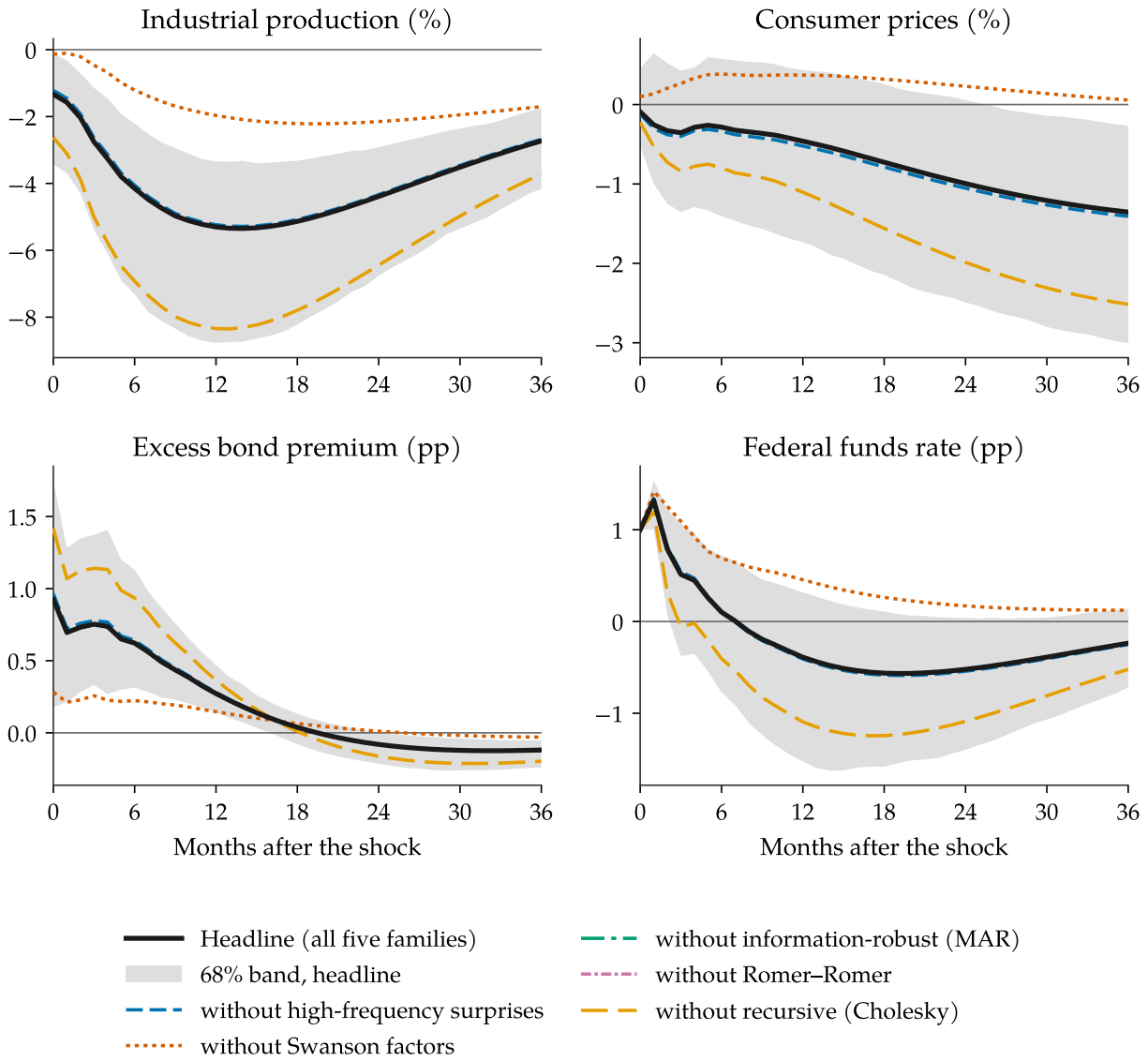


Figure 15: Leaving one family out. Headline (solid black, with its 68% band) and the estimate with each family dropped in turn, precision-weighted, same VAR and windows. The estimates without the information-robust and without the Romer-Romer family lie on the headline (angles 0.2° and 0.1°).

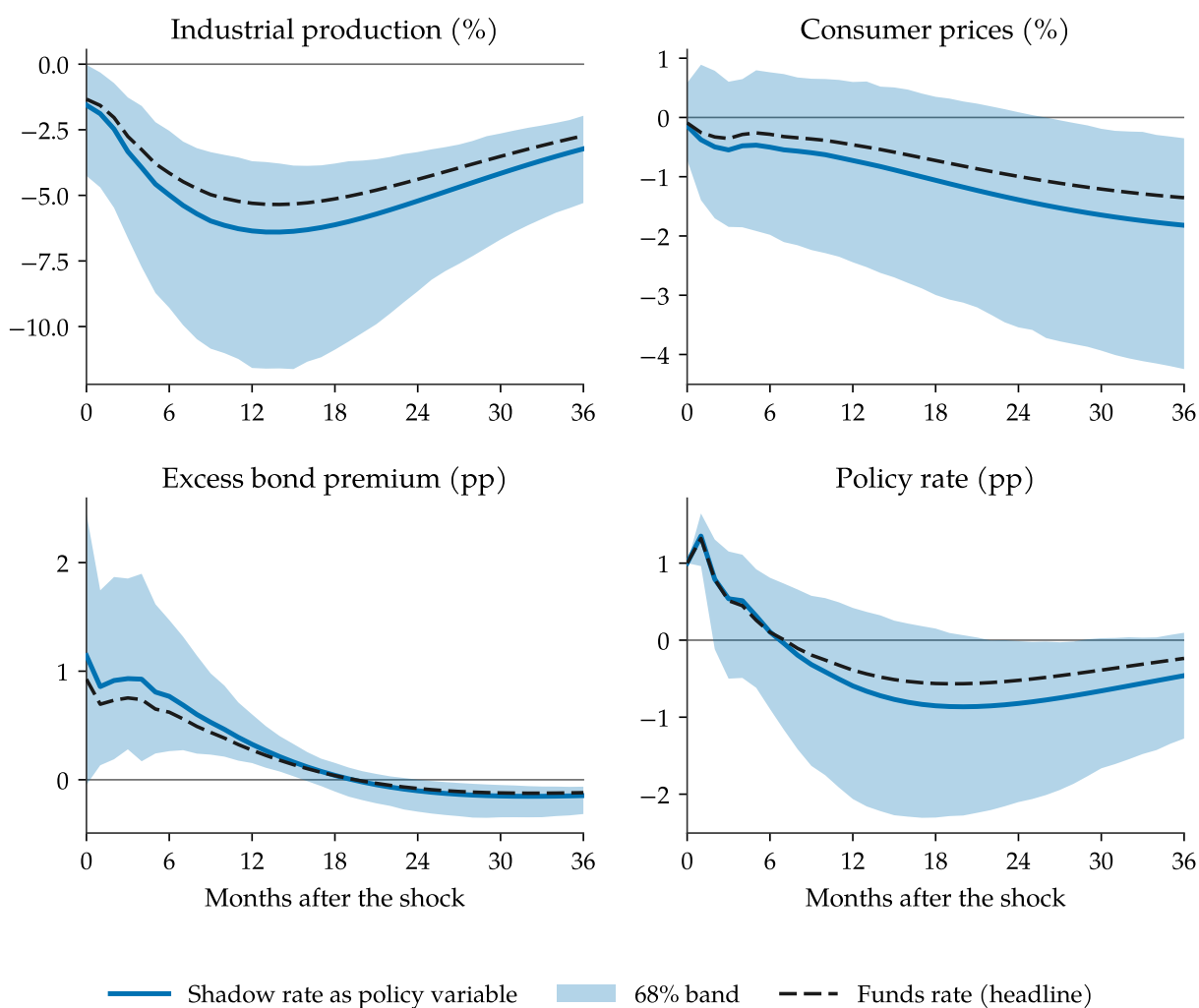


Figure 16: The shadow rate as policy indicator. Responses to a +1 percentage-point shock to the shadow rate (the funds rate away from the zero lower bound, the [Wu and Xia \(2016\)](#) shadow rate at it), with 68% bands; the funds-rate headline is dashed. The bottom-right panel is the response of each specification's own policy rate.

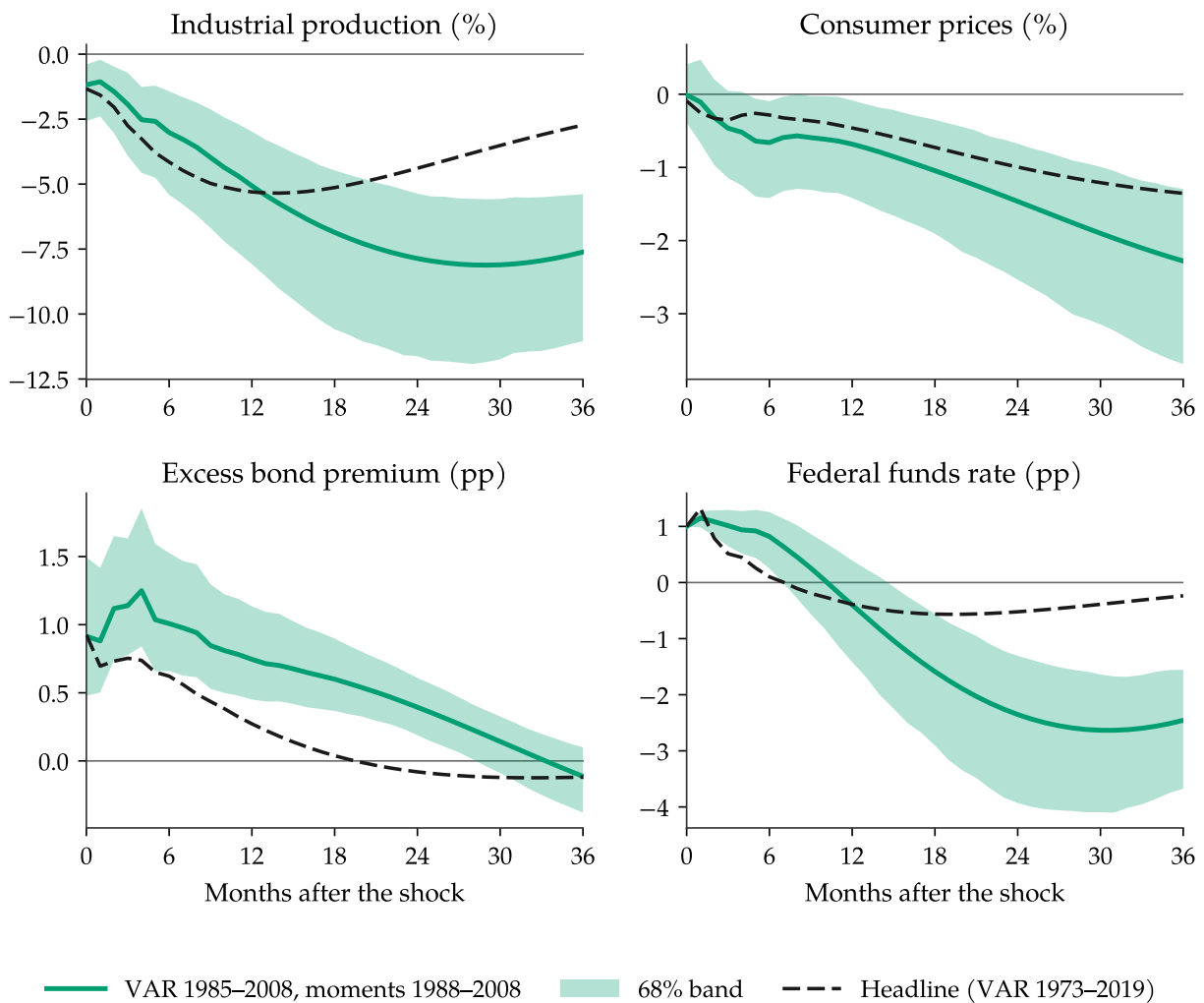


Figure 17: **A pre-zero-lower-bound VAR.** VAR on 1985m1–2008m12 and moments on 1988m1–2008m12 (solid, with 68% band) against the headline (dashed).

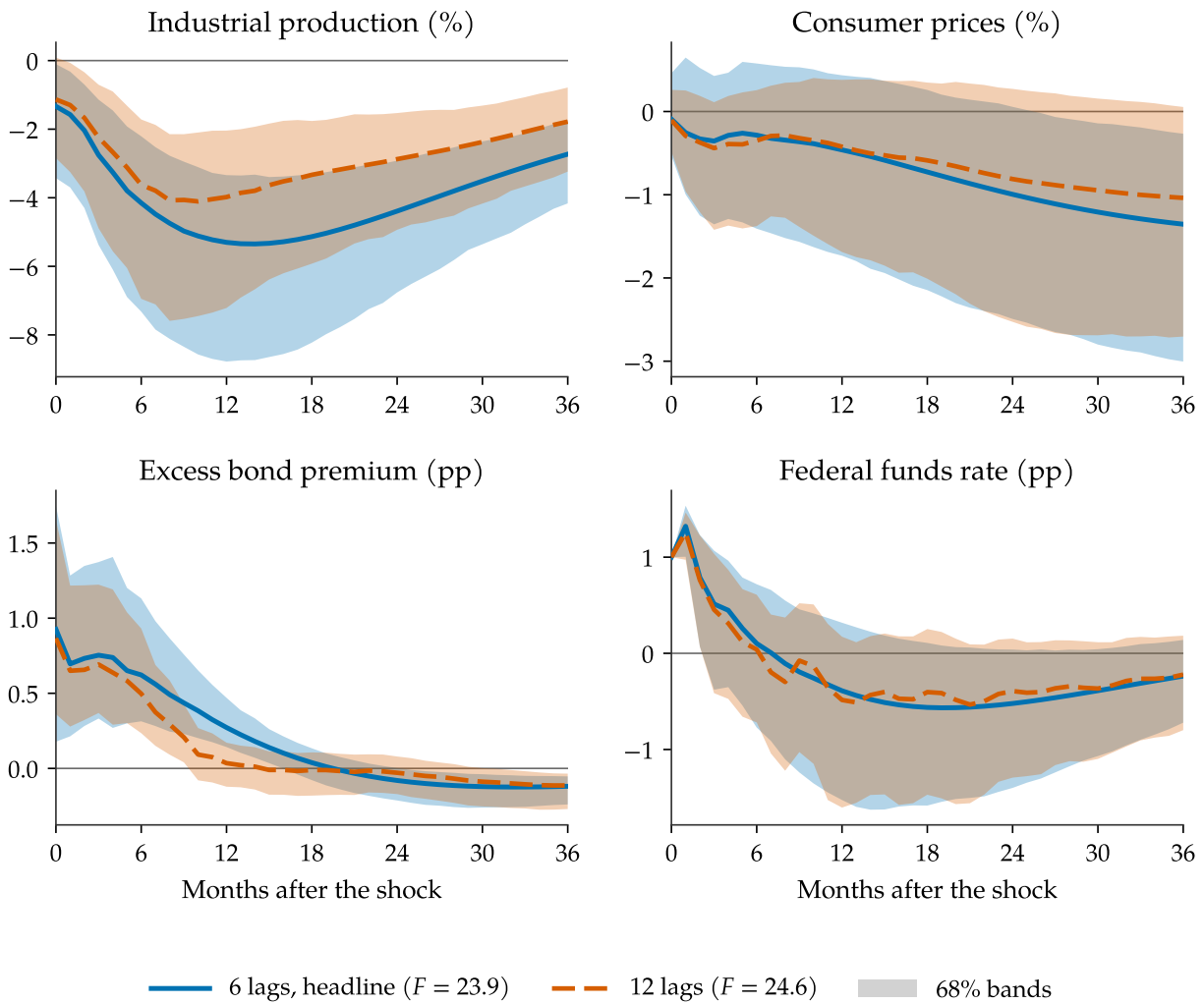


Figure 18: Lag length. The headline (six lags, solid) and the same estimator with twelve lags (dashed), each with its 68% band.

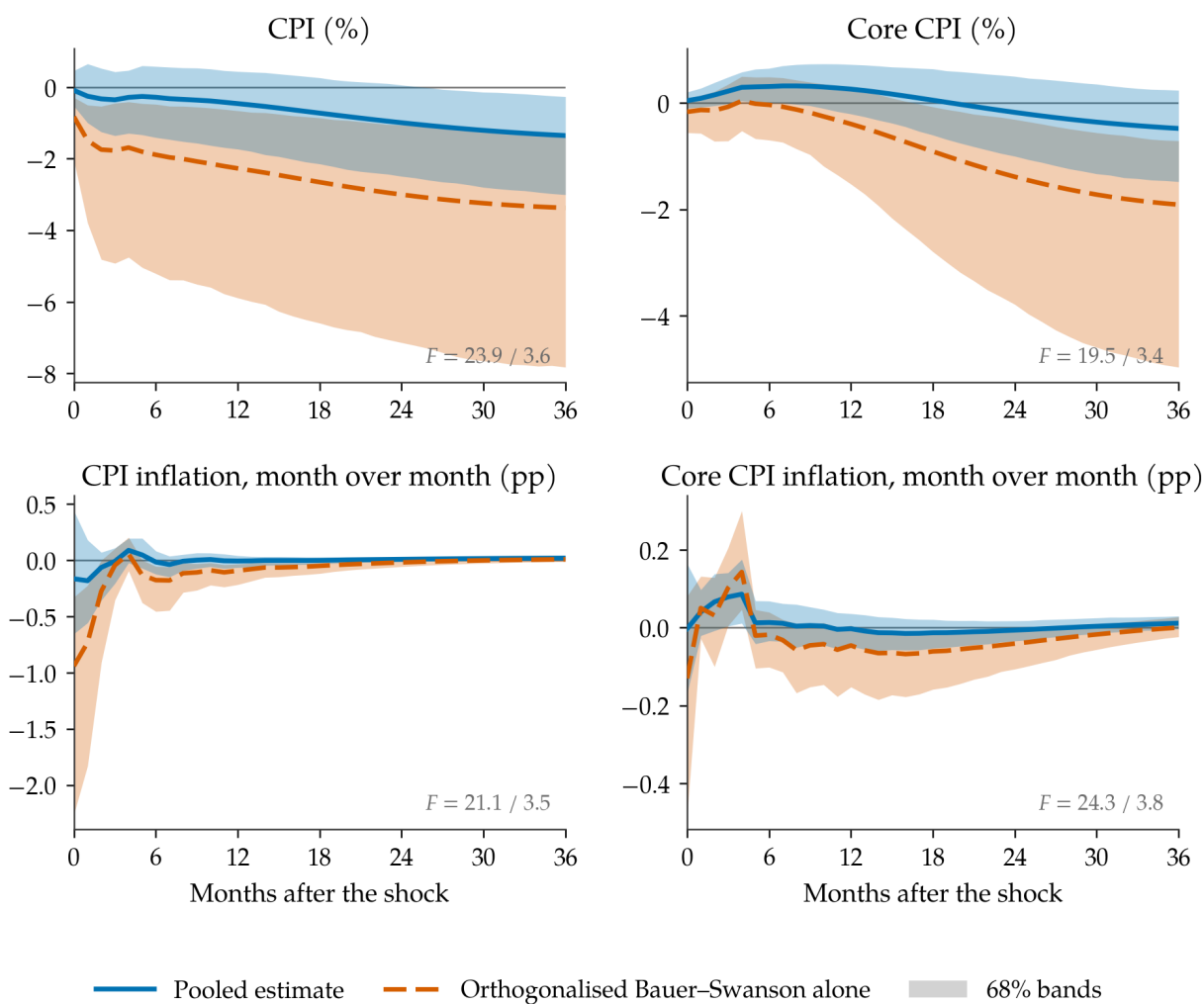


Figure 19: The price measure: levels and month-over-month inflation. Responses to a +1 percentage-point funds-rate shock of the CPI and core CPI (top, $100 \times \log$ level) and of month-over-month headline and core inflation (bottom, percentage points), each from a separate four-variable VAR in which the price variable is the one shown. Pooled estimate (solid) and the orthogonalised [Bauer and Swanson \(2023b\)](#) surprise alone (dashed), each with its 68% band. Same families, precision weights, windows and bootstrap as the headline. F : first stage of the pooled instrument / of the orthogonalised surprise.

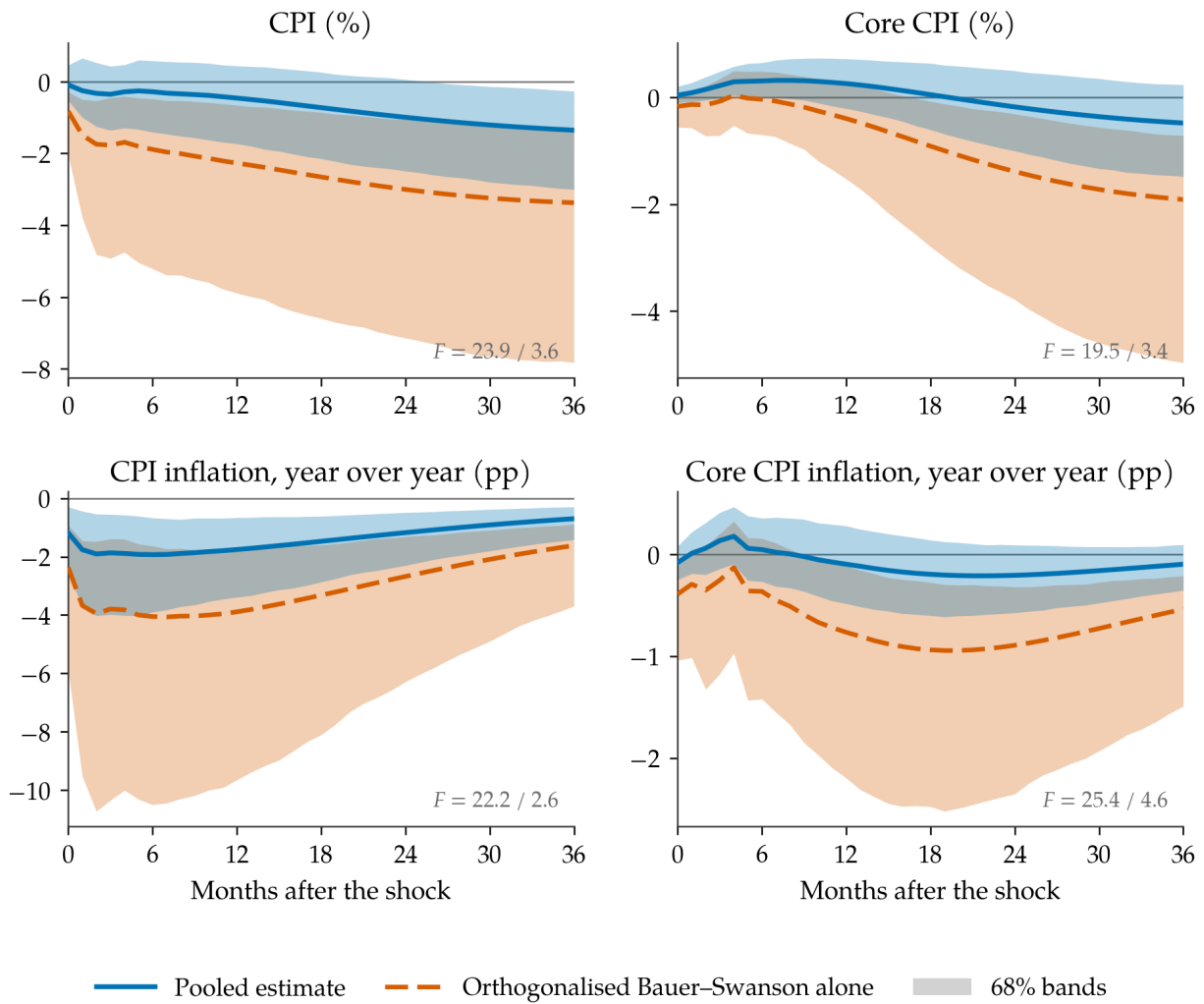


Figure 20: The price measure: levels and year-over-year inflation. As Figure 19, with year-over-year headline and core inflation, $100(\log P_t - \log P_{t-12})$, in the bottom row.